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MASTER'S DEGREE THESIS

Multi-Modal Drone Detection with Trust-Aware Sensor Fusion and Confuser-Aware Gating

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Dedicated to...

Abstract

Protecting a fixed site against consumer drones demands high recall and a low false-alarm rate at once. The single-stage detectors trained in this work are already strong on ordinary footage; the false-alarm problem is specific to scenes containing aerial *confusers* (birds, airplanes, helicopters), where the production RGB detector fires on 30.4% of an out-of-distribution confuser corpus, and suppressing confusers by retraining the detector collapses small-drone recall (0.961 → 0.306). This thesis builds and evaluates, under one declared protocol, a drone-detection *system* that moves confuser discrimination off the detector: an RGB and a thermal detector for day and night, a per-frame trust classifier that routes between them, a per-detection MLP *confuser filter* that reads the detector's own internal features and detection metadata to suppress the false-positive classes, and a temporal aggregator that rejects transient hallucinations, behind an operator GUI.

Because the bare detectors are already good, the pipeline must prove itself where they fail. Under one standard with bootstrap confidence intervals, the full pipeline lifts Svanström paired drone F1 from 0.742 to 0.946 while *raising* recall (0.948 → 0.991); on the out-of-distribution confuser corpus it cuts frame fire from 30.4% to 1.4%; and on the saturated Anti-UAV control it does no harm (0.973 → 0.984). An ablation isolates the contribution of every stage. Two methodological contributions produce the components: a statistics-before-training instrument (the Model MRI) that measures what a detector's feature space can separate and then trains the confuser filters directly (positives = drone detections, negatives = confuser detections), and a human-in-the-loop data engine that co-develops the scarce-data thermal detector with its corpus. One finding emerged unplanned: the thermal-trained detector, run zero-shot on grayscale-converted RGB, still detects drones when the visible channel fails.

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Chapter 1

Introduction

1.1 Background

Consumer unmanned aerial vehicles (UAVs) are inexpensive, widely available, and increasingly implicated in unauthorised surveillance, perimeter incursion, and disruption of civil airspace operations [1], [2], [3]. Protecting a fixed site against them imposes two simultaneous requirements on any sensing system: *high recall*, because a missed drone is an unrecoverable failure of the system's purpose, and a *low false-alarm rate*, because frequent alerts for birds and passing aircraft risk desensitising the operator the system is meant to serve. The Counter-UAS (C-UAS) sensing surveys [1], [2], [3] document the limitations of each sensing modality: radar suffers on the small cross-section of quadcopter-class drones; radio-frequency detection requires an active command link and is defeated by autonomous pre-programmed flight, and more recently by fiber-optic-guided drones, fielded at scale since 2024, whose tethered control link emits no RF signature at all [4]; acoustic arrays degrade under wind, traffic, and aircraft noise. Visual detection with deep neural detectors is passive, low-cost, and context-rich, and is the modality adopted in this work.

Modern single-stage detectors are genuinely strong on this task. The YOLO family [5], as packaged in the Ultralytics framework [6], delivers high drone-class precision and recall at real-time rates: the production RGB detector of this thesis reaches $P = 0.989$, $R = 0.982$ on the Anti-UAV benchmark [7] (4,000 even-spread paired frames at `imgsz=640`; Anti-UAV frames appear in the detector's training corpus, so this is an in-distribution sanity floor rather than a generalisation claim, Section 2.2), raising only 41 false positives across those 4,000 frames, and its hallucination rate on its composite RGB test corpus is 2.8% of frames. On *ordinary* footage (sky, buildings, traffic, vegetation), the false-alarm problem is, for the detectors trained in this work, largely solved at the detector; Section 4.3.5 quantifies this with a background-tagged failure profile.

1.2 Problem Statement

The false-alarm problem is not generic; on much footage the detector never fires falsely at all. It is specific to scenes containing the three aerial confuser categories (birds, fixed-wing aircraft, and helicopters). On the `rgb_confusers_merged` test split — a 2,633-image out-of-distribution confuser corpus assembled for this thesis, two-thirds Svanström-sourced and fully unseen by the production detector's training (Section 3.1.2) — the same detector family fires on 30.4% of frames, rising to as much as 94% on bird-only Svanström frames at that benchmark's canonical `imgsz=1280` (the per-category breakdown is in Table 4.1; Figure 3.3 shows representative hallucinations). At surveillance ranges all four object classes present as small, low-texture silhouettes against sky, and the appearance gap that separates them is too narrow for the detector to exploit *at its decision layer* without collateral damage: as the next paragraph shows, retraining against confusers trades away drone recall. The separation does, however, demonstrably exist deeper in the detector, in its own intermediate feature space (Section 3.7), and exploiting it *there* instead of at the detection head is precisely the design of this thesis.

The obvious mitigation, retraining the detector with confuser images as hard negatives, has diminishing and *class-specific* returns: `hardneg_v3more`, trained with Svanström's confuser splits as explicit negatives, lowers helicopter and airplane fire but leaves bird fire essentially unchanged (Table 4.1). A more aggressive retraining (`retrained_v2`) does suppress bird fire, but collapses drone recall on Svanström from $R = 0.961$ to $R = 0.306$ at the same precision. The collapse is out-of-distribution-specific rather than a general defect: on its own in-domain test split the same weights score $F1 = 0.949$, *above* the baseline's 0.942, so in-domain selection would pick exactly the detector that fails on deployment-like small-drone footage. Detector recall, once lost, cannot be recovered downstream: no later stage adds detections the detector never proposed.

Two further gaps separate benchmark performance from field readiness. *Resolution*: small or distant drones fall below the resolvable floor at standard inference sizes and disappear from the detector entirely (Section 3.8.6); the per-size breakdown of Section 4.3.4 makes this concrete (on Svanström, where the median drone ground-truth $\sqrt{\text{area}}$ is 29.8 px in the native 640×512 frame, bare RGB recall falls from 0.90–0.96 in the 16–64 px buckets to 0.63 below 16 px). *Modality complementarity*: the visible-light channel degrades at night and under glare and low contrast, which is exactly where the thermal channel is strongest, so the system is designed dual-modality from the start, and which modality dominates *reverses* with the scene (Section 4.1.1): on Svanström the thermal detector leads (own-GT F1 0.940 vs the visible detector's 0.607), on Anti-UAV the visible detector leads (0.985 vs 0.961), and the routed system matches the stronger one on each (Table 4.9).

Any discrimination the detector cannot learn without costing recall must therefore be moved *off* the detector. This thesis relocates confuser discrimination, modality arbitration, and false-positive filtering into a downstream dual-modality system: a trust classifier that routes between an RGB and a thermal detector per frame, a per-detection *MLP confuser filter* (a small MLP that re-reads the detector's own 517-dimensional internal features and detection metadata, already computed for each detection, Section 3.8.11) that removes confuser false positives from the surviving detections, and a temporal aggregator that turns per-frame decisions into alerts. The base detector is left tuned purely for recall.

One evaluation surface comes from a deployment partner: a fixed perimeter-CCTV installation (SelCom) whose 1080p stream renders drones at a median $\sqrt{\text{area}}$ of $\approx$37 px in the native 1920×1080 frame amid urban clutter. The system is not deployed there; the partner provided RGB footage on which the RGB branch was fine-tuned and evaluated (the `ft4` fine-tune lifted SelCom validation F1 from the baseline's 0.145 to 0.591 while leaving the composite in-domain test corpus essentially unchanged, 0.942 $\rightarrow$ 0.929). The field gaps above were observed across *all* of the evaluation surfaces of this thesis, with the CCTV footage contributing the small-drone-in-clutter case. Two engineering constraints hold throughout: every component must run interactively on modest hardware with no specialised accelerators beyond a consumer GPU (the stage-cost budget of Section 4.3.7 quantifies this), and every reported number must correspond to a configuration the system can actually ship.

1.3 Research Questions

The thesis is organised around three research questions; answers are mapped to evidence in Section 5.1.

RQ1 (capability) Retraining the base detector is the cheap first answer to false alarms, and this work tried it first: hard-negative retraining either leaves the dominant confuser class untouched or collapses small-drone recall (Section 1.2). *Where retraining fails*, can a layered detection pipeline maintain or improve drone-detection performance while suppressing false positives, on both ordinary and confuser-rich scenes?

RQ2 (attribution) What does each stage of the pipeline contribute? Measured by ablation: the trust classifier, the *confuser filter* (the per-detection MLP defined above), their composition order, and temporal evidence aggregation, each isolated on the same frames and with the runtime cost of each stage accounted for.

RQ3 (dual-modality value) Does per-frame routing between two modalities outperform either modality alone? The scoring protocol this question requires is defined in Section 3.2.1.

A methodological discipline supports the downstream stages: *statistics before training*. For the trust classifiers and the confuser filters (and only for those; the detectors were trained conventionally), a model-agnostic feature-space instrument (the Model MRI, Section 3.7) measured what the detectors' features can separate *before* any training run, and a leakage-ratio statistic, applied to the candidate routing features, excluded scene-fingerprint features from the trust classifier. Separately, a human-in-the-loop review process co-develops the IR detector with its training corpus.

1.4 Contributions

This thesis makes four contributions, and reports one finding that emerged from them.

1. A *deployable dual-modality drone-detection system with an operator GUI*. An RGB and a thermal-infrared YOLO detector are fused by an XGBoost trust classifier that makes a per-frame four-way modality decision (`reject` / `trust_rgb` / `trust_ir` / `trust_both`); the trusted modality's detections are then filtered by the per-detection MLP confuser filter (Section 3.8.11), and a temporal smoother converts per-frame decisions into alerts. Because earlier project numbers had mixed inference sizes, matching rules, and splits and were not mutually comparable, the system is held to one declared evaluation standard (Section 3.2); under it the full pipeline lifts Svanström paired drone $F1$ from 0.742 to 0.946 while *raising* recall, holds the saturated Anti-UAV control to no harm, and cuts out-of-distribution confuser fire by more than an order of magnitude, every stage running on every frame within budget. Chapter 4 reports the per-stage ablation and the full numbers, and Appendix D maps each headline figure to its results file and replay command in the `runs/` directory.
2. The *Model MRI*, a statistics-before-training instrument for detector feature spaces. It hooks a YOLO detector's pyramid features, reads a 517-D per-detection embedding (internal activations plus detection metadata), and answers design questions with classical statistics before any model is trained: linear-discriminant separability, per-feature ANOVA, single-feature AUROC, a leakage ratio that exposes scene-fingerprint features (applied in this work to select the trust classifier's routing features), and per-modality standardisation diagnostics. Its measurements then train the filters directly: where the

feature space is separable, the MRI trains a lightweight MLP confuser filter on the detector's own internal-feature embeddings (positives = drone detections, negatives = confuser detections). Drone/confuser linear separability measured 0.952 (RGB) and 0.981 (IR) on the detectors' own cached feature corpora; these are in-corpus diagnoses that license a training attempt, and the filter's out-of-distribution evals (Chapter 4) are what validate it.

3. A *human-in-the-loop data engine*: a purpose-built label-reviewer GUI and a disciplined dataset–model co-evolution loop, reported as a six-revision case study of the IR detector. Every revision, first to last, is scored on the same fixed held-out test split of the final IR corpus ($F1$ 0.503 $\to$ 0.967, n = 9,612), so the trajectory is one comparable series rather than each model on its own contemporary split. The loop exists because clean thermal drone data is scarce: public drone datasets are difficult to obtain in RGB, far more so in thermal or paired form, and the available material needed frame-level review. The trace includes the one revision where the protocol was violated (V5: a bulk positive ingestion bypassed review and cost 12.7 points of precision), and the recovery was itself a product of the loop: the same review process that the ingestion bypassed is what located and corrected the regression.
4. A set of *curated data artifacts*, the product of the manual review work behind the system. On the corpus side: the 129,130-frame thermal corpus, frame-level reviewed across the six dataset revisions that co-evolved with the IR detector (Section 3.5); the 27,024-frame RGB confuser hard-negative corpus with sequence-disjoint splits (Section 3.1.2); and the paired Svanström conversion, 28,710 aligned visible/thermal frame pairs converted from the dataset's MATLAB `groundTruth` annotations to YOLO format. On the benchmark side: a 19-clip real-video diagnostic, 2,609 frames with 1,234 ground-truth drone instances annotated frame by frame, the only evaluation surface in this thesis that is fully out-of-distribution for *every* pipeline component (Section 3.1.8). The source datasets' licences do not permit redistributing the corpora, so they are claimed as reviewed artifacts rather than released data; the real-video benchmark, by contrast, is largely reproducible outside this project: its labels are publishable, and 17 of the 19 clips carry recoverable source identifiers in Appendix A (two confuser compilations' identifiers were not preserved).

One finding emerged that was not designed: a *cross-modal transfer* result. The thermal-trained IR detector, run zero-shot on grayscale-converted RGB, performs within a few points of the dedicated RGB detector on the same frames ($F1$ 0.580 vs 0.607 on the Svanström visible channel, n = 4,000, matched configuration) and ties it on the hardest bird-cluttered video clip (`flock_of_seagulls_attack_drone_beach`, $F1$ 0.837 vs 0.840). (The near-linear separability of drone from confuser inside the detector's own internal features, which lets the 517-D MLP outperform the MobileNetV3 patch classifier, is the premise behind the Model MRI contribution above and is shown in Section 4.1.3.)

Evaluation is held to a single declared standard: one canonical configuration per dataset (inference size, matching rule, frame budget), per-modality trust-aware scoring, frame counts printed with every number, and bootstrap confidence intervals on headline results (Section 3.2). A reproducibility layer (per-run JSON manifests, weight-hash-tagged inference caches, and a unified detection cache that makes every downstream ablation a deterministic replay) ties every reported number to the command that produced it, and Appendix D gives the reader the concrete map: headline number → canonical results file in `runs/` → replay command → automated audit check. Each evaluated model additionally ships a co-located `*.model_card.yaml` naming its train and test surfaces, so a reader can open the provenance of any weight directly (Appendix B).

1.5 Ethical Considerations and Dual-Use

Counter-UAS sensing systems are dual-use technology. The same passive visual-detection capability that flags a hostile drone over critical infrastructure can, in a different deployment, contribute to mass aerial surveillance or armed-response targeting. This thesis develops the detection component only; it neither integrates with effectors (jammers, kinetic intercept, take-down systems) nor classifies operator behaviour. The training corpora contain no personally identifying information, no identifiable third-party imagery beyond what is publicly available in the cited datasets and YouTube clips, and no acoustic or RF data that could enable operator identification. The deployment scope assumed throughout is protection of fixed sites (perimeter monitoring, airport approaches, industrial facilities) where a detected drone is reported to a human operator who decides on response. The reproducibility layer (manifests, hash-tagged caches, and the number-provenance map of Appendix D) is intended in part to make any future adaptation auditable: an auditor can re-run the replay commands against the published caches and verify that the reported behaviour is the shipped behaviour.

1.6 Thesis Outline

Chapter 2 reviews the literature on real-time object detection, drone detection and the confuser problem, thermal imaging, multi-modal fusion, detection scoring, cross-modal transfer, cascaded rejection, and the two methodological literatures this work builds on: model-assisted annotation and feature-space analysis. Chapter 3 establishes the empirical infrastructure: the datasets and their roles, the evaluation protocol and its justifications, the system architecture, each component's design and training, the Model MRI, and the human-in-the-loop method. Chapter 4 reports the empirical evaluation: the full-pipeline ablation that answers RQ1–RQ3, the temporal segment-level evidence, the per-component deep dives (including the IR detector's six-revision evolution), the Model MRI findings, the grayscale cross-modal finding, and threats to validity. Chapter 5 answers the research questions, states the production stack with its honest carve-outs, and lists future work. Appendix A details the datasets; Appendix B the models evaluated; Appendix E the glossary.

Chapter 2

Related Work

2.1 Real-Time Object Detection

The detector used in this thesis is YOLO [5] as packaged in the Ultralytics framework [6]. The nano variant was chosen at the start of the project for first-order reasons: real-time throughput on modest hardware, a mature training and deployment ecosystem, and strong small-object performance at the chosen inference resolutions; the cascade then grew around it, running RGB and IR detectors back-to-back plus a trust classifier and a per-frame distilled MLP confuser filter, leaving little room for a heavier backbone. Two-stage architectures such as Faster R-CNN [8] and Cascade R-CNN [9] reach higher accuracy at a latency cost this budget does not allow. The transformer-based detection family, DETR [10] and its real-time variant RT-DETR [11], is a plausible drop-in for future work. It was not adopted mid-project for a switching-cost reason that follows from the build order: the downstream stages were trained *after* and *against* the YOLO detectors, so the trust classifier is calibrated to the YOLO confidence-score distribution and the confuser filter reads YOLO's pyramid features; replacing the backbone now would mean re-deriving both, making it a long-horizon decision rather than a free tuning knob.

A decisive hyperparameter in this work is the inference resolution `imgsz`. YOLO letterboxes the input to a square canvas; objects smaller than a few pixels on that canvas are unresolvable regardless of model quality, which is why the small-object literature resorts to up-scaled or sliced inference [12]. On Svanström the effect is stark, and it has a concrete geometric cause: the corpus is natively 640×512 while the production RGB detector is trained at `imgsz=1280`, so inference at `imgsz=640` renders the median 29.8-px drone at half its trained-for scale. A within-model sweep (Section 3.8.6, Figure 3.12) quantifies it: the baseline loses 28 pp of drone recall when inference drops to `imgsz=640` from the trained-for 1280, and the confuser-suppressing `retrained_v2` variant collapses further on the same frames.

2.2 Drone Detection and the Confuser Problem

At the ranges typical of aerial surveillance, drones, birds, fixed-wing aircraft, and helicopters all present as small, low-texture silhouettes against sky, with similar aspect ratios and little fine detail to separate them. The resulting fire rates are corpus- and category-dependent rather than uniform: on the merged OOD confuser corpus at `imgsz=640` the baseline detector family fires on 23–58% of frames per category (Section 1.2), and the worst case is bird-only Svanström footage at that benchmark's canonical `imgsz=1280`, where a YOLO detector trained *with* bird negatives still fires on 94.4% of frames. This is the *confuser problem*: training-time hard-negative mining suppresses helicopters and partially airplanes but cannot drive the bird fire rate down without sacrificing recall on small-drone targets that share the same silhouette; the one stance that suppresses birds (`retrained_v2`) collapses Svanström drone recall to $R = 0.306$.

The C-UAS sensing surveys agree that visual detection is operationally attractive (passive, inexpensive, and able to identify what it sees) and that small targets at range and confusion with birds are among its recognised limits [1], [2], [3]; Taha and Shoufan [2] add an observation this thesis's evaluation protocol responds to directly, that published drone-detection results are experimental and "hardly comparable" across papers, with reference datasets and requirement-driven problem specifications still missing. Early deep-learning approaches [13], [14] treated drone detection as a single-class small-object problem; subsequent work explicitly added confuser classes. Schumann et al. [15] detect UAVs as a two-stage pipeline (region proposals from background subtraction or a proposal network, then a CNN classifier separating UAVs from distractors such as birds) and show the classifier transfers across a substantial domain gap, an early detect-then-discriminate structure this thesis's cascade echoes. The Drone-vs-Bird challenge [16] consolidated the bird-confuser benchmark; its winning entries relied on explicit negative training images and synthetic data to control bird false positives, and bird confusion remained the challenge's central difficulty across editions.

Two public benchmarks anchor the empirical work. The Svanström RGB+IR dataset [17], [18] provides per-frame box annotations with three explicit confuser classes (birds, airplanes, helicopters); its native 640×512 resolution forces the small-drone regime, making it the discriminating benchmark. The Anti-UAV RGBT benchmark [7], [19] provides large-scale paired RGB+IR sequences but is saturated for this work's detectors, at both evaluation sizes: baseline and `retrained_v2` RGB both reach $P = 0.9922, R = 0.9950, F1 = 0.9936$ at `imgsz=1280` (3,178 TP, 25 FP, 16 FN), the production detector reaches $F1 = 0.986$ at the Tier-1 `imgsz=640` configuration, and IR-only reaches $F1 = 0.9654$. The reason is disclosed rather than hidden: Anti-UAV

RGB frames appear in both variants' training corpus (59,413 `anti`-prefixed frames, Table 3.2), so no train/test split is claimed; Anti-UAV serves as an in-distribution sanity floor, not a discriminating benchmark. Section 2.9 compares the production stack to published numbers on both datasets.

The cascade is the response to the residual confuser fire rate that training-time mining cannot eliminate; the trust classifier and per-frame confuser filter catch what the detector cannot.

2.3 Thermal Infrared Imaging

Thermal infrared cameras image surface emission rather than reflected visible light, with three properties relevant here. First, drone motors, electronic speed controllers (ESCs), and batteries produce localised hot spots against sky, whereas birds tend to maintain more uniform body temperatures, a discriminative cue absent in the visible spectrum and a likely reason the IR detector hallucinates less on bird confusers (Section 4.5). Second, IR is largely illumination-invariant, aiding night and low-visibility operation. Third, single-channel IR imagery shares low-level statistical properties with grayscale RGB: both render small drones as silhouettes against sky. This shared structure underwrites the cross-modal transfer result in Section 4.5: on the YouTube drone-clip diagnostic surface (9 clips, 1,359 frames, Appendix A) the IR detector reaches an aggregate drone $F1 = 0.636$ on grayscale-converted RGB video, and on its single best clip, `flock_of_seagulls_attack_drone_beach`, it ties the RGB baseline (0.837 vs 0.840); the aggregate, not the best clip, is the representative number. The transfer was emergent, not designed; the grayscale-RGB mode exists as a fallback for when thermal hardware is unavailable.

IR imaging also has limitations: lower spatial resolution, thermal contrast collapse when ambient approaches drone surface temperature, and smaller labelled datasets complicating training and OOD evaluation; the IR-side OOD evaluation of this thesis is therefore carried by the thermal confuser corpus and the cross-modal surfaces of Chapter 4.

2.4 Multi-Modal Fusion and Hard-Negative Mining

Multi-modal C-UAS systems combine RGB and thermal streams through early, intermediate, or decision-level fusion [20]. Intermediate-level fusion (the canonical example being multispectral pedestrian detection [21]) typically requires paired-modality training data. This thesis instead adopts decision-level fusion with a per-frame learned XGBoost trust classifier [22], [23]. The motivation is a measured complementarity in *what* the two detectors hallucinate on: the RGB detector's residual fire is bird- and helicopter-driven while the thermal detector's is airplane-dominated, and the thermal channel barely fires on the visible channel's confuser corpus at all (Section 4.3.5). The production router makes its four-way trust decision from eight detection-evidence features only (per-modality confidences, box geometry, and cross-modal ratios; Section 3.8.9); a superseded variant that additionally consumed per-frame scene statistics is retained as a comparison (Section 4.1.4). Hard-negative mining at training time (OHEM [24], Focal Loss [25], explicit confuser augmentation) was the first thing tried and is class-asymmetric, as quantified in Section 1.2: helicopters and partially airplanes respond, birds essentially do not, and the stance that does suppress birds collapses small-drone recall. Birds and small drones are not separable at the detection stage without an unacceptable recall cost; the downstream cascade is the response to that case. The mining experiments reported there are RGB-side by design; the thermal branch's confuser handling is the thermal-native IR filter of Section 3.8.8.

2.5 Detection Scoring

The choice of matching rule is consequential. Standard IoU under-counts predictions that are correct in centre but imprecise in scale; for small drones whose GT boxes exceed the visible target (Svanström's annotators drew loose boxes), this is routine, so this thesis uses Intersection over Prediction (IoP) at threshold 0.5 for Svanström RGB scoring. The second scoring axis, how detections from two modalities are aggregated into one F1, is defined and justified as evaluation protocol in Section 3.2.2; every multi-modal number in this thesis declares its rule.

2.6 Cross-Modal Transfer and Cascaded Rejection

Two literatures bear directly on the architectural pieces that do not fit neatly under object detection, thermal imaging, or fusion.

Cross-modal transfer. The closest published setting is *modality hallucination* [26]: training on one modality so the model can be queried at test time on another. RGB-to-thermal style translation [27], [28] addresses the inverse direction, and domain-adversarial training [29] provides the general feature-alignment toolkit. All of these provide or construct cross-modal information at training time; we are not aware of published work matching the setup here, in which no cross-modal information is provided at training time and transfer emerges from shared low-level statistics (single-channel intensity, sky-background silhouettes). The result is presented as emergent in Section 4.5.

Cascaded rejection. The architectural pattern of an initial high-recall stage followed by progressively more discriminative rejection predates this application by two decades: Viola and Jones [30] for face detection, Cascade R-CNN [9] for general objects. One framing difference matters: in those designs the first stage is deliberately recall-tuned and weak on precision, whereas the detectors here are strong on both axes standalone (Anti-UAV $P = 0.989/R = 0.982$; 2.8% hallucination on the composite RGB test corpus) and the cascade exists for a *concentrated* residual failure mode, confuser scenes, that training cannot fix without recall cost. Beyond that, both classical cascades reuse the same backbone across stages, and this design differs in two further ways: (i) the rejection stages are heterogeneous (an XGBoost router on detection-evidence features, then a separate MLP on detection features), making it a refined-decision pipeline, not a refined-bounding-box pipeline; and (ii) the confuser filter can be applied per frame or at the alert-emit boundary (Section 3.8.4), and the deployed GUI uses the alert gate. The Viola-Jones intent (early rejection cheap, late rejection expensive) is preserved.

2.7 Model-Assisted Annotation and Data-Centric Development

The thesis's human-in-the-loop data engine (Section 3.5) sits in the active-learning and data-centric-AI lineage. Classical active learning [31] selects which samples to label next by model uncertainty; its object-detection instantiations [32] rank candidate frames by detection-level criteria. The loop used here inverts the emphasis: rather than selecting unlabelled data for fresh annotation, it routes *model–ground-truth disagreements* to a human adjudicator, treating every disagreement as either a model error (a future hard example) or a label error (a correction). That second branch connects to the label-error literature: confident learning [33] shows that benchmark label noise materially distorts model rankings, and the corrections surfaced by this thesis's reviewer (missing boxes, oversized boxes, mis-classed confusers) are exactly the error classes it formalises. The broader data-centric framing [34], [35], that data work rather than model work dominates applied ML outcomes, is borne out quantitatively here: the IR detector's F1 trajectory from 0.503 to 0.967 (Section 4.1.2) is driven almost entirely by corpus operations, and its one regression by a corpus operation performed without review.

2.8 Probing and Reusing Detector Representations

The Model MRI (Section 3.7) belongs to the representation-analysis lineage: linear probes [36] established reading a frozen network's intermediate features with shallow classifiers as a measurement of what the network already encodes, and feature-space statistics on penultimate layers underpin Mahalanobis-style OOD detection [37]. The MRI applies the same stance to a detection FPN (`p3/p5` features; classical statistics: LDA, ANOVA, AUROC, and a leakage ratio) but uses the measurements *prescriptively*: where the separation exists, the MRI trains the confuser filter directly on the same detector features (drone detections as positives, confuser detections as negatives). The filter itself is related to knowledge distillation [38] in spirit, a small model trained from a larger model's internals, though here the large model is not a teacher producing soft labels but a feature extractor whose representation is reused unchanged, closer to classical transferable-features reuse [39]. Representation-similarity tools such as CKA [40] ask a related question (how similar are two networks' representations); the MRI asks the deployment-facing one: is *this* network's representation sufficient for *that* downstream decision.

2.9 Comparison to Prior Work

Direct numerical comparison to other systems is unusually difficult: papers vary in dataset split, inference resolution, scoring rule, and headline metric (mAP, F1, success rate), and this work is not evaluated on the same test splits as any published number. The comparison below is therefore offered as *context, not head-to-head*: a qualitative architectural map (Table 2.1) and a numerical panel for studies reporting metrics on the same public benchmarks (Table 2.2), with the non-comparability caveats stated per row and below the table.

2.9.1 Architectural Comparison

Prior work uses either a single detector or a tightly-coupled multi-modal one and treats confuser discrimination as a training-time problem. This thesis keeps the detectors strong on both precision and recall in their own right and adds a separate discrimination stage (trust classifier + confuser filter, run per frame or at the alert gate) for the concentrated confuser failure mode, making the modality-trust decision per frame.

Table 2.1: Architectural map of representative drone-detection systems. "Modality" covers what the system consumes; "Discrimination" covers how confusers are rejected. The cascade in this thesis is, to our knowledge, the only one that combines learned modality-trust fusion with a downstream per-frame distilled confuser filter.

System / paper	Modality	Discrimination
Svanström 2021 [17]	RGB + IR (parallel)	Per-modality YOLO only
Anti-UAV challenge baselines [7]	RGB + IR (parallel)	Trackers + YOLO; no confuser stage
Coluccia 2021 drone-vs-bird [16]	RGB only	Single-stage YOLO retrained on bird negativ
Cross-modal attention fusion (e.g., [21])	RGB + IR (intermediate)	End-to-end CNN, no explicit confuser stage
This thesis	RGB + IR (decision-level)	Trust classifier + per-frame distilled (`mlp_v5`

2.9.2 Numerical Comparison

Table 2.2: Numerical comparison on public benchmarks. **All comparisons require the caveats below the table.** "Scoring" indicates the matching rule used in the cited paper where it can be determined; where it cannot, the cell is marked `ND` (not disclosed).

Dataset	System	Scoring
Svanström (visible video)	Svanström 2021 (avg. over distance bins)	IoU@0.5
	This thesis (baseline RGB, S1)	IoP@0.5
Svanström (thermal video)	Svanström 2021 (avg. over distance bins)	IoU@0.5
	This thesis (IR `v3b`, `imgsz=640`)	IoP@0.5
Anti-UAV (drone, RGB)	Anti-UAV baselines (range)	IoU
	This thesis (`ft4`, @640)	IoU@0.5
Anti-UAV (drone, IR)	Anti-UAV baselines (range)	IoU
	This thesis (IR `v3b`, @640)	IoU@0.5
Confuser corpus (no GT, RGB)	*No published baseline*	ND
	This thesis (confuser filter, @640)	fire-rate
Thermal confuser corpus (no GT, IR)	*No published baseline*	ND
	This thesis (best ablation, `robust6`, @640)	fire-rate

Three caveats apply to every row of Table 2.2.

Scoring rule. Svanström *et al.* report per-sensor precision/recall via MATLAB's `bboxPrecisionRecall` (an IoU-overlap matcher) at IoU 0.5 and a detection threshold of 0.5, with YOLOv2 at input sizes 416×416 (visible) and 256×256 (thermal); their table values are averages over distance bins, and their close-range bins are higher (0.868 visible, 0.885 IR). The Anti-UAV baselines rarely disclose their IoU threshold, and neither source discloses a multi-modal aggregation rule. IoP@0.5, used here for Svanström RGB, is more lenient than IoU@0.5; an IoU@0.5 re-score would produce lower numbers.

Dataset split. The Svanström paper held out an evaluation set of 120 IR and 120 visible clips (five per class and distance bin), but the clip list is not published with the dataset, so it cannot be adopted here. This thesis instead evaluates the full Svanström corpus, which is leakage-free on the RGB side (no Svanström frames of any class appear in any RGB training corpus of this thesis, Section 3.3) and disclosed as in-distribution on the IR side. One further conversion caveat: the dataset's annotations ship as MATLAB `groundTruth` objects, and this thesis converted them to YOLO format with its own script, so box geometry may differ marginally from the authors' MATLAB pipeline. Anti-UAV has a more canonical split but uses sequence-level tracking metrics not directly equivalent to per-frame F1.

Task. Anti-UAV baselines are tracking systems. Their state-accuracy numbers (0.90–0.97 range across years) include temporal aggregation absent from this thesis's frame-level numbers. The Anti-UAV row is included for orientation only.

Two claims survive these caveats. First, this thesis's per-modality numbers *exceed* the published Svanström averages (0.950 vs 0.785 visible, 0.961 vs 0.760 IR); the caveats above carry the qualification, since part of that gap is attributable to detector generation (YOLOv26n vs YOLOv2), inference resolution (1280/640 vs 416/256), and matcher leniency (IoP vs IoU) rather than to a like-for-like superiority. At minimum, the detectors are competitive baselines rather than weak strawmen. Second, no published system reports a separately-evaluated OOD confuser fire rate at the granularity of Table 4.11, because no published system explicitly separates a confuser-rejection stage from drone detection. The suppression numbers of Section 4.3.2 (30.4% $\rightarrow$ 4.9% at the router, $\rightarrow$ 1.4% at the per-frame filter, at the canonical configuration) therefore cannot be directly benchmarked. The contribution is the architectural separation that makes such numbers measurable.

Chapter 3

Methodology and System Design

This chapter establishes everything the evaluation chapter consumes: the datasets and their roles (Section 3.1), the evaluation protocol and its justifications (Section 3.2), the data-leakage audit (Section 3.3), the training recipes (Section 3.4), the human-in-the-loop data engine (Section 3.5), the reproducibility layer (Section 3.6), the Model MRI instrument (Section 3.7), and, building on these, the system architecture and the design of each component (Section 3.8). Results live in Chapter 4; where a design decision below was driven by a measurement, the key number is quoted inline with a forward reference to its table: the IoP matching rule by the loose-box annotation audit (Section 3.2.1), the `imgsz=1280` requirement by the Svanström resolution sweep (Section 3.8.6), and the alert-gate placement of the confuser filter by the segment-level gain on real video (Section 3.8.4).

3.1 Datasets and Their Roles

The evaluation uses eight labelled corpora and a 19-clip YouTube real-video set, drawn from public benchmarks, Roboflow / Kaggle sources, and the deployment partner's CCTV stream. Each corpus plays one declared role: a training surface, an in-distribution test surface, or an out-of-distribution (OOD) stress surface. Full provenance is in Appendix A.

3.1.1 Composite RGB Training Corpus (`rgb_dataset`)

The RGB detector's training and in-distribution test corpus is a composite of nine public sources totalling 172,022 frames split 80/10/10 (137,506 train, 17,307 val, 17,209 test), sequence-disjoint. Of these, 134,339 (78.1%) carry at least one drone box and 37,683 (21.9%) are background negatives; the corpus contains 146,540 drone-class boxes. Per-source breakdown:

The bird hard negatives that make the baseline meaningful rather than a strawman come from three channels: the dedicated `AirBird` subset, the bird-focused `FBD-SV` video frames, and the bird instances of `WosDetC`, all treated as background during training. *No Svanström frames of any class appear in this corpus*, nor in any other RGB training corpus of this thesis (the fine-tune chain adds SelCom and

Table 3.1: Evaluation surfaces and their roles. "Role" distinguishes training surfaces, in-distribution evaluation surfaces, and OOD stress surfaces; no OOD surface contributes any training data. Benchmark and stress surfaces are evaluation-only and carry no split.

Tag	Frames	Train / val / test	Modality	Role
`rgb_dataset` (composite RGB corpus)	172,022	137,506 / 17,307 / 17,209	RGB	RGB-detector train + in-distribution test
`ir_dset_final`	129,130	107,809 / 11,709 / 9,612	IR	IR-detector train + in-distribution test
`rgb_confusers_merged`	27,024	21,784 / 2,607 / 2,633	RGB	Confuser hard-negative training source + OOD confuser test (test split; see Section 3.1.2 for the per-model OOD status)
`IR_confusers`	5,938	— (eval-only)	IR	OOD thermal-confuser test (airplane 4,281 / bird 1,200 / helicopter 457; evaluated split 5,237); no training contribution
`svanstrom` (paired)	28,710	— (eval-only)	RGB+IR	Discriminating benchmark (small drones; labelled confusers); IR-side training overlap audited in Section 3.3
`antiuav` (RGBT)	85,374	— (eval-only)	RGB+IR	Saturated benchmark / no-harm control; training overlap audited in Section 3.3
`selcom_cctv`	2,076	80 / 20 split (val 311 evaluated)	RGB	Deployment-partner CCTV (fine-tune surface + held-out val)
YouTube real-video diagnostic	2,609	— (eval-only)	RGB	Real-world clips: drone-positive (9) + confuser-only (10); consecutive frames carry the temporal evaluation

Figure 3.1: Ground-truth drone crops across datasets and modalities (red box = GT; window ≈ 4× the box). Svanström drones are 19–29 px small silhouettes against open sky (the regime that makes birds and distant aircraft hard to separate and that drives the cascade), while Anti-UAV drones are an order of magnitude larger; the thermal (IR) view renders the same target as a bright blob rather than a dark silhouette (Section 3.8.2).

Table 3.2: Per-source composition of the composite RGB training corpus. Negatives include VIRAT surveillance, UA-DETRAC traffic, and BDD100K driving frames as background contrast.

Source prefix	Count	Description
`anti`	59,413	Anti-UAV benchmark (CVPR 2020 / 2021) [7] RGB frames
`wosdetc`	53,259	WosDetC drone corpus
`anti-muav-roboflow`	14,652	Anti-mUAV Roboflow community variant
`AirBird`	10,000	AirBird drone-vs-bird dataset (supplies bird negatives at training time)
`FBD-SV`	9,754	Fine-grained Bird Detection (Sky View)
`mav`	9,744	MAV (Micro Aerial Vehicle) frames
`dut`	5,200	DUT Anti-UAV aerial imagery [41]
`VIRAT`	3,334	VIRAT surveillance benchmark (background negatives)
`UA-DETRAC`	3,333	UA-DETRAC traffic benchmark (background negatives)
`BDD100K`	3,333	BDD100K driving dataset (background negatives)
Total	**172,022**	80/10/10 split; `drone` is the only positive class

non-Svanström confuser material only, Section 3.8.7); Svanström is therefore fully out-of-distribution for every RGB detector variant, drones and confusers alike (Section 3.3).

3.1.2 Confuser Hard-Negative Corpus (`rgb_confusers_merged`)

The hard-negative training source for the `hardneg_v3more` and `retrained_v2` variants, and the OOD RGB-confuser test split. 27,024 images, all with empty YOLO labels, from seven sources split 80/10/10 (21,784 train, 2,607 val, 2,633 test). Sources without existing splits were divided by video / sequence ID to prevent temporal leakage (per-source split policy in Appendix A).

Hard-negative mining for `retrained_v2` (Section 3.8.7) ran the baseline detector over all 27,024 confuser images, retained the 11,729 (43.4%) on which it hallucinated, and merged those with the composite RGB corpus to produce the 183,751-image `retrain_dataset`.

What "out-of-distribution" means on this corpus's test split is *per-model*, and the tally matters. Of the 2,633 test frames, 1,754 (66.6%) are Svanström-sourced (airplane 1,034, helicopter 311, bird 409) and 879 (33.4%) come from the Roboflow/Kaggle sources. For the `baseline` the whole split is fully OOD (its only confuser exposure is bird negatives, Section 3.1.1). For `hardneg_v3more` and `retrained_v2` the cor-

Table 3.3: Per-source composition of the RGB confuser hard-negative corpus. The test split (right column) is the *OOD RGB-confuser* evaluation surface used in Chapter 4.

Source	Category	Train	Val	Test
Svanström paired	Airplane	4,528	528	1,034
Svanström paired	Helicopter	4,489	827	311
Svanström paired	Bird	4,586	303	409
New_Dataset (Roboflow)	Mixed	3,514	439	439
Airplane (Roboflow)	Airplane	2,093	200	99
Helicopter (Kaggle)	Helicopter	1,387	188	175
raihanrsd (Kaggle)	Bird	1,187	122	166
Total		**21,784**	**2,607**	**2,633**

pus is their hard-negative *training* source, so the test split is held-out in-distribution material for them, not OOD. For the production `ft4`, whose confuser hard-negatives explicitly excluded Svanström (a $\leq$8% injection mined only from frames the model hallucinated on), the Svanström two-thirds is fully OOD and the remaining third shares source pools with its hard-negative set but is sequence-disjoint. The headline suppression numbers of Chapter 4 are `ft4` numbers and carry this status.

Because the corpora are single-class (drone), confuser images carry no boxes; the category of every confuser frame lives in its *filename* (source prefix), which is how the per-category fire rates of Chapter 4 are computed. Figure 3.2 shows representative frames from both confuser corpora, picked as the highest-confidence detector-fired frame per category; Figure 3.3 shows the resulting hallucinations themselves, with the production confuser filter's verdict on each.

Figure 3.2: What the confuser corpora look like. Top: `rgb_confusers_merged` (airplane / bird / helicopter / mixed `New_Dataset` source, labelled "other"). Bottom: `IR_confusers` thermal frames. Each frame is the highest-confidence frame the bare detector fired on in that category (deterministic pick from the Tier-1 cache), i.e. representative hard negatives rather than easy samples.

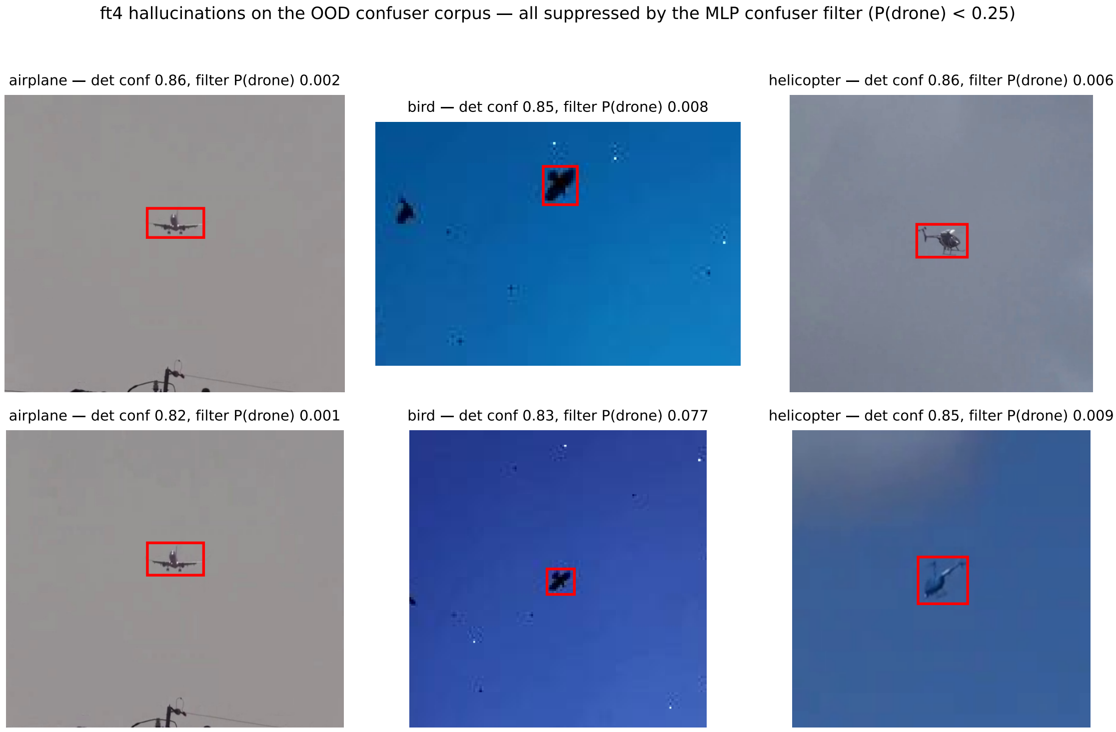


Figure 3.3: Representative `ft4` hallucinations on the OOD confuser corpus (context crops around the detection; red box = false positive). Detector confidences are high (0.82–0.86) while the confuser filter's $P(\text{drone})$ on the same detections is 0.001–0.077: every example shown is suppressed at the production threshold (0.25). This is the per-detection discrimination the detector's decision head cannot express without recall cost.

3.1.3 IR Detector Corpus (`ir_dset_final`)

129,130 thermal frames split 83.5 / 9.1 / 7.4 (107,809 train, 11,709 val, 9,612 test); 72.5% positive / 27.5% negative (94,142 drone-class boxes). Single-class (`drone`). Sources by filename prefix:

Bird, airplane, and helicopter thermal frames enter across HITL revisions (Section 3.5) as the IR detector hallucinates on confuser scenes. The thermal confuser problem is real but differently shaped from the visible one: on the shared Svanström confuser sequences the production IR detector fires on only 0–1.7% of frames where the RGB detector fires on 65–74% (Section 4.3.5), while on the dedicated thermal confuser corpus its residual fire is 29.4% and airplane-dominated (35.2% airplanes, 12.2% birds, 0% helicopters; Table 4.11). The `sea` and sky/building negatives are similarly mined, and they hold: on the 1,400 negative frames of the evaluated test split (the same mined scene types) the production detector fires on 1.6% of frames (38 false detections). Source prefixes of the form `dv`N record the *dataset version* at which material entered: the corpus co-evolved with the model, each HITL round graduating a new dataset version alongside its model revision, so `dv5` is the corpus state assembled in the V5-era round. This table is the state at the production `v3b` checkpoint; earlier revisions (V2–V5) operated on smaller variants of the same source families.

Table 3.4: Per-source composition of `ir_dset_final`. The Svanström IR partition (`svan`) is the largest single source and is the reason Svanström IR evaluation in Chapter 4 is in-distribution for the IR detector (Table 3.6).

Prefix	Count	Description
`dv5`	53,512	Thermal drone sequences, dataset version 5 (the corpus state graduated by the V5-era HITL round); includes the Anti-UAV-derived material audited in Section 3.3
`cst`	22,957	CST thermal captures
`svan`	21,637	Svanström IR drone partition
`czoom`	9,808	CZoom thermal zoom sequences (train only)
`sea`	8,398	Sea / maritime thermal backgrounds (negatives)
`flir`	6,360	FLIR thermal imagery
`ovh2`	2,898	Overhead thermal set 2
`ovh1`	2,866	Overhead thermal set 1
`gemini`	392	Gemini thermal captures (synthetic-augmented)
`yt`	302	YouTube thermal clips (train only)
Total	**129,130**	83.5 / 9.1 / 7.4 split; 72.5% positive / 27.5% negative

3.1.4 Thermal Confuser Corpus (`IR_confusers`)

The thermal counterpart of the RGB confuser zoo: a 5,938-frame corpus of thermal aerial-confuser frames with no drones (4,281 airplane, 1,200 bird, 457 helicopter), exported from Roboflow-hosted thermal confuser video; source streams fully disjoint from every `ir_dset_final` source family. The evaluated split contains 5,237 frames, sampled to 4,000 by even spread (3,043 airplane / 871 bird / 86 helicopter). Every detection on this surface is a false positive by construction. It contributes no training data to any model and serves exactly one purpose: the OOD stress test of the IR branch (the thermal analogue of the `rgb_confusers_merged` test split). It supersedes the 180-frame CBAM probe used during development: CBAM is a Roboflow-hosted thermal aerial-confuser dataset of ∼1,775 frames (bird / drone / airplane; the name comes from the CBAM TF-Net model the dataset ships with), from which a 180-frame confuser probe was sampled. That probe's `valid` split remains the *held-out* recall-recovery gate for the thermal-native filter's training (`mlp_aligned_thermalonly`, Section 3.8.8), since `IR_confusers` postdates that training and the CBAM `valid` split is disjoint from the CBAM `train` drones the filter is trained on.

3.1.5 Svanström Paired (`svanstrom`)

The Svanström dataset [17], [18] is the discriminating benchmark. The paired version contains 28,710 frames at stride 3 from 279 sequences, at 640×512 native resolution. Drones are routinely a few tens of pixels across, the regime driving the imgsz 640 vs. 1280 swing of Figure 3.12. Per-class counts: 11,695 drone, 6,090 airplane, 5,298 bird, 5,627 helicopter. Svanström RGB scoring uses IoP at 0.5 (Section 3.2.1) because annotators drew loose boxes and IoU systematically under-counts useful predictions.

3.1.6 Anti-UAV RGBT (`antiuav`)

The Anti-UAV RGBT benchmark [7], [19] provides large-scale paired RGB+IR sequences (1920×1080 RGB, 640×512 IR). Both modalities perform near ceiling on it (Chapter 4); 59,413 Anti-UAV RGB frames also appear in the RGB training corpus. Anti-UAV is therefore used as a *no-harm control*: a strong result here is necessary but not sufficient, and the pipeline must not degrade it. Svanström and the confuser surfaces do the discriminating work.

3.1.7 SelCom CCTV (`selcom_cctv`)

The deployment-partner corpus: a fixed-camera RGB stream (1920×1080, 25 fps, ∼1:30 source clip) covering an urban perimeter. 2,076 frames (1,953 drone-positive + 123 true negatives), median drone $\sqrt{\text{area}} \approx 36.8$ px (∼12 px at `imgsz=640`, ∼24 px at 1280). Drones blend into urban clutter, making SelCom a clutter-bound rather than purely pixel-bound regime. The `selcom_mixed_ft2_val` surface (311 images, 295 GT boxes) is a pure-SelCom 15% held-out split (seed 0); the SelCom fine-tunes (Section 3.8.7) train on the remaining frames mixed 20:80 with general RGB. The footage was provided by the partner for development; the system itself is not deployed at the site, and no thermal channel exists there.

3.1.8 YouTube Real-Video Diagnostic (19 Clips)

19 curated YouTube clips totalling 2,609 evaluated frames: 9 drone-positive clips (1,359 frames; 1,234 GT drone instances, several with concurrent birds) and 10 confuser-only clips (1,250 frames; every detection is a false positive). The drone-positive set was annotated frame-by-frame; frames are consecutive as extracted, which makes this the surface that carries the *temporal* (segment-level) evaluation. The full per-clip table is in Appendix A. This is the only fully OOD surface for every pipeline component: no clip appears in any training set; the footage is hobbyist material from handheld or consumer cameras, with unknown optics and codec-level

compression, in contrast to the fixed-mount research and surveillance sensors behind every other surface; and the drone-positive clips include bird-vs-drone scenes absent from Svanström.

3.2 Evaluation Protocol

Every number in this thesis is produced under one declared protocol: a single canonical configuration per dataset, per-modality trust-aware scoring, frame counts printed with every result, and bootstrap confidence intervals on headline cells. Evaluation uses held-out test splits, and the out-of-distribution surfaces (the confuser corpora and the real-video benchmark) are held out of all training. This section states the protocol once so the result tables do not have to re-justify it.

3.2.1 Metrics and the IoP Rule

Detection scoring uses **Intersection over Prediction (IoP)** at threshold 0.5 for all Svanström and SelCom RGB scoring:

$$\text{IoP} = \frac{|B_{\text{pred}} \cap B_{\text{gt}}|}{|B_{\text{pred}}|}$$

A prediction is a true positive iff IoP ≥ 0.5 against some GT box; otherwise it is a false positive. IoP is *required* by Svanström: its GT boxes are routinely larger than the drone (loose annotator boxes), so IoU systematically under-counts useful predictions. SelCom does not share that annotation problem; it is scored under the same rule for cross-surface consistency, and an IoU@0.5 re-score was not separately run for it. Anti-UAV and the in-distribution test splits use standard IoU@0.5; their GT boxes are tight and IoU/IoP differ negligibly.

The headline metrics are *precision*, *recall*, and *F1*; mAP50 and mAP50-95 appear only as supplementary numbers (e.g., Table 4.3). Precision and recall directly reflect false-alarm and missed-drone rates; a single mAP obscures both. For confuser-only surfaces the headline is the *frame fire rate* (fraction of frames with at least one detection) and the absolute FP count, since no drone GT exists.

3.2.2 Scoring-Rule Sensitivity (why trust-aware scoring)

A multi-modal detection can be scored two ways, and the choice is consequential. Under *dual scoring*, each modality is scored independently and FN/FP counts aggregated, so a frame silent on RGB but detected on IR still books an RGB false negative: the system is penalised on a channel it deliberately did not trust. Under *trust-aware scoring*, only the modality the trust classifier chose is scored, each against its *own* ground truth, and the two are never merged into a union. The user sees one per-frame alert; trust-aware scoring matches that.

The size of the difference is the reason this is protocol rather than a footnote. On Svanström, a representative cascade (the reject-class `robust8`) scored both ways moves by 2.8 pp ($F1 = 0.921$ dual vs. 0.949 trust-aware, at identical precision 0.939, since the predictions are identical and only the booked false negatives differ). The magnitude scales with how often the router rejects a modality: an earlier measurement on the May-10 stack at `imgsz=640`, where the weak visible channel was rejected on most frames, showed a 27.7 pp swing (0.663 dual vs 0.940 trust-aware). A scoring choice that can move a result by anywhere from 3 to 28 points depending on the operating point is large against most inter-system differences in the literature, so all cascade-level F1 in this thesis uses trust-aware scoring, and any cross-paper comparison must state its rule. The same discipline matters for RQ3: the thermal channel's value concentrates on frames where the visible channel fails (night, glare), which a union-based score would mis-attribute.

3.2.3 Canonical Configurations and Frame Budgets

Inference resolution. The standard is `imgsz=640`. Exactly two surfaces use `imgsz=1280`, each for a documented reason of *source* resolution: Svanström-RGB (native 640×512; Figure 3.4 shows its drones clustered near the 640 resolvable floor, where the baseline loses 28 pp of recall, 0.964 → 0.684 at 640, Section 3.8.6) and SelCom-RGB (low-fidelity CCTV; doubling `imgsz` doubles recall *and* raises precision, Section 3.8.7). Everything with normal high-resolution sources runs at 640, including Anti-UAV (1920×1080) and all confuser surfaces. The IR detector runs at its native 640 everywhere.

Frame budgets. Surfaces are evaluated in full when they contain ≤4,000 frames; larger surfaces are sampled by an *even spread* across the whole corpus (every frame index on a uniform grid, never a truncated prefix, which silently biases toward the alphabetical head of a sorted directory) capped at ≈4,000 frames. The realised n and the source size are printed with every table.

Uncertainty. Headline cells carry 95% bootstrap confidence intervals. Each interval is computed by re-drawing the evaluated frames at random with replacement 1,000 times and recomputing the metric on every redraw; the central 95% of those recomputed values is the interval. It answers one question: how much could this number have moved had the evaluation drawn a different sample of frames from the same surface. When two cells' intervals overlap substantially, the difference is within sampling noise, and this thesis treats the comparison as a tie rather than declaring a winner.

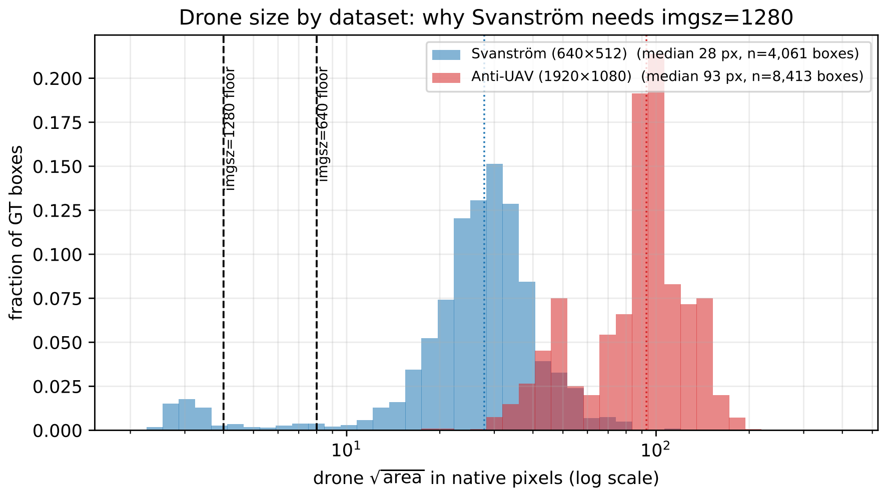


Figure 3.4: Drone size by dataset: the distribution of ground-truth *box* sizes ($\sqrt{\text{area}}$ in native pixels; a sample of up to 9,000 label files per dataset, n in the legend counts boxes, not frames; each histogram shows the fraction of that dataset's boxes per bin). Svanström drones (median 28 px) sit close above the YOLOv26n `imgsz=640` resolvable floor ($\sqrt{\text{area}} \approx 8$ px), with a visible sub-floor tail; small targets near the floor are where detection degrades long before the hard limit, which is why Svanström recall degrades sharply at `imgsz=640` (baseline 0.964 → 0.684) and resolves at `imgsz=1280` (floor ≈ 4 px; Section 3.8.6). Anti-UAV drones (median 93 px) are an order of magnitude larger and supra-floor at either resolution.

Table 3.5: Canonical evaluation configuration per surface. Numbers measured at any other configuration are labelled as such and never share a table with canonical numbers.

Surface	n eval'd	of	`imgsz`	Scoring
Svanström paired	4,000	28,710	1280 (RGB) / 640 (IR)	IoP@0.5, trust-aware
Anti-UAV paired	4,000	85,374	640 / 640	IoU@0.5, trust-aware
RGB test split	4,000	17,209	640	IoU@0.5
IR test split	4,000	9,612	640	IoU@0.5
SelCom val	311	311	1280	IoP@0.5
RGB confusers	2,633	2,633	640	fire rate / FP
IR confusers	4,000	5,237	640	fire rate / FP
Grayscale (Svanström)	4,000	28,710	640 (IR model)	IoP@0.5
Real video (19 clips)	2,609	2,609	640	IoP@0.5; segment-level
Svanström clean split	5,557	5,557	1280 (RGB) / 640 (IR)	IoP@0.5, trust-aware; Sectio
Anti-UAV clean split	57,542	57,542	640 / 640	IoU@0.5, trust-aware; Sectio

3.2.4 Unified Detection Cache

All pipeline evaluations replay one *unified detection cache*: the two production detectors run exactly once per surface (at the canonical configuration, detector confidence 0.25), and for every retained detection the cache stores the box, the confidence, the detector's own 517-D fused `p3+p5` feature vector that the MLP confuser filter consumes (stored as float32: float16 rounding shifts the standardised features enough that the filter vetoes essentially everything, a documented failure), and the patch-CNN confuser probability, kept so the superseded patch baseline can be replayed for comparison; per frame it stores the trust classifier's feature rows and the ground truth of each modality separately (RGB-frame boxes and IR-frame boxes), since trust-aware scoring evaluates each channel against its own labels. Every downstream ablation (router choice, filter choice, thresholds, composition order, scoring variants) is then a deterministic CPU replay of the same detections, which removes the largest source of silent incomparability (re-detection under drifted settings) and makes the full ablation matrix reproducible in minutes.

3.3 Training–Evaluation Overlap Audit

The two paired evaluation surfaces share data with the training corpora of some pipeline components. This section measures that overlap per component, then evaluates the pipeline on a *held-out clean split* to quantify what the overlap is worth. The headline numbers of this thesis remain the full-surface figures; the clean split is the control that bounds their inflation.

Table 3.6: Measured training–evaluation overlap per component and paired surface. "Clean" means the component's training data was verified to contain no frames from that evaluation surface; every non-clean cell is quantified.

Component	Svanström eval	Anti-UAV eval
RGB detector (`ft4`)	Clean (no Svanström in any RGB corpus)	In-distribution: R from all 91 eval s
IR detector (`v3b`)	17,314 train frames (+1,325 zoom-aug.); 37.3% of eval frames are exact train images	22,603 train fram segments; 6.3% ex
Trust router (`robust8`/`robust8-nr`)	Trained on 214 of 273 eval sequences	Trained on 61 of 9
Patch filter (MobileNetV3)	In-distribution (crop sources incl. Svanström)	In-distribution (c
RGB filter (`mlp_v5_v4`)	Clean (no Svanström; trained on `rgb_dataset` train+val, `rgb_confusers` train+val, pure-SelCom, bird.v1i train)	Clean (Anti-UAV
IR filter (`mlp_aligned_thermalonly`)	In-distribution (Svanström IR crops in thermal-head training)	Clean (Anti-UAV

The held-out clean split. The overlap is at *sequence* level, so excluding individual frames is not a sufficient control: the IR training corpus holds stride-sampled siblings of most evaluation frames. The control is therefore a sequence-level split: every evaluation sequence with *zero* frames in IR training, evaluated in full (every frame, no sampling). For Svanström this gives 54 of 279 sequences (n = 5,557 frames), held out from *both* detectors' training, since no Svanström frames exist in any RGB corpus. For Anti-UAV it gives 61 of 91 segments (n = 57,542 frames), held out from IR training only: as the table shows, the composite RGB corpus contains material from every Anti-UAV segment, so no RGB-clean Anti-UAV subset exists. The sequence lists, derivation procedure, and frozen results are versioned in the repository (`runs/clean_split/`).

Table 3.7: Held-out clean split versus the full-surface headline. Clean cells carry 95% bootstrap CIs. The cascade arm is the reject-class `robust8` at classifier → filter; this leakage control predates the no-reject switch, and the shipped `robust8-nr` shows the same pattern (Svanström 0.946 → 0.913, discussed below).

Surface / arm	Clean $F1$ [95% CI]	Headline $F1$	Δ
Svanström (54 seqs, $n = 5{,}557$)			
RGB detector solo (own GT)	0.572 [0.557–0.585]	0.607	−3.5 pp
IR detector solo (own GT)	0.867 [0.854–0.880]	0.940	−7.3 pp
bare paired	0.684 [0.672–0.695]	0.742	−5.7 pp
cascade (`robust8`)	**0.935** [0.928–0.941]	0.949	−1.4 pp
Anti-UAV (61 segments, $n = 57{,}542$)			
RGB detector solo (own GT)	0.988 [0.987–0.989]	0.985	+0.3 pp
IR detector solo (own GT)	0.966 [0.964–0.967]	0.961	+0.5 pp
bare paired	0.977 [0.976–0.977]	0.973	+0.4 pp
cascade (`robust8`)	**0.986** [0.986–0.987]	0.984	+0.2 pp

Three conclusions follow. First, *Anti-UAV shows no inflation at all*: every arm scores at or slightly above its full-surface headline on the clean segments, so the overlapped segments were, if anything, the harder ones; the no-harm conclusion and the production decisions anchored on Anti-UAV hold on 57,542 held-out frames. Second, *the Svanström IR detector was inflated, boundedly*: its solo $F1$ drops 7.3 pp on clean sequences, but the RGB detector, which has no Svanström anywhere in its training, drops 3.5 pp on the same sequences, so part of the gap is sequence difficulty and 7.3 pp is an upper bound on the leakage component. Third, *the cascade conclusion is robust*: the `robust8` cascade moves only −1.4 pp, the smallest change of any arm, and still lifts paired $F1$ from 0.684 to 0.935 on sequences nothing in the detectors ever saw; the shipped no-reject `robust8-nr` shows the same pattern (0.946 → 0.913, −3.3 pp), still far above the bare 0.684. The relative ordering of all router variants is unchanged on clean data, so no production decision is affected. One residual overlap remains disclosed rather than removed: the trust router's training rows were mined from both surfaces, so cascade evaluation runs on router-training sequences even in the clean split; the router's own train/test protocol is sequence-stratified, and a stricter subset excluding router-training sequences (measured on the cached evaluation sample) showed the pipeline gain persists.

3.4 Training Recipes

All detectors are YOLOv26n (nano) single-class drone detectors; the IR detector's per-revision configuration is in Section 3.5.

Table 3.8: Detector training configurations. The baseline RGB and `retrained_v2` share a recipe but differ in training corpus (Section 3.8.7). All training was on an NVIDIA RTX 3090 (24 GB VRAM); evaluation was on a GTX 1050 Ti (4 GB VRAM).

Parameter	RGB (baseline / `retrained_v2`)	SelCom fine-tune (`ft2_1280`)
Pretrained init	`yolo26n.pt` (COCO)	`Yolo26n_trained` (baseline)
Epochs	70	up to 50 with patience 10
Patience	10	10
Batch size	48	4
`imgsz`	640	1280
Optimiser	AdamW	AdamW
Initial LR	0.001	0.001
Final LR ratio	0.01	0.01
Warmup	3 epochs	3 epochs
Weight decay	0.0005	0.0005
Mosaic	0.33 (off after epoch 50)	0.33
Erasing	0.4	0.4
Scale / Translate / Flip-LR	0.5 / 0.1 / 0.5	0.5 / 0.1 / 0.5
HSV (H, S, V)	0.015 / 0.7 / 0.4	0.015 / 0.7 / 0.4

At the end of baseline training, the standard held-out validation (`yolo detect val` on the corpus's own test split) recorded $P = 0.978$, $R = 0.915$, mAP@50 = 0.951; these training-time numbers are the model's own acceptance check and predate the canonical evaluation protocol of Section 3.2, under which the corresponding Tier-1 figures appear in Chapter 4.

Backbone-freeze depth in confuser-injection fine-tuning. Injecting hard-negative confuser crops risks catastrophic forgetting. For FT4, the production RGB detector and the feature source for the distilled filter (Section 3.8.11), backbone-freeze depth is the controlling knob: freezing 15 layers cleared every drone-recall regression gate, whereas freezing only 12 regressed Svanström recall and $F1$ at the same 300 hard negatives. Five configurations were tried (Table 3.9); the production recipe (FT4 R3: 300 hard negatives, `freeze`= 15, 3 epochs) was the only one that cleared all gates: R1 and A1 failed the Svanström drone-recall gate, R2 and A2 failed the SelCom validation gate. R3 achieves a 16-pp confuser-hallucination cut without recall loss.

Table 3.9: The FT4 confuser-injection recipe grid. Gates: Svanström drone recall must not regress materially, and SelCom validation $F1$ must not drop. Only R3 passes both.

Variant	Hard negatives	Freeze	Epochs	Svanström R	Gate
R1	600	12	3	0.898 (−2.9 pp)	fail (recall)
R2	300	12	3	—	fail (SelCo
R3 (production)	**300**	**15**	**3**	**0.919** (−0.8 pp)	**pass**
A1	600	15	3	0.897 (−3.0 pp)	fail (recall)
A2	600 + 4,000 extra positives	15	3	0.919 (−0.8 pp)	fail (SelCo

3.5 Human-in-the-Loop Data Engine

3.5.1 The Co-Evolution Problem

A static-corpus detector has a fixed relationship with its training distribution; deployment surfaces shifts that cannot be anticipated at training time. Making the model's own failure modes the primary driver of corpus expansion concentrates annotation effort where marginal value is highest.

Dataset and model state are coupled: a model on corpus k generates errors that define corpus $k+1$, and corpus $k+1$ trains model $k+2$. The coupling is productive when the loop is disciplined, since reviewed errors become clean labels or hard negatives. It turns regressive when it is not, and the failure mode is broader than missed labels: incoming source material arrives with un-assessed annotation quality, so un-reviewed batches can carry unlabelled positives (which then train as confuser-class negatives, as the V5 regression in Section 4.1.2 illustrates; distinct from the `mlp_v5_v4` filter of Section 3.8.11), ground-truth boxes so oversized they span most of the frame, missing boxes, and inconsistent class conventions. No formal quality assessment of incoming sources was performed in this project; the frame-level review loop below is the mitigation, applied at the point where errors become training data.

3.5.2 Label Reviewer GUI

The label reviewer (`label_reviewer/`) is a Tkinter desktop tool for dataset curation, expanded as the IR-development loop surfaced new failure modes. It is organised as two windows: a setup window where the session is configured (review mode, image/label/model paths, confidence and `imgsz`, disagreement filters), and the review window proper, an OpenCV canvas where the labels are actually edited. Its working unit is a candidate frame: an image, the existing GT box(es), and the current model's predicted box(es). For each frame, the reviewer can:

- **Accept GT, reject model**: GT is correct; the model's disagreement is a FP

/ FN case logged for analysis.

- **Accept model, override GT**: the model is correct; the existing GT is missing a box or contains an over-sized / mis-located one. Common in the IR set, where many "missing" GT boxes correspond to small drones the original annotator missed.
- **Adjust the box geometry**: click-drag editing of corners; addresses the loose-box pattern common in scraped datasets.
- **Demote to negative**: deleting all drone boxes from a frame turns it into a hard negative; this is both the channel feeding the loop's negative-mining branch and the mechanism used for confuser-corpus cleanup, where a frame initially flagged as “drone” is in fact a bird, airplane, or helicopter.
- **Export to dataset**: at session end, write the reviewed labels back into the dataset layout in YOLO format; the originals are automatically backed up to a timestamped folder before anything is overwritten.

Batch operations correct a contiguous segment in one step (range deletion, page-level thumbnail review); filtered views surface only disagreement frames (FP-only / FN-only / all-mismatches), with the model acting as a triage layer.

[**PLACEHOLDER: Label reviewer setup window screenshot**] The Tkinter setup window: review-mode selector (Review / Review+Model / Compare Labels / Auto-Label), dataset paths, confidence and `imgsz` fields, disagreement filter, and the Launch and Export-to-Dataset actions.

Figure 3.5: Label reviewer, setup window: the session is configured here (what to review, against which model, under which disagreement filter) before the review canvas is launched.

3.5.3 The Co-Evolution Loop

Each revision follows the same loop:

1. **Train.** Train the next model version on the current dataset state.
2. **Predict.** Run the new model on a candidate pool: held-out validation frames, freshly acquired video, and frames the previous model failed on.
3. **Disagree.** Identify model-vs-GT disagreements: (i) model fires on a GT-less frame and is right (missing-annotation FN); (ii) model is silent on a GT frame

[**PLACEHOLDER: Label reviewer review-canvas screenshot**] The OpenCV review window on a Svanström frame: GT and model boxes in two colours, box-edit handles, and the keybinding bar (add / delete / delete-image / range-delete / save).

Figure 3.6: Label reviewer, review canvas, with a Svanström frame on which V3 fires and the existing GT is missing. The reviewer's resolution path turns the disagreement into either a corrected label (positive ingest) or a hard negative (negative-mining branch).

and is right (spurious-GT FP); (iii) model and GT agree on centre but disagree on box size (loose annotator boxes).

4. **Review.** Adjudicate each disagreement via the label reviewer (Section 3.5.2). Accept-correct-defer.
5. **Identify negative gaps.** Model firings on non-drone scenes (textured clouds, buildings, hot-pixel rows) are promoted to hard negatives in the next training set.
6. **Augment.** Apply the corrected labels, append the new positives and negatives, and emit the next dataset version.

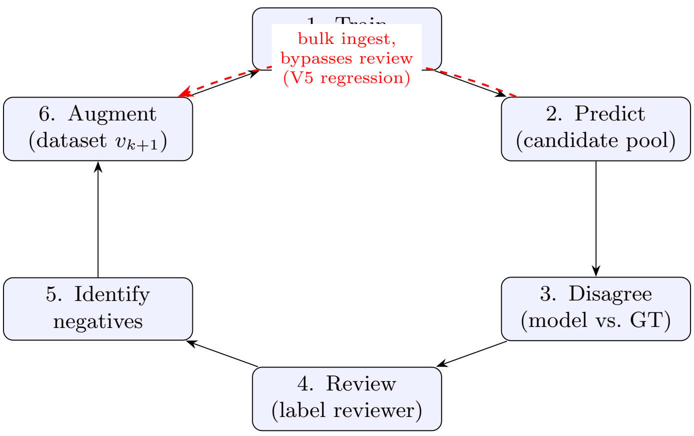


Figure 3.7: The HITL co-evolution loop. The disciplined path (blue) routes every candidate batch through Disagree→Review before it is augmented into the next dataset. The V5 regression (Section 4.1.2) took the red shortcut: a large positive batch was ingested directly, bypassing review, so unlabelled drones entered training and were scored as false positives.

A model FP is either a real error (hard negative needed) or an annotation error (correct the GT); a model FN is either a real miss (hard positive) or a spurious GT box (remove it). Every disagreement is actionable. The empirical record of this method, the IR detector's six-revision precision/recall trace on a fixed test split including the one revision where the loop was bypassed, is the case study of Section 4.1.2.

3.6 Reproducibility Infrastructure

Every run writes a manifest.json recording: git commit hash (and dirty-tree flag); SHA-256 of loaded weights; library and CUDA versions with the device; the full command line; per-frame conf threshold and imgsz; and run timestamps. The writer is eval/run_manifest.py; the per-run files live beside their results under eval/results/<run>/manifest.json, and each detection cache carries a sidecar manifest in eval/cache/. The cache layer builds a deterministic tag from (rgb_weights_hash, ir_weights_hash, imgsz, stride), preventing silent cache reuse across configurations; the unified detection cache (Section 3.2.4) extends this to the whole pipeline. Every quantitative claim is registered with its source file, reproduction command, and status. Detector inference is deterministic at fixed weights and conf; the trust classifier is deterministic at fixed feature input. The only stochastic step is the trust classifier's XGBoost *training* (the production robust8 and its comparisons), controlled by a fixed-seed sequence-stratified group split.

3.7 Model MRI: Feature-Space Analysis

This thesis follows a *statistics-before-training* discipline for its downstream learned components: design decisions are settled by measuring what a detector's features encode before any training run. The discipline applies to exactly three models: the feature-reuse RGB filter mlp_v5_v4 (Section 3.8.11), the thermal-native IR confuser filter mlp_aligned_thermalonly justified by IR feature separability (Section 3.8.8), and the trust router's feature selection that produced robust6/robust8 (Section 3.8.9). The detectors themselves were trained conventionally, and the hand-engineered sa32 comparison classifier predates the discipline. The feature-space evidence is produced by the *Model MRI* tool (python -m mri; modules mri/{extract,classifier,hol

3.7.1 What the Instrument Does

The tool hooks the detector's FPN layers (p3, stride 8; p5, stride 32) and pools each detection's region of those activations into a 517-feature embedding — 512 internal pyramid features (p3 pooled to a 2×2 grid, p5 to 1×1) plus 5 detection-metadata scalars (confidence, log-area, aspect ratio, and box centre) — then emits separability statistics, per-layer rankings, spatial activation overlays, and a stats.json keyed to the weight hash and corpus, written under mri/results/<run>/ in the project repository (e.g. `mri/results/v5_report_regen/stats.json`). Each statistic answers a specific sub-question; the two alignment diagnostics at the end of the list are produced by the MRI's modality-alignment module (`mri/modality_align.py`).

Linear Discriminant Analysis (LDA). Projects features onto the Fisher axis maximising between- to within-class variance for drone vs. confuser. Threshold accuracy on that axis quantifies linear separability; it is also the visualisation axis of the MRI findings in Section 4.2.

ANOVA F-statistic. Per-feature between- to within-class variance ratio; a large F marks a near-binary drone/non-drone switch neuron.

AUROC. Probability a random drone outranks a random confuser (0.5 chance, 1.0 perfect); a threshold-free separability measure, used for the IR drone-vs-confuser check of Section 3.8.8.

Leakage ratio. A second F-statistic computed *within* the drone class exposes features that discriminate *which scene* rather than *whether a drone*; the ratio of the two is the feature-selection gate of Section 3.8.9.

3.7.2 From Statistics to a Trained Filter

The MRI is not only a diagnostic; its output *is* the filter's training pipeline. The same 517-D per-detection embeddings it extracts for analysis become the training matrix for a small multi-layer perceptron (MLP) over two classes — positives are drone detections, negatives are confuser detections — trained with 5-fold cross-validation (`mri/classifier.py`); held-out evaluation (`mri/holdout.py`) gates the result on a corpus excluded from training. The production RGB filter (`mlp_v5_v4`) and the thermal-native IR filter (`mlp_aligned_thermalonly`) are both trained this way: measure first, then train the MLP on the MRI's own positive (drone) and negative (confuser) detection embeddings, and verify on a held-out gate.

Every run renders a human-readable report. Figure 3.8 reproduces the verdict block of the production RGB run verbatim; the full report is in Appendix C.

Headline IR-filter numbers come from a held-out evaluation with the CBAM probe set (Section 3.1.4) excluded from training. Re-running the MRI on the shipped V5 corpus corrected two figures and one headline claim in an earlier hand-written report: the report had quoted a mis-merged corpus size (35,098 where the shipped corpus holds 32,931 detections), rendered its evidence figures from a non-shipped sibling corpus, and named the detector's confidence score as the strongest single discriminator ($F \approx 15{,}000$), where the regenerated statistics place a `p5` channel first ($F = 42{,}346$) with confidence only sixth ($F = 10{,}696$). The instrument audits its own paper trail.

```
Model MRI -- Yolo26n_selcom_confuser_ft4_1280
Corpus:  19,334 drone / 13,597 confuser detections
Verdict:  Classifier strongly recommended -- large FP cut at low
recall cost.
Signal                   Value   Meaning
Raw hallucination rate   54.4%   bare detector, FP per confuser image
LDA separability         0.952   linear split of drone vs confuser
Max ANOVA F              42,346  strongest single feature
Projected FP cut         97.4%   confusers the classifier would reject
Recall retention         98.9%   true drones the classifier keeps
```

Figure 3.8: Excerpt (verbatim) from the Model MRI's auto-generated report for the production RGB detector: the instrument states a verdict, the evidence for it, and the projected operating point of the filter it then trains. Source: `mri/results/v5_report_regen/report.md`.

3.8 System Architecture and Component Design

3.8.1 Overview

The pipeline processes paired RGB and thermal-IR streams through a four-stage cascade (Figure 3.9). The production choice for each component is justified by the ablations of Chapter 4.

1. **RGB detector**: a YOLOv26-class single-stage detector trained on the 172,022-frame composite RGB corpus of Table 3.2 (drone positives plus bird confuser negatives from the AirBird, FBD-SV, and WosDetC sources and surveillance/traffic/driving background negatives); the production variant `ft4` adds the SelCom and 300-hard-negative confuser fine-tune of Section 3.8.7. Strong on drones; residually hallucinates on small bird-on-sky frames.
2. **IR detector**: a YOLOv26-class detector trained on iteratively curated thermal imagery (Section 3.5), with bird, airplane, and helicopter frames as hard negatives. Strong and selective on in-distribution thermal; on the OOD thermal-confuser corpus its residual fire is airplane-dominated, and on the paired benchmarks it is the stronger single-modality detector on Svanström and the weaker one on Anti-UAV, the asymmetry the trust classifier exploits. Also operates as a *grayscale-RGB fallback* when no thermal camera is available (Section 3.8.8).
3. **Trust classifier**: an XGBoost model that maps a frame's detection and scene features to a 4-way modality decision (`reject_both`, `trust_rgb`, `trust_ir`, `trust_both`).
4. **Confuser filter**: in production, the distilled `mlp_v5_v4` RGB feature-space filter (and its thermal-native IR counterpart) (Section 3.8.11). Because it reuses the detectors' already-computed internal features rather than re-processing crops, it is cheap enough to run *per frame* on every detection. It supersedes a 4-class MobileNetV3 patch filter which, being a separate CNN over crop

pixels, was expensive enough to require alert-gating (Section 3.8.10).

A temporal smoother sits on top of the cascade: it raises an alert only when the per-frame decision has persisted across a configurable sliding window. In the predecessor design the smoother also gated the (expensive) patch filter. The per-frame `mlp_v5_v4` filter removes that coupling, so the alert-gate analysis of Section 3.8.4 now describes the superseded patch path.

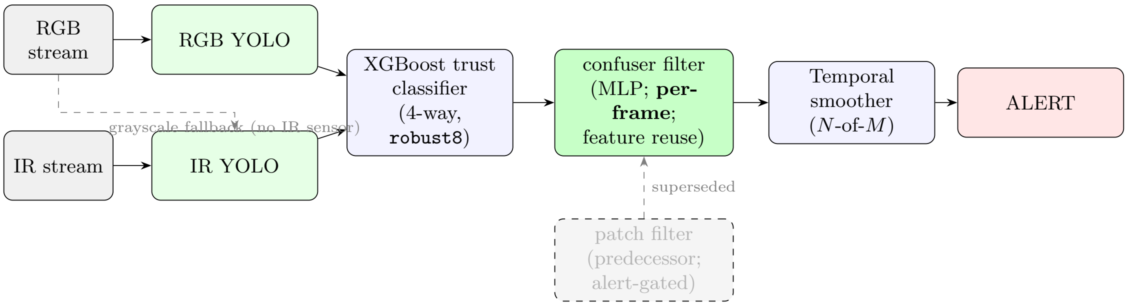


Figure 3.9: Multi-stage drone detection pipeline in the production composition: detectors, then the confuser filter, then the trust classifier. The order itself was ablated (Section 4.3): filter-first ships because it edges classifier-first on Svanström ($F1 = 0.946$ vs 0.931), and the two are within about a point on the paired benchmarks. The production confuser filter is the distilled `mlp_v5_v4` (RGB) with its paired thermal-native IR head, which runs **per frame** because it reuses the detectors' already-computed internal features rather than re-processing crops (Section 3.8.11). It supersedes the MobileNetV3 patch filter (greyed), which was expensive enough that it had to be *alert-gated*: consulted only when the temporal smoother was about to fire (Section 3.8.4).

The downstream stages contribute the majority of confuser suppression without retraining the detectors; Figure 3.10 shows the failure mode this addresses and the per-detection separation that fixes it. Three architectural choices make this possible: *deferred suppression* at the detectors (the detectors hold standard operating points and are never made stricter to fight confusers; precision is owned downstream, Section 3.8.2), trust-aware fusion (Section 3.8.3), and a downstream per-frame confuser filter. The production filter is a confidence-gated veto: a detection survives only if the MLP assigns it P(drone) at or above the production threshold (RGB 0.25, IR-thermal 0.05). A fail-open filter variant that releases uncertain detections was evaluated and rejected, because the detections it releases on cluttered surfaces are exactly the false positives the filter exists to remove.

Two-filter fusion (filter-first). With both the RGB (`mlp_v5_v4`) and IR filters running per-frame, the production composition is *filter-first*: each modality's confuser filter screens its detections, then the trust classifier (Section 3.8.3) routes among the survivors. `reject_both` drops the frame; `trust_rgb` keeps it only if the RGB filter passed; `trust_ir` keeps it only if the IR filter passed; `trust_both` is resolved per-modality (recall-first: the detection survives if either trusted filter passed). The IR filter is a single thermal-native CBAM net (`mlp_aligned_thermalonly`, $P(\text{drone}) \geq 0.05$; Section 3.8.8). The composition order (filter$\rightarrow$classifier vs classifier$\rightarrow$filter) is itself ablated in Section 4.3.

Figure 3.10: The confuser problem, and why a downstream filter solves it. *Left*: the production RGB detector (`ft4`) fires on a bird amid its flock (red box, detector confidence 0.46). *Right*: it fires on a drone (confidence 0.85). The detector cannot tell them apart, but the feature-reuse filter reads the detector's own internal features and assigns $P(\text{drone}) = 0.00$ to the bird (vetoed) and 0.96 to the drone (kept). This separation, not a better detector, is what the cascade provides.

3.8.2 Design Rationale: Deferred Suppression

The cascade's central design principle is asymmetric recoverability: a confuser false positive raised by the detector can be vetoed downstream, but a drone the detector missed cannot be recovered by any later stage. Suppression is therefore never the detector's job. This does not mean the detectors run loose: their floors are standard operating points (RGB 0.25, IR 0.40 in deployment; the canonical evaluation caches both at 0.25), and the confidence sweep shows those are normal optima rather than opened gates: Anti-UAV bare $F1$ is flat (0.959–0.963) across floors 0.05–0.5, and the in-distribution RGB test split peaks at exactly 0.25 ($F1 = 0.926$). What the principle forbids is moving the floor *up*, or retraining the detector into suppression, when confusers appear. Both detectors are trained with confuser negatives (Section 3.8.7, Section 3.8.8), but training-time mining does not eliminate all categories: bird hallucination at the RGB detector is irreducible at small drone sizes (Figure 3.1), as the three-stance comparison shows: the stance that suppresses birds (`retrained_v2`, bird fire 94.4% $\rightarrow$ 3.4%) collapses Svanström drone recall 0.961 $\rightarrow$ 0.306 (Table 4.1, Section 4.1.1). The cascade is built around that residual: the downstream stages catch what training cannot, at a cost and benefit quantified stage-by-stage in Section 4.3. The same deferral is what licenses the low-confidence operating mode (Section 4.3.6): on a recall-starved OOD surface the floor can drop to 0.05 *because* precision ownership has already moved downstream (SelCom 0.591 bare at the default floor $\rightarrow$ 0.692 with the filter at floor 0.05).

3.8.3 Trust-Aware Fusion

The fusion stage is a learned per-frame decision rather than a fixed vote. The classifier outputs one of four labels:

- **`reject_both`**, neither modality's detection is trustworthy (likely a confuser).
- **`trust_rgb`**, only RGB is reliable for this frame.
- **`trust_ir`**, only IR is reliable for this frame.
- **`trust_both`**, both modalities agree and are trustworthy.

The 4-way formulation captures *when* each modality is reliable. The asymmetry the classifier exploits is that *the two detectors hallucinate on different scenes*: visual confusers fool the RGB detector more reliably; thermally degraded conditions fool the IR detector more reliably. The router does not read the scene to detect this. It sees *detection evidence only*: its eight features are the per-modality maximum detection confidence, the mean RGB confidence, and the best detection's log box area and aspect ratio for each modality, plus the binary grayscale-regime flag. A frame on which one detector emits low-confidence, oddly-shaped boxes while the other emits a single confident, well-proportioned one carries the failure-profile signal in the detections themselves; scene statistics appear only in the hand-engineered `sa32` comparison classifier. A per-frame classifier can pick the modality whose failure profile is least active; a fixed-weight rule cannot. The per-surface evidence, including the modality reversal between Svanström (IR stronger) and Anti-UAV (RGB stronger) that makes routing pay, is in Section 4.3.

The production router is the no-reject `robust8-nr`, the eight-feature `robust8` with the `reject` class dropped (Section 4.1.4), whose feature set was screened by the scene-fingerprint ratio of Section 3.7: candidate features that discriminated *which scene* a frame came from, rather than *whether the detections are trustworthy*, were excluded, so the router cannot pass its evaluation by memorising surface identity. The 32-feature `sa32` serves as the strongest classical comparison throughout the evaluation; because its features leak scene statistics, its in-domain lead is read as an upper bound that is not trusted to generalise to unseen surfaces, not as a deployable target.

3.8.4 Alert-Gate Cascade (superseded patch path)

This section describes the MobileNetV3 *patch* filter's position in the predecessor design, retained because part of the historical evidence was measured under it. Alert-gating was forced by the patch filter's cost (a second CNN over crop pixels): the filter was consulted only when the temporal smoother was about to emit an alert. Two alternative cascades were considered and rejected at the time (`filter_then_classifier` and `classifier_then_filter` as *per-frame patch* stages), because the patch CNN's per-frame cost and its veto of marginal drone TPs cost ≈1 pp Svanström F1. The per-frame `mlp_v5_v4` filter removes the cost argument entirely, and the composition-order question is re-answered with it in Section 4.3. The production GUI path is implemented in `ir_gui/pyside_engine.py` (`should_suppress_alert`).

3.8.5 Temporal Smoother

The temporal smoother is an N-of-M sliding-window aggregator with configurable cooldown. An alert fires only when the trust classifier's per-frame decision has been positive on at least N of the last M frames, absorbing isolated firings characteristic of birds and aircraft transiting the field of view. Its measured contribution, at the segment grain on the real-video clips, is in Section 4.4.

The same temporal state drives a *ROI fallback* in the PySide engine (`_run_with_roi`). When the full-frame YOLO returns nothing and a recent track is still alive, YOLO is re-run on an expanded crop around the previous box at reduced confidence. This is a GUI-only feature, and the frame-mode evaluation harness does not use it; it accounts for part of the GUI-versus-eval performance gap on small distant drones. The GUI maintains tracks and renders a path image, but does not yet classify on track-level features (flight dynamics); track-level classification is a future-work direction.

[**PLACEHOLDER: PySide6 deployment GUI**] Annotated screenshot of the deployed GUI: dual-modality view (RGB + IR side by side), per-frame trust-label readout, sliding temporal window indicator, alert banner, ROI fallback crop inset, drone-path overlay across recent frames, threshold sliders for RGB / IR / filter.

Figure 3.11: PySide6 operator GUI. The temporal smoother's state, the trust classifier's per-frame decision, and the ROI fallback crop are all surfaced for the human operator.

3.8.6 Resolution Dependency

Resolution is the single largest hyperparameter affecting small-drone recall. Svanström's native resolution is 640×512; a dedicated sweep evaluating both RGB variants at both sizes under one harness (stride-7 sample, $n = 4{,}102$, IoP@0.5) isolates the resolution effect within each model (Figure 3.12): doubling the inference size buys both models roughly the same recall increment (-28 pp baseline, -25 pp `retrained_v2` at 640), while the absolute level is set by the training stance, with the confuser-suppressing `retrained_v2` unusable on Svanström at either size. On the full corpus the baseline's canonical 1280 figure is $R = 0.961$. Anti-UAV imagery (1920×1080) saturates at `imgsz=640` because drones already span enough pixels. All Svanström RGB evaluation uses `imgsz=1280` (Section 3.2.3). SelCom CCTV is a third regime: clutter is the binding constraint, and the imgsz sweep identifies `imgsz` = 960 as the F1-optimal setting for that camera's deployment operating point.

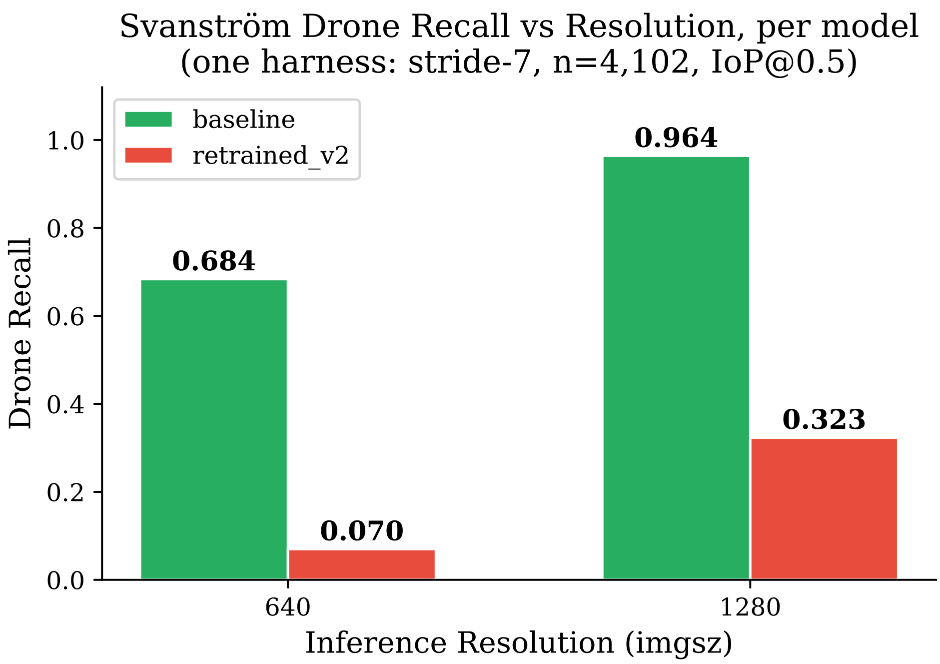


Figure 3.12: Inference-resolution dependency of Svanström drone recall, isolated *within* each model (one harness: stride-7 sample, $n = 4{,}102$, IoP@0.5, conf 0.25). Doubling `imgsz` buys both variants a similar recall increment (+28 pp baseline, +25 pp `retrained_v2`); the absolute level is set by the training stance. The confuser-suppressing `retrained_v2` is unusable on Svanström at either size.

3.8.7 RGB Drone Detector

The RGB detector is a YOLOv26n (nano) single-class detector trained on the composite corpus of Section 3.1.1, which includes a drone-vs-bird subset as deliberate confuser negatives. Svanström imagery is held out from training.

Three Training Stances

Three RGB-detector variants explore how much additional confuser-specific data to add on top of the baseline corpus; they occupy three operating points on the recall–hallucination axis, and their full comparison is Table 4.1 (Section 4.1.1).

1. **Baseline** (`Yolo26n_trained`). Trained on the composite drone corpus. On Svanström@1280 the bare detector, no downstream stage applied, reaches $P = 0.940$, $R = 0.961$ on drones, with a 94.4% bird-frame fire rate.
2. **`hardneg_v3more`**. Same backbone, retrained with the Svanström bird/airplane/helicopter confuser splits supplied as additional negatives. Helicopter fire drops 66.2% $\rightarrow$ 41.9% and airplane 74.6% $\rightarrow$ 64.7%, while bird fire is essentially unchanged (94.2%): hard-negative mining is class-specific.
3. **`retrained_v2`**. Aggressive retraining with heavy confuser augmentation. Bird fire collapses to 3.4%, and Svanström drone recall collapses with it, to $R = 0.306$. The lesson that motivated the per-detection confuser filter: at small drone scales, bird suppression and drone recall cannot both be had at the detector. (The trust router is needed regardless; its job is modality choice, not confuser rejection.)

The production RGB detector, `ft4` (`Yolo26n_selcom_confuser_ft4_1280`), is a fourth stance built on the lessons of the first three: the SelCom fine-tune line (Section 3.8.7) plus a *regression-gated* confuser injection (300 hard negatives at `freeze`=15; Section 3.4) that cut confuser hallucination 16 pp without recall loss.

SelCom CCTV Fine-Tune

A separate fine-tune line targets the deployment partner's CCTV stream (Section 3.1.7). The fine-tune is on top of the *baseline* weights (not `retrained_v2`). Two mixed-training fine-tunes (80% general RGB + 20% selcom; pure-selcom val split, seed 0) were trained at 640 and 1280; the 1280 variant doubles SelCom recall *and* raises precision over its 640 sibling (Table 4.2, Section 4.1.1), the clearest single demonstration that resolution, not architecture, is the binding constraint on small-drone CCTV. A preprocessing sweep established that no OpenCV variant (denoise, CLAHE, sharpen, contrast) improves detection independently of fine-tuning and `imgsz`; the imgsz sweep identifies 960 as the F1-optimal deployment point for this camera. The regression on the standard RGB dataset is within -0.008 F1 of baseline.

3.8.8 IR Drone Detector

The IR detector uses the same YOLOv26n architecture finetuned on iteratively curated thermal data via the HITL engine (Section 3.5). Bird, airplane, and helicopter thermal frames are added as hard negatives across revisions. The production weights (`v3b`) are a 2-epoch corrective fine-tune on the `Final` checkpoint addressing specific deployment-time failure modes. The full evolution across six numbered revisions (including the V5 regression and its recovery) is the case study of Section 4.1.2; on the paired benchmarks the IR detector is the stronger single-modality detector on Svanström and the weaker one on Anti-UAV (Section 4.3), which is precisely the asymmetry the trust classifier exploits.

Grayscale-RGB Operating Mode

The deployed PySide GUI exposes a *grayscale-RGB* operating mode (`_infer_grayscale`). When no thermal camera is available, the GUI converts the RGB stream to grayscale via `cvtColor(BGR2GRAY)`, replicates to three channels, and feeds the result to the IR detector at `imgsz=640`. This was built as a conservative hardware fallback; its measured behaviour (competitive with the dedicated RGB detector on bird-cluttered scenes, a few points behind it overall) turned out to be one of the thesis's findings (Section 4.5).

Cross-Modal Feature Alignment: a Thermal-Native Confuser Filter

The IR branch carries a per-detection confuser filter of the same kind as the RGB branch: a thermal-native net trained directly on thermal features. A Model MRI measurement (evidence figures in Section 4.2) establishes that the signal it needs is already present in the detector:

The confuser-rejection signal already lives inside the IR detector. A Model MRI scan (the feature-space instrument of Section 3.7) of the `v3b` fused `p3`+`p5` features, over 14,697 drone and 1,386 confuser detections, finds the two classes almost linearly separable (Table 3.10): a single linear discriminant splits them at 0.981 train accuracy. The IR feature space is the thermal twin of the RGB filter's, readable by the same kind of lightweight MLP with no detector retraining. The IR detector is also, unlike the RGB detector, already clean on near-domain confusers: on its Model-MRI thermal-confuser corpus (Section 3.7, the IR mining set) it fires on only 1.8% of images, against the RGB detector's 30.4% frame-fire on the `rgb_confusers_merged` out-of-distribution corpus (Section 4.3.2), so a thermal confuser filter is not strictly required. It earns its place on the hardest out-of-distribution thermal-aerial confusers: on the held-out CBAM `valid` split (Section 3.1.4; the TEST split disjoint from the CBAM `train` drones the head is trained on), the thermal-native head recovers drone recall to $R = 0.967$ while cutting false positives to 6 (bare 48), at no recall cost (Section 4.1.3).

Production therefore ships a single 517-D thermal-native head (`mlp_aligned_thermalonly`, a CBAM-trained net), trained on 8,112 thermal drones and 2,045 `IR_confusers/train` confusers (Svanström-IR, Anti-UAV-val-IR, `ir_dset_final`-train thermal drones plus the CBAM `train` drones, GT-aware). The thermal-native head holds drone recall on held-out thermal surfaces (antiuav_ir 0.937, ir_video 0.971, svanstrom_ir

Table 3.10: Model MRI of the `v3b` IR detector's fused `p3+p5` features (14,697 drone / 1,386 confuser detections): the confuser-rejection signal is already linearly present in the detector's own features. The projected columns are in-pool 5-fold estimates of what a reader of those features could recover.

MRI signal	value
LDA train separability (drone vs confuser)	0.981
Max ANOVA F (strongest single neuron)	5,370 (`p5`)
Median ANOVA F	256
Raw detector hallucination rate	1.8%/img
Projected FP cut ($\leq$10% recall cost)	89%
Projected recall retention	99.7%

0.966); the only cost is `ir_dset_final` 0.965 $\rightarrow$ 0.928 (-3.7 pp), on genuinely airplane-like drones that no threshold separates. It cuts CBAM confuser false positives from 48 to 6 (held-out CBAM `valid` split). The thermal head is trained on `IR_confusers/train` confusers plus Svanström-IR / Anti-UAV-val / `ir_dset_final`-train thermal drones, and is evaluated on the held-out CBAM `valid` split and the `IR_confusers` val/test re-mine (TEST split, disjoint from training); the held-out results are in Section 4.1.3.

3.8.9 Trust Classifier

Feature Set

The shipped trust classifier consumes per-frame features computed from the two detectors' raw outputs alone (detection flags, confidences, box geometry, cross-modal agreement); it does not read the source frames. The shipped classifier is the eight-feature no-reject `robust8-nr` (the eight `robust8` features with the `reject` class dropped; the ship rationale and ablations are in Section 4.1.4); the hand-engineered comparison set is `sa32` (32 features) and two 40-feature variants, which additionally drew on source-frame scene statistics (the family Section 3.8.9 shows to be a leakage source and drops). The feature families:

- **Detector signals**: per-modality binary detection flags, top-k confidence scores, detection counts.
- **Scene features**: image intensity statistics, edge density, frequency-domain summaries; intended to detect "confuser-rich" scenes vs "clean sky" scenes; as Section 3.8.9 shows, also the family that leaks scene identity.
- **Target features**: bounding-box geometry (aspect ratio, normalised area, position), confidence distribution per modality.
- **Agreement features**: cross-modal IoU, centre distance, confidence ratio.

The label space is the 4-way trust decision of Section 3.8.3. The model is an XGBoost [22] gradient-boosted classifier. Training uses sequence-stratified group shuffling so that frames from a given sequence are not split across train and test. The Svanström overlap this leaves at the cascade level is flagged in Section 3.3.

Hand-Engineered Variants (the comparison set)

Three hand-engineered variants preceded the statistical selection: `sa32` (`scene_aware_v3more_32fea` the strongest, and the working production pick through the real-video comparison, though it leaks scene statistics so its lead is not trusted to generalise), `fusion_no_fn_v1.1` (the most conservative on OOD confusers; rejects most drone TPs on operational video; retained as an open-world fallback where missed drones are operationally tolerable), and `control_v3more_40feat` (deprecated; no surface favours it). Their quantitative comparison, and the surface reversal between Svanström and real video that it revealed, is in Section 4.1.4.

The early `retrained_v2_32feat` classifier is superseded because the production RGB detector is no longer `retrained_v2`; a training-time/production confidence-distribution mismatch accounts for most of its gap. A trust classifier is not portable across RGB detector swaps without re-checking calibration.

Scene-fingerprint over-fitting. Under sequence-stratified splitting, per-clip image statistics act as *scene fingerprints*: the classifier learns which clip a frame came from rather than whether a drone is present, inflating in-distribution scores while degrading transfer. Dropping these features (the `lean13`→`lean10` pruning) recovers 18–26 pp drone $F1$ on held-out clips. Section 3.8.9 turns this hindsight observation into a statistic computed before training.

Statistical Feature Selection: `robust6` and `robust8`

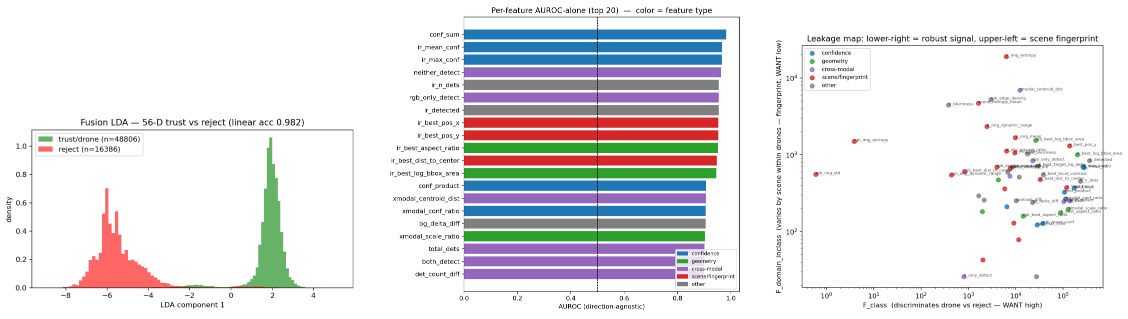
The variants above were assembled by hand; this subsection replaces that procedure with a statistical one. Two practical pressures sharpen the motivating question: `sa32` was trained on detections from a superseded detector (its inputs no longer match the production `ft4/v3b`), and several of its features require an expensive per-frame OpenCV pass.

Method. From a cached 56-feature matrix of 65,192 frames we ran four analyses. Linear Discriminant Analysis (LDA), which projects features onto the single axis that best separates the two classes, asks whether the signal is present: a single Fisher axis separates trust-positive from reject at 98.2% training accuracy (Figure 3.13a), so the problem is *selecting* features, not a signal shortage. Principal Component Analysis (PCA) asks whether the structure is trivial: classes overlap in the top components, meaning scene-to-scene variance dominates over the drone signal. A per-feature ANOVA F-test and single-feature AUROC (area under the ROC curve) then rank each feature individually (Figure 3.13b).

A leakage statistic. A scene-fingerprint feature can score as discriminative in a single pool simply because it correlates with which scene a sample came from; ANOVA and AUROC would recommend keeping it. To expose this we compute a second F-statistic *within the drone class only* and form the ratio

$$leakage\_ratio = \frac{F_{\text{domain-in-class}}}{F_{\text{class}}}, \tag{3.1}$$

which is low for a robust signal feature and high for a scene fingerprint. Equation 3.1 converts the earlier hand-tuned deprecations into a number computed before any training run.



(a) (b) (c)

Figure 3.13: Statistical anatomy of the 56-feature fusion space. (a) Linear discriminant: the trust signal is linearly separable (98.2% train accuracy). (b) Per-feature single-feature AUROC. (c) Leakage map (Eq. 3.1): the lower-right region is robust signal (high AUROC, near-zero leakage); the upper-left is scene fingerprint (chance-level AUROC, leakage in the hundreds). Source: `classifier/fusion_feature_stats.py`.

What the statistic finds. The leakage map separates two groups cleanly (Table 3.11). Scene-statistic features score at chance on the class label yet have leakage ratios in the hundreds: they discriminate *which scene*, not whether a drone is present, and are also the expensive per-frame features. The robust core (detection confidences, box geometry, and cross-modal ratios) scores AUROC above 0.9 with leakage below 0.006. A within-source check catches what the leakage ratio misses: `rgb_blurriness` scores AUROC 0.872 in the pooled set but collapses to 0.516 within each source (a cross-clip mean-shift artefact), so the gate drops it alongside the scene fingerprints. Two caveats. Per-modality detection-presence flags top the AUROC chart but are excluded by reasoning, since the trust label is *derived from* those detections. And the corpus is IR-dominant, so IR features rank above their RGB counterparts on RGB-fallback surfaces.

`robust6`. Applying the keep rule yields `robust6` = {`rgb_max_conf`, `ir_max_conf`, `rgb_best_log_bbox_area`, `ir_best_log_bbox_area`, `rgb_best_aspect_ratio`, `ir_best_aspect_r`

Table 3.11: Leakage analysis splits the feature space. *Top*: scene fingerprints, chance-level class AUROC but very high leakage; drop. *Bottom*: the robust core, high AUROC, near-zero leakage; keep. Source: `feature_stats_ranked.csv`.

feature	family	AUROC-alone	leakage ratio
Scene fingerprints (drop)			
`rgb_img_std`	scene	0.502	349.6
`rgb_img_entropy`	scene	0.510	307.4
`ir_blurriness`	scene	0.795	11.7
`rgb_edge_density`	scene	0.525	1.75
Robust core (keep)			
`conf_sum`	confidence	0.983	0.002
`ir_max_conf`	confidence	0.965	0.002
`ir_best_aspect_ratio`	geometry	0.952	0.002
`ir_best_log_bbox_area`	geometry	0.946	0.005
`xmodal_conf_ratio`	cross-modal	0.905	0.002
`rgb_max_conf`	confidence	0.816	0.004
`rgb_best_log_bbox_area`	geometry	0.806	0.059
`rgb_best_aspect_ratio`	geometry	0.778	0.011

the six independent per-modality primitives, all free detector outputs and box geometry, all near-zero leakage. The two remaining keep-set entries in Table 3.11, `conf_sum` and `xmodal_conf_ratio`, clear the same gate but are summary aggregates of those per-modality confidences, so `robust6` does not carry them separately; the table lists a representative slice of the 56-feature pool, not the final feature set. Training data is re-mined from the current `ft4/v3b` detectors, fixing the drift in `sa32`'s inputs. Two ablation levels confirm the design: in-domain $F1$ is flat as the feature count grows from 6 to 32 (count is not the lever), but on OOD drone video the set *with* fingerprint features collapses to $F1 = 0.262$, climbing to 0.578 for `robust6`; dropping fingerprints roughly doubles OOD generalisation. Dropping IR geometry also loses confuser rejection, so box-geometry features are load-bearing. `robust8` extends this six-feature core with two further free detector features (`rgb_mean_conf` and `is_grayscale`); the shipped router uses those same eight features but drops the `reject` class (`robust8-nr`), for the reasons given in Section 4.1.4.

A further free lever. The strongest unused lever is the filter's own output: the RGB filter's drone probability separates trust-positive from reject at AUROC 0.949 vs 0.842 for raw confidence. It is withheld from `robust8` because coupling the filter into the classifier introduces stage dependency; it is recorded as the next free lever.

3.8.10 Patch Filter (Superseded Predecessor)

This section documents the *predecessor* confuser filter, retained because part of the historical evidence was measured against it; the production filter is `mlp_v5_v4` (Section 3.8.11).

The patch filter is a 4-class image classifier (`airplane` / `helicopter` / `bird` / `other`) on a MobileNetV3-Small backbone [42], chosen because it fits within a YOLO-forward-pass budget on the GTX 1050 Ti development hardware. It was trained on **45,917 RGB+IR patches**: drone crops taken from the detector training corpora's ground-truth boxes, and bird/airplane/helicopter crops hard-negative-mined from detector false positives over the confuser corpus (Section 3.1.2) and OOD YouTube video (`classifier/extract_patches_v2.py`); no CCTV crops are included. Per-modality twins are trained (`confuser_filter4_{rgb,ir}_v2`). On its own validation split it reaches accuracy 0.975, with per-class precision/recall of 0.907/0.971 (airplane), 0.988/0.938 (helicopter), and 0.893/0.712 (bird); birds were already the weakest class at training time, foreshadowing the deployment gap.

At inference the filter consumes the YOLO crop and emits a softmax over the four classes; the veto uses $p_{\text{confuser}} = \max(p_{\text{airplane}}, p_{\text{helicopter}}, p_{\text{bird}})$ thresholded at `patch_thr`. The `other` class is an explicit OOD outlet: crops that look neither like a trained confuser nor like a drone route to `other` instead of being forced into a confuser class. Four numbered versions exist; v2 is the audited production version (v3 over-vetoes and is never shipped; v4 ties v2). Its threshold sweep, per-category catch rates, and the cost analysis that motivated its replacement are in Section 4.1.3.

3.8.11 Feature-Reuse Filter (`mlp_v5_v4`)

The patch filter re-processes each detection crop with a network entirely separate from the detector; on unseen surfaces it abstains to `other` (on SelCom CCTV it is bit-for-bit identical to the bare detector, having never seen CCTV-scale crops). This motivates a structurally different filter.

The *feature-reuse filter* replaces pixel re-processing with feature reuse: a small MLP consumes the RGB detector's own internal `p3+p5` activations and detection metadata — the same 517-D per-detection embedding the Model MRI extracts (Section 3.7) — already computed for each candidate box. The production `mlp_v5_v4` (built by `build_balanced_v4_birdsplit.py`) extends this corpus with an in-distribution bird.v1i train split (birdsplit) and a rebalanced size×source drone manifold, closing a 22% coverage gap whereby the predecessor over-vetoed real `rgb_dataset_test` drones. The Model MRI characterised the parent feature space on the 32,931-detection corpus the shipped predecessor was trained on (19,334 drone / 13,597 confuser detections mined across the RGB surfaces): a single linear discriminant separates drone from confuser at ≈95% train accuracy, while unsupervised PCA shows near-zero cluster separation (silhouette 0.067): the signal is real but lives in a supervised subspace, so it needs a trained classifier rather than a distance threshold. The MRI findings, including the activation "brain scans" of what the filter reads, are in Section 4.2; the filter's measured effect on every surface is in Section 4.1.3. `mlp_v5_v4` is trained on `rgb_dataset` train+val, `rgb_confusers` train+val, pure-SelCom (the 311 `selcom_val` files blocklisted), and the bird.v1i 728-image *train* split, and is evaluated on the held-out bird.v1i 484-image *test* split (TEST split, disjoint by name from training): it keeps only 30/230 unseen-bird fires against the shipped filter's 91/230, and lifts `rgb_dataset_test` recall 0.691 → 0.887. All numbers are reported on each dataset's TEST split; Svanström-RGB is the in-sample exception (no held-out split exists, so its absolute number is in-sample and fair only for shipped-vs-`v4` Δ).

Feature reuse makes the filter 46–72× faster per detection than the patch CNN (1.3–2.1 ms vs 59–112 ms; 1–4% pipeline overhead), which is what makes *per-frame* verification affordable and removes the alert-gate coupling of the predecessor design. On the shipped corpus the MLP reaches 5-fold CV $F1$ of 0.9857 ± 0.0004, rejecting 97% of confuser detections while retaining 98.9% of true drones. Its IR counterpart is the thermal-native `mlp_aligned_thermalonly` (Section 3.8.8).

Chapter 4

Empirical Evaluation

This chapter answers the three research questions, building from the components up to the system. The component deep-dives (Section 4.1) examine each stage in turn — the RGB detector's three training stances, the IR detector's six-revision HITL case study, the per-detection confuser filter, and the trust-classifier comparison — each isolated on the job it owns, and including the design-evolution evidence measured under earlier configurations, always labelled as such. The Model MRI findings (Section 4.2) then show *why* those components work: the discrimination they exploit is already linearly present in the detectors' own features. The full-pipeline ablation (Section 4.3) composes them, isolating what every stage contributes on every canonical surface under the single protocol of Section 3.2, with temporal (segment-level) evidence on the consecutive-frame video surface (Section 4.4). The grayscale cross-modal finding (Section 4.5), threats to validity, and limitations close the chapter. Every table states its n; headline cells carry 95% bootstrap confidence intervals in brackets: the central 95% range of the metric over 1,000 resamples of the evaluation frames drawn with replacement, so two cells whose intervals do not overlap differ by more than sampling noise.

4.1 Component Deep-Dives

4.1.1 RGB Detector

Table 4.1: RGB variants on Svanström @ `imgsz=1280` (IoP@0.5, drone class; full 28,710 frames, the pre-protocol full-set run, internally consistent). Aggressive retraining destroys recall on small Svanström drones while leaving its own in-domain test performance intact.

Model	Drone P	Drone R	Bird halluc	Airplane halluc	Heli halluc
baseline	0.940	**0.961**	94.4%	74.6%	66.2%
`hardneg_v3more`	0.941	0.950	94.2%	64.7%	41.9%
`retrained_v2`	0.943	0.306	**3.4%**	**5.6%**	**4.5%**

The three stances trace the lever that motivated the whole architecture (Section 3.8.7): hard-negative mining is class-specific (helicopters yield, birds do not), and the one stance that suppresses birds collapses drone recall. The dedicated resolution sweep (Section 3.8.6, Figure 3.12) puts the `imgsz=640` regime in within-model terms: the baseline loses 28 pp of recall at 640 and `retrained_v2` drops further from its already-collapsed level. (That sweep reports `retrained_v2` at $R = 0.323$ at 1280 against the 0.306 here: the same already-collapsed level measured on its stride-7 harness, $n = 4{,}102$, rather than this full 28,710-frame set, a sampling difference of 1.7 pp, not an inconsistency.)

Table 4.2: SelCom CCTV fine-tune; IoP@0.5, conf=0.25, val split 311 imgs / 295 GT. Doubling `imgsz` doubles recall *and* raises precision: a resolution win, not a trade.

Model	imgsz	TP	FP	FN	P	R
baseline (`Yolo26n_trained`)	640	2	6	293	0.250	0.007
baseline	1280	26	37	269	0.413	0.088
`Yolo26n_selcom_mixed_ft2`	640	72	50	223	0.590	0.244
`Yolo26n_selcom_mixed_ft2_1280`	**1280**	**138**	**43**	**157**	**0.762**	**0.468**

At `imgsz=960`, `selcom_960` is the best SelCom-holdout drone-first RGB variant tested ($F1 = 0.585$ on the hold-out); it is deliberately not in the production stack (it loses to the baseline on Svanström medium-range drones) but is recorded as the deployment operating point for this camera.

Real-video cameo: the three-stance continuum generalises. On the 19-clip diagnostic (per-clip table in Appendix A), the three stances occupy the same continuum (`retrained_v2` high-precision/low-recall at $P = 0.909, R = 0.453$; `selcom_1280` high-recall/low-precision at $P = 0.649, R = 0.812$; baseline between), and no single RGB variant dominates across scenes. The full six-mode table (including the IR-on-grayscale rows it introduced) is Table 4.19 in Section 4.5.

4.1.2 IR Detector: the Six-Revision HITL Case Study

The IR detector was developed across five numbered revisions (V2 through V6) plus a `Final` retrain and a `v3b` corrective fine-tune, by the loop of Section 3.5. Earlier per-version best-epoch mAP50 numbers are not directly comparable because the val split changed with the dataset. Every version was therefore re-evaluated on the same fixed `IR_dset_final` test split at `imgsz=640` (script: `eval/eval_ir_versions.py`).

The trajectory reads in four passes. *V2 → V3*: a narrower curated subset trades a little precision for a large recall gain (0.406 → 0.579), the premise for the mining cycles that follow. *V3 → V4*: precision jumps 0.648 → 0.895 with recall barely moving, the signature of an FP-mining cycle. *V5, the regression*: precision drops to 0.768 because a large positive batch was injected all at once, bypassing the review loop. The un-reviewed batch carried unlabelled drones; V5 fires on them and is scored as FP. The model did not get worse; the dataset state did, which is precisely the coupling Section 3.5.1 warns about. *V6/`Final`*: corpus cleanup recovers and

Table 4.3: IR detector versions evaluated on the fixed `IR_dset_final` test split @ `imgsz=640`. Comparable across versions; supersedes the per-version best-epoch mAP narrative. `v3b` is the production weight.

Version	mAP50	mAP50-95	P	R	F1	Status
V2	0.458	0.219	0.661	0.406	0.503	superseded
V3	0.571	0.242	0.648	0.579	0.611	superseded
V4	0.722	0.393	0.895	0.669	0.765	superseded
V5	0.694	0.446	0.768	0.709	0.737	superseded (regression)
V6	0.956	0.571	0.921	0.941	0.931	superseded
Final	**0.977**	**0.602**	0.955	**0.980**	0.967	base for prod
v3b (production)	0.972	0.591	**0.957**	0.977	**0.967**	current

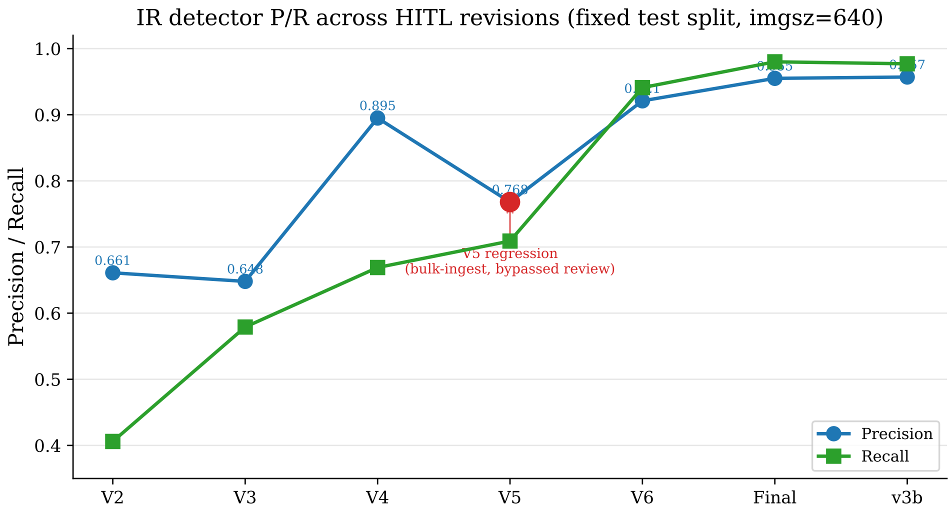


Figure 4.1: IR detector P/R trajectory across the five numbered revisions (V2–V6) plus the `Final` retrain and production `v3b`, on the fixed `IR_dset_final` test split. The V5 regression (precision 0.895 → 0.768) is recovered by V6 and the production `v3b`.

surpasses ($F1 = 0.967$). `Final` and `v3b` are indistinguishable on this split; `v3b` is retained as production because its 2-epoch corrective fine-tune addresses deployment failures (GUI regressions, hot-pixel scenes) not represented here.

OOD bound. The IR detector's strength is in-distribution. Its off-distribution *recall* on independent thermal drone footage is untested on verified data (exploratory runs on community thermal sets suggested substantial degradation, but those sets' label quality could not be verified and their numbers are not reported); this is a stated limitation (Section 4.6), and one of the reasons the RGB detector is retained as a parallel channel rather than treating thermal as primary.

4.1.3 Confuser Filter

The patch generation: useful, gated, and bounded. The patch filter's operating point was set by a threshold sweep on Svanström (stride 9, 3,190 frames): at `patch_thr`=0.9 it catches ∼63% of the classifier stage's residual confuser FPs at −4.0 pp drone F1, the production choice of its era; at 0.5 the recall cost doubles. Its veto distribution is bimodal (confusers $\gtrsim 0.9$, drone TPs near 0), which is what makes the threshold a usable knob.

The audit's verdict (every confuser bucket below the decisiveness bar, airplanes worst, and a hard distribution bound: on SelCom-scale crops it is bit-for-bit the bare detector) is what motivated replacing pixel re-processing with feature reuse.

Table 4.4: Feature-reuse filter (the production `mlp_v5_v4`) vs the MobileNetV3 patch filter (v2), both on the FT4 RGB detector. $F1$ is drone-class (IoP@0.5 on Svanström/SelCom, IoU@0.5 on Anti-UAV/`rgb_dataset`); *halluc* is the confuser hallucination rate. `mlp_v5_v4` wins every confuser-rich surface and, with the bird-split re-mine, also closes the prior `rgb_dataset` carve-out (0.792 → 0.916), now leading the patch filter there too. (The Svanström bare $F1$ shown here, 0.596, is on this table's stride-9 comparison sample; the canonical even-strided n=4,000 value is 0.607, Table 4.7 — a sampling difference, not an inconsistency.)

	Drone $F1$			Halluc. rate		
Surface	bare FT4	+ patch v2	+ `mlp_v5`	bare	patch v2	`mlp_v5`
Svanström (IoP@0.5)	0.596	0.768	**0.861**	0.470	0.163	**0.044**
Anti-UAV (IoU@0.5)	0.986	0.986	0.984	0.011	0.011	**0.012**
SelCom CCTV (IoP@0.5)	0.591	0.591	**0.612**	0.071	0.071	**0.022**
`rgb_dataset` (IoU)	**0.929**	0.904	**0.916**	0.028	0.026	**0.029**
confuser-only	—	—	—	0.317	0.107	**0.021**

The feature-reuse generation.

The carve-out, diagnosed and then closed. The earlier `mlp_v5` filter regressed on the photo-style `rgb_dataset` split to $F1 = 0.792$ (−11 pp), driven by a drone recall collapse (0.899 → 0.691). The Model MRI localised why: the falsely-vetoed drones were not low-confidence (confidence-scalar difference between vetoed and kept: 0.000). They were *smaller*, with a distinct deep-feature signature, and sat *farther from the confuser distribution than the kept drones did* (centroid distance 16.5 vs 11.1) while being far from the filter's training-drone cluster (15.4). The filter rejected them because they did not resemble its training drones, not because they resembled birds: an OOD *coverage* gap, not a structural overlap.

That diagnosis prescribed its own fix, and the production filter implements it. `mlp_v5_v4` re-mines the filter's training corpus with the missing manifold filled in: a size×source-balanced drone re-sample (small `rgb_dataset` drones from the `train+val` splits, the cells the original stride-8 quota left empty) plus an in-distribution bird-confuser split, the same diverse-data lever that made the IR filter recall-safe. The coverage gap closes directly: `rgb_dataset_test` recall recovers 0.691 → 0.887 (held out; F1 0.792 → 0.916, Table 4.4) without retreating on any other RGB drone surface, and the per-size cells confirm the recovery is exactly where the gap was (Table 4.13, sub-32 px). The earlier no-retrain fail-open gate (k-NN distance to the confuser reference) is therefore superseded: it recovered 91–100% of the vetoed drones at a 5–10% confuser leak but backfired on cluttered Svanström (precision 0.887 → 0.631), and is retained only as design-history (Figure F.4) for the operating point the re-mine made unnecessary.

Held-out bird suppression (TEST split). The RGB filter `mlp_v5_v4` is *trained* on the v2 balanced corpus (19,334 drone / 13,597 confuser: balanced `rgb_dataset` drones from `dataset/{train,val}` + Anti-UAV `val`/RGB; pure-SelCom mined crops with the 311 `selcom_val` files blocklisted; `rgb_confusers_merged/{train,val}`) *plus the bird.v1i TRAIN split* (728 of 1,212 images, 60/40 seed-0 name split). It is *evaluated* on the held-out **bird.v1i TEST split** (484 unseen images, 230 bird fires): `mlp_v5_v4` keeps only 30/230 (13%) where the shipped `mlp_v5` kept 91/230 (40%), so the production filter *beats* its predecessor on *unseen* birds (AUROC 0.981). The four cached drone/confuser surfaces above are genuine disjoint test splits (`rgb_dataset_test`, `antiuav_rgb`, `selcom_val` by blocklist, `rgb_confuser`); the Svanström RGB numbers are read as a shipped-vs-candidate Δ. The single residual cost is +5 non-bird FP on the cached `rgb_confuser` surface (Table 4.11, 29 → 39).

The thermal-native IR filter. The production IR filter is a single thermal-native head (`mlp_aligned_thermalonly`), trained directly on thermal confuser crops (Section 3.8.8). It is evaluated on a held-out gate: CBAM, a novel thermal aerial-confuser set excluded from its training.

Two properties define the thermal head: a drone-diversity re-mine (∼8k thermal

Table 4.5: Thermal-native deploy filter (`mlp_aligned_thermalonly`, `conf`=0.40, `thr`=0.05) vs the bare IR detector; FP in parentheses. CBAM is held out of training. Trained directly on thermal confuser crops, the filter cuts held-out novel-confuser FPs 48 → 6 (−88%) *and* recovers the thermal drone recall the earlier grayscale-harvested filter collapsed (CBAM recall 0.967, no collapse); its only cost is −3.7 pp recall on the genuinely airplane-like `ir_dset` drones no threshold separates. The MobileNetV3 patch filter on the same held-out CBAM cuts only 7 FP.

Surface	n	bare IR (`v3b`) $P/R/F1$ (FP)	+ aligned MLP $P/R/F1$ (FP)	
CBAM (held out)	180	0.547/0.967/0.699 (48)	**0.905/0.967/0.935 (6)**	0.0
`ir_dset_final`	4806	0.965/0.965/0.965 (109)	0.969/0.928/0.948 (98)	−0.
`ir_video` test	831	0.909/0.977/0.942 (80)	0.909/0.977/0.942 (80)	0
Anti-UAV test	4269	0.983/0.942/0.962 (68)	0.983/0.942/0.962 (68)	0
patch filter on CBAM	180	—	0.564/0.883/0.688 (41)	(cu

drones from Svanström IR, Anti-UAV `val`/IR, `IR_dset_final/train`, `IR_video/train`, plus GT-aware CBAM-`train` drones) keeps it recall-safe on held-out thermal, and *thermal-native* confuser crops (`IR_confusers/train`, category×size-balanced, plus GT-aware CBAM-`train` confusers) provide its rejection signal: this is the change that closed the airplane gap the earlier filter could not. Held-out generalisation is genuine: on the `IR_confusers val+test` re-mine (388 disjoint fires), the thermal head keeps only 22 (94% removed) against the shipped filter's 90 (77%); ir_dset_final, antiuav_ir, and ir_video are all evaluated on their `test` splits (Svanström IR drone recall holds flat at 0.966). This held-out 90 → 22 is distinct from the on-cache `IR_confusers` fire in Table 4.11, whose surface is the filter's own `train` split. The airplane category remains the relatively weakest, now as an in-distribution recall trade rather than an open suppression hole.

Operating points: the recall/false-positive dial. Each filter's threshold is a tunable operating point, not a fixed property of the network. Figure 4.2 sweeps the per-detection $P(\text{drone})$ threshold of both production filters, plotting drone-detection recall (the fraction of true-drone detections retained, pooled over that filter's drone surfaces) against the confuser fire-rate on the matching confuser surface. Two shapes emerge. The RGB `mlp_v5_v4` sits on a flat high-recall shoulder: at the shipped 0.25 it keeps 95.6% of true-drone detections at a ∼1% confuser fire-rate, and stays recall-transparent ($\geq$ 97%) well below that, so its recall is essentially free to recover. The thermal-native head now *suppresses* thermal confusers (the airplane gap the earlier filter could not reach): its fire-rate falls with the threshold rather than staying flat, and the residual is the genuinely-ambiguous airplane-like fraction the filter keeps recall-safe rather than a threshold artefact (Section 4.7). The reading for a deployer is that the shipped points sit deliberately on the high-precision side. Where the limit is the threshold (RGB), recall is recoverable almost for free by lowering it; where the limit is coverage (the small in-distribution airplane-like thermal fraction), lowering helps only partially and at a steeper false-positive cost. The shipped configuration chooses precision where confusers dominate; a recall-first deployment moves left along these curves.

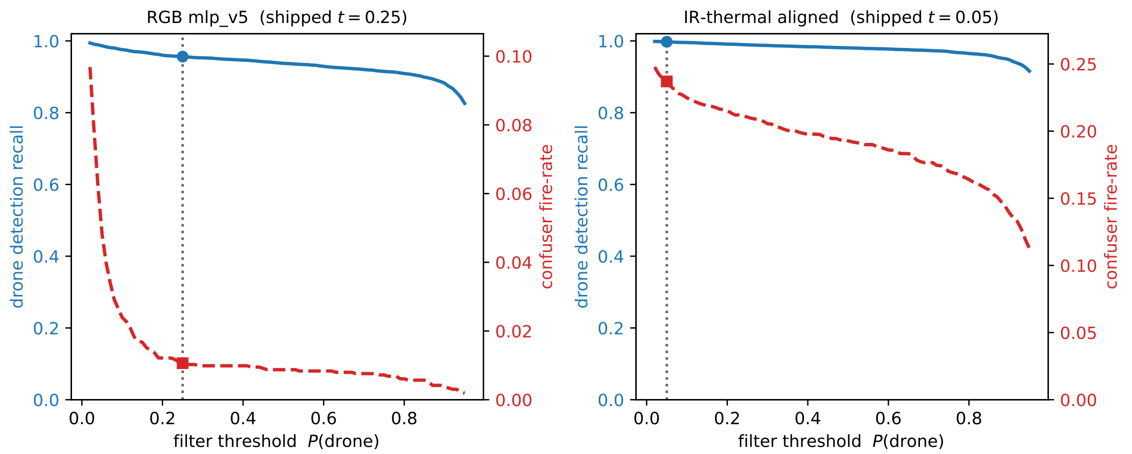


Figure 4.2: Filter operating points (detection level). For each production confuser filter, drone detection recall (blue, left axis; fraction of GT-matched true-drone detections retained, pooled over the drone surfaces) and confuser fire-rate (red dashed, right axis; matching confuser surface) as the per-detection P(drone) threshold varies; the shipped point ($t = 0.25$ RGB, 0.05 IR-thermal) is marked. RGB sits on a flat high-recall shoulder; the thermal-native head now suppresses thermal confusers (fire falls with the threshold; the residual is the airplane-like in-distribution fraction). Source: `eval/filter_operating_sweep.py`.

4.1.4 Trust Classifier

The production router is the no-reject `robust8-nr`; this section reports the design path that produced it, from statistical feature selection through the reject-class variants to the shipped no-reject form. The lineage is one feature set with one decision rule changed. Feature selection (Section 3.8.9) produced the six-feature `robust6` and its eight-feature extension `robust8`, both *four-class* routers that can label a frame `reject` (trust neither modality). The shipped router keeps those eight features but *drops the reject class*, delegating all false-positive removal to the per-frame filter. The reject-class variants flank each other in the pipeline ablation (Tables 4.7, 4.11) exactly as the feature statistics predicted. On the measured surfaces `sa32` actually edges the statistical generation, both in-domain and on raw confuser suppression (clf-only RGB-confuser fire 0.003 vs `robust6`'s 0.035); but it does so by *leaking scene statistics* — discriminating which scene a frame came from rather than whether its detections are trustworthy (Section 3.8.9) — so its lead is not trusted to generalise to unseen surfaces, and it costs 404× more per frame and was trained against a superseded detector's calibration. The statistical `robust6`/`robust8` generation is therefore preferred not because it suppresses more, but because it is leakage-free, cheap, and re-mined on the production `ft4`/`v3b` detectors; the shipped `robust8-nr` then drops the reject class, trading a little of `robust8`'s confuser suppression for the recall it recovers on clean, single-modality, and degraded-channel surfaces.

The shipped router: dropping the reject class. The reject class is a liability on any surface that is not a confuser-rich paired stream. When a modality is absent (single-camera deployment), degraded (the grayscale-IR fallback), or the surface is simply clean (DUT), `reject` fires on recoverable drone frames and recall collapses: clf-only $R\,0.383$ on the RGB test split, 0.125 on SelCom, and a 24% reject rate on confuser-free DUT (Tables 4.12, 4.10). `robust8-nr` removes the class at its source: it retrains the eight-feature `robust8` with the neither-correct rows dropped, so the router always routes to RGB, IR, or both (argmax, no threshold) and the per-frame filter owns all false-positive rejection. This recovers the lost recall in full (composed router+filter F1, the only configuration deployed: DUT 0.701 → 0.790, RGB test split 0.551 → 0.922, SelCom 0.220 → 0.612; mean across drone surfaces 0.744 → 0.850; the no-reject router suppresses no confusers on its own, so these are never router-only numbers) and dissolves two artefacts of the reject class at once: the grayscale recall *hole* (below) and the per-class threshold τ that patched it. The cost is confuser fire the reject net no longer compounds away: RGB-confuser frame fire 0.0011 → 0.014 (still near 1%, and damped by temporal voting, Section 4.4), thermal 0.024 → 0.028, real-video-confuser 0.098 → 0.236. We ship the no-reject router and keep `robust8` as the ablation that quantifies the reject net; on confuser-rich *paired* streams (the Svanström flagship), where both modalities are live and a `reject` is rarely wrong, `robust8` remains the stronger choice (at the shipped `filt→clf` order, 0.964 vs 0.946) and is the recommended configuration there. (The composed-F1 recovery figures above compare `robust8` and `robust8-nr` at a fixed `clf→filt` composition, isolating the reject variable; the shipped `filt→clf` order, chosen in Section 4.3.1, lifts the no-reject numbers further still, e.g. Svanström 0.931 → 0.946 and DUT 0.790 → 0.835.)

The grayscale hole, and why the reject class caused it. `robust6` inherits one honest weakness from the four-class formulation: in the grayscale-IR fallback the thermal features go uninformative, so drone-only-in-RGB frames are routed to `reject` and grayscale recall collapses. `robust8` patched this *within* the reject formulation by adding `rgb_mean_conf` and a per-class threshold τ on $P(\text{trust\_rgb})$: at $\tau = 0.20$ it lifts Svanström-grayscale recall 0.577 → 0.681 (Figure F.1), at the structural cost of 7–10× real-video-confuser fire on the grayscale slice (0.006 → 0.046), because `rgb_mean_conf` reads as a drone on Svanström but as a confident bird on real-motion video. Dropping the reject class removes the hole at its source: a router that never rejects cannot route a grayscale drone to `reject`, so `robust8-nr` needs neither the τ dial nor the precision/recall trade the figure illustrates. The sweep is retained as the design-history record of why the reject formulation was awkward to operate.

A dual-classifier extension (validated, not shipped). Splitting the router into a paired model and a separate grayscale model strictly dominates single `sa32` on missing-modality cases (RGB-fallback bird-confuser drone $F1$ 0.247 $\rightarrow$ 1.000; `rgb_dataset_test` 0.690 $\rightarrow$ 0.960) with paired-core performance unchanged; grayscale airplane/helicopter cases remain weak ($F1 \approx 0.50$). It is recorded as a validated extension rather than the shipped default; `robust8`'s two extra features close most of the same gap without a second model.

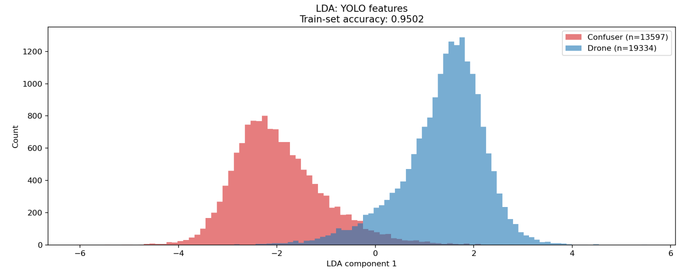
Design evolution: the OOD confuser zoo (predecessor configuration). The first quantitative router comparison ran the predecessor cascade (baseline RGB, alert-gated patch v2, `imgsz=1280`) over the confuser zoo. The open-world `fusion_no_fn_v1.1` suppressed the bare 52.1% fire to 1.6% at the classifier stage (0.8% after the filter); `sa32` reached only 20.5% (10.3%), a 13× gap on this surface.

The surface reversal. On real video the ranking inverted: `sa32` dominated segment drone $F1$ (0.826 baseline RGB) while `fusion_no_fn_v1.1` accepted only about one in eight of the correct RGB drone TPs (segment $R = 0.128$; seven of eight drone events missed) and `control_v3more_40feat` trailed `sa32` by 18 pp with no compensating win anywhere. The Svanström/zoo ordering does not select the production classifier; the operational surface does. `fusion_no_fn_v1.1` is retained as the open-world fallback for deployments where missed drones cost less than false alarms. This reversal is what discredited hand-ranking classifiers on single benchmarks and motivated the statistical selection that produced `robust6`/`robust8` above. Even `sa32`'s real-video win is not bankable: its scene-statistic leakage (Section 3.8.9) lets it score partly by memorising surface identity, so its lead is not trusted to generalise beyond the surfaces measured.

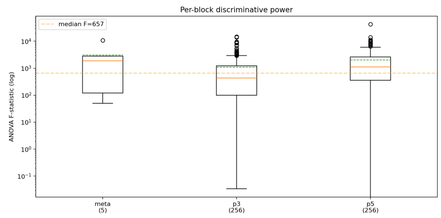
4.2 Model MRI Findings

The instrument is described in Section 3.7; this section reports what it found. The through-line: the discrimination the pipeline needs already exists inside the detectors' feature spaces, linearly accessible, in both modalities. That is why the production filters are 517-D MLPs rather than CNNs, and why the trust classifier needs eight features rather than thirty-two.

The signal is linear, supervised, and deep. On the RGB detector, a single Fisher axis separates drone from confuser at ≈95% threshold accuracy while unsupervised PCA finds near-zero cluster structure (silhouette 0.067): the signal is real but invisible to distance-based methods, which is why a small trained MLP



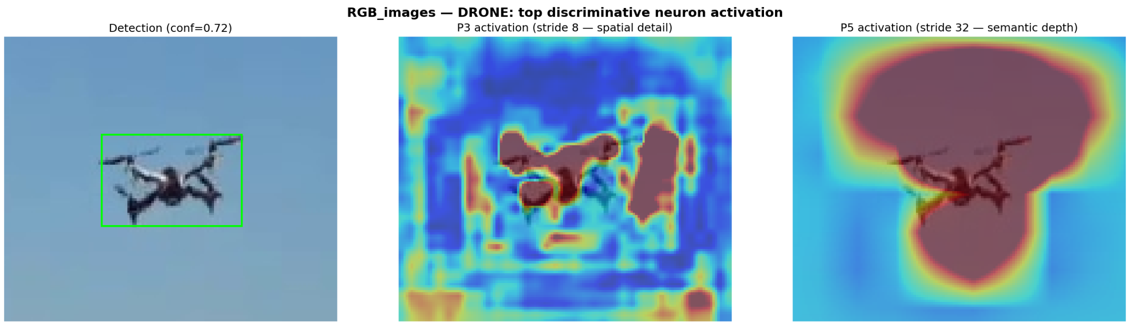
(a)



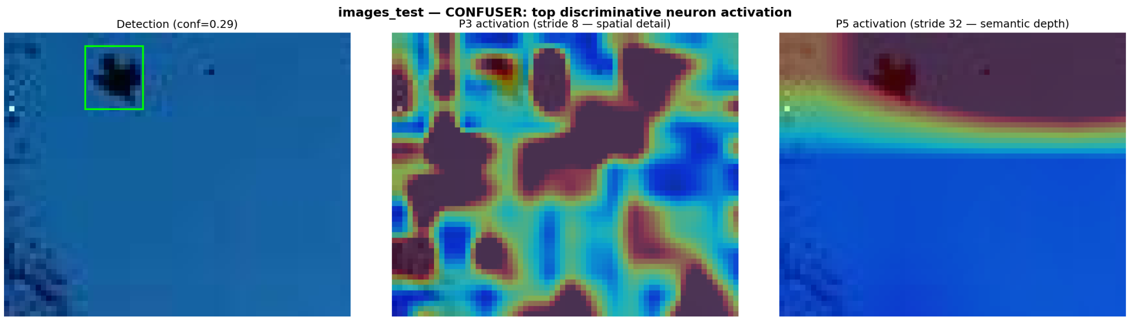
(b)

Figure 4.3: Model MRI of the FT4 detector's fused p3+p5 features on the shipped 32,931-detection corpus. (a) The linear-discriminant projection separates drone (blue) from confuser (red) into two cleanly-resolved peaks at $\approx 95\%$ single-threshold accuracy. (b) Per-block ANOVA F-statistic: p5 is the dominant discriminative layer (max $F = 42{,}346$, median 657), ahead of p3 and the detection metadata; the confidence scalar ranks only 6th.

is the right reader. The strongest single discriminator is a p5 semantic channel ($F = 42{,}346$), $4\times$ stronger than the detection-confidence scalar: the filter reads *what the detector saw*, not *how sure it was*. The IR detector is the thermal twin: LDA 0.981 over 14,697 drone / 1,386 confuser detections, strongest neuron $F = 5{,}370$, top discriminators again p5 channels with `meta:conf` far down the ranking.



(a) Drone (Svanström, conf= 0.72)



(b) Confuser (`rgb_confusers` test, conf= 0.29)

Figure 4.4: Model MRI spatial "brain scan" of the top discriminative neurons. Each row: the detection crop, the p3 activation (stride 8, fine spatial detail), and the p5 activation (stride 32, semantic depth). On the drone (a) p3 binds to the airframe and p5 forms a centred semantic blob; on the confuser (b) the same neurons fire diffusely across background structure. This activation-signature difference is what the feature-reuse filter reads.

The instrument audits its own paper trail. Re-running the MRI on the shipped V5 corpus corrected two stale figures and one headline claim in an earlier hand-written report (the dominant discriminator is p5, not the confidence score). The same diagnostic pattern (localise the failure in feature space, prescribe the recall-safe fix) recovered both the RGB carve-out diagnosis (Section 4.1.3) and the IR filter (Section 3.8.8).

4.3 Full-Pipeline Ablation

Stack under test. The production stack: RGB detector `ft4`, IR detector `v3b`, the no-reject trust router `robust8-nr`, and the two per-frame confuser filters, composed in the filter-then-router order (`filt`→`clf`; the design is detailed in Section 4.1.4). The router carries *no* reject class: false-positive rejection is delegated to the per-frame filter, which is the better tool for it (per-detection confuser discrimination on the detector's own internal features), leaving the router only to choose which modality to trust. Two different threshold stages should not be conflated. A *detector confidence floor* discards a detection before it enters the pipeline; these caches run both detectors at `conf`=0.25 (the floor at which the router's features were mined). Dropping the reject class does not change this floor: it is a detector setting, independent of the router's class set. The deployment IR floor is higher, 0.40, the IR detector's F1-optimal operating point (Section 3.8.2); the evaluation is insensitive to it because the filter rescores every surviving detection, so one floor is used for both detectors. A *filter threshold* is then applied to the filter's own P(drone) output, the probability a surviving detection must *clear to be kept*: `mlp_v5` (RGB) at 0.25, and the thermal-native IR filter at 0.05 on thermal input (Section 4.1.3). A *lower* bar keeps more detections, so the thermal 0.05 is the more permissive of the two: it is deliberately low because the IR detector barely hallucinates on thermal input (1.8% near-domain confuser fire, Section 3.7), so its filter should intervene as little as possible (recall-safe), not because thermal needs aggressive filtering. It is a filter threshold on thermal input, not the IR detector's deployment floor. Ablation arms swap exactly one element at a time: the router (`robust8`, the reject-class version of the same eight features; `sa32`, the strongest hand-engineered classifier; `robust6`, the six-feature statistical base the router was derived from), the filter (the superseded MobileNetV3 patch CNN at `patch_thr`=0.5), the composition order (`clf`→`filt` against the shipped `filt`→`clf`), or the stage's presence. All cells replay the same cached detections (Section 3.2.4); scoring is per-modality trust-aware, each modality against its own ground truth and never unioned (Section 3.2.2).

4.3.1 Paired Drone Benchmarks: the Cost Side

On drone-positive paired surfaces the question is what the pipeline *costs*: every vetoed detection here is potentially a missed drone.

Five readings, each attributable to one element.

Table 4.6: Pipeline ablation in aggregate, across all evaluation surfaces: bare detectors vs the shipped pipeline (per-frame confuser filter then the no-reject router `robust8-nr`, `filt`→`clf`). Drone surfaces report drone $F1$ (higher better; IoP@0.5 on Svanström/SelCom, IoU@0.5 elsewhere); the two confuser surfaces report frame fire rate (lower better). On the single-modality surfaces (IR test, SelCom) the router is inactive, so the shipped column there is detector + filter. The component-by-component isolation behind these numbers is in the deep-dives of this chapter; the full per-surface, per-stage cell matrix is given per surface below.

	Drone $F1$ ↑					Confuser fire ↓	
Stage	Svan	Anti-UAV	DUT	IR-test	SelCom	RGB	IR
bare detectors	0.742	0.973	0.758	0.961	0.591	0.304	0.294
shipped pipeline	**0.946**	**0.984**	**0.835**	0.942	0.612	**0.014**	**0.028**

Table 4.7: Full-pipeline ablation, Svanström paired (n = 4,000 frames, even-strided across the full 28,710-frame set, not the first 4,000; RGB @1280 / IR @640, IoP@0.5). *RGB only*/*IR only* score that detector alone against its own ground truth; every row below the second rule is trust-aware and *sums both modalities* (RGB-side scored vs RGB GT + IR-side vs IR GT), so its TP/FP/FN base is both modalities' drones combined, not the frame count. `clf` = +trust router; `filt` = +per-frame filter; `clf`→`filt` = router then filter. The shipped production cell is `filt`→`clf[robust8-nr]` (filter then the no-reject router; `robust8` with the reject class is the adjacent ablation, Section 4.1.4); the pipeline *raises* recall over bare (0.948 → 0.991) while adding ∼29 pp precision. `clf`→`filt` is the same router in the other order (Section 4.1.4 on why `filt`→`clf` is shipped).

cell	TP	FP	FN	P	R	F1 [95% CI]
RGB only (`ft4`, vs RGB GT)	1532	1845	141	0.454	0.916	0.607 [0.590–0.623]
IR only (`v3b`, vs IR GT)	1610	174	31	0.903	0.981	0.940 [0.931–0.949]
bare (both detectors)	3142	2019	172	0.609	0.948	0.742 [0.730–0.754]
filt only (`mlp_v5`, RGB)	3043	337	271	0.900	0.915	0.908 [0.900–0.914]
filt only (`aligned`, IR)	3142	2018	172	0.609	0.948	0.742 [0.730–0.754]
filt only (patch)	3004	782	310	0.793	0.907	0.846 [0.836–0.855]
clf only [robust8-nr]	3136	1997	27	0.611	0.992	0.756 [0.745–0.768]
clf only [robust8]	3092	354	31	0.897	0.990	0.941 [0.935–0.947]
clf only [robust6]	3132	286	34	0.916	0.989	0.951 [0.946–0.957]
clf only [sa32]	3137	187	30	0.944	0.991	0.967 [0.962–0.971]
clf→filt [robust8-nr]	3037	314	126	0.906	0.957	0.931 [0.923–0.937]
clf→filt [robust8]	2993	187	130	0.941	0.955	0.948 [0.943–0.954]
clf→filt [sa32]	3039	162	128	0.949	0.960	0.955 [0.949–0.960]
filt→clf [robust8-nr]	3037	317	29	**0.905**	**0.991**	**0.946 [0.938–0.950]**
filt→clf [robust8]	2996	191	35	0.940	0.989	0.964 [0.957–0.968]

Table 4.8: Full-pipeline ablation, Anti-UAV paired ($n = 4{,}000$ frames, even-strided across the 85,374-frame set, both modalities @640, IoU@0.5). Row conventions as in Table 4.7 (*RGB/IR only* score one detector vs its own GT; the trust-aware rows sum both modalities). The no-harm control: every pipeline cell is within ∼1 pp of bare or above it, the shipped no-reject `robust8-nr` included (this saturated surface rarely rejects, so dropping the reject class changes nothing here).

cell	TP	FP	FN	P	R	F1 [95% CI]
RGB only (`ft4`, vs RGB GT)	3657	41	68	0.989	0.982	0.985 [0.982–0.988]
IR only (`v3b`, vs IR GT)	3775	133	173	0.966	0.956	0.961 [0.956–0.966]
bare (both detectors)	7432	174	241	0.977	0.969	0.973 [0.970–0.976]
filt only (`mlp_v5`, RGB)	7430	172	243	0.977	0.968	0.973 [0.970–0.976]
filt only (`aligned`, IR)	7432	173	241	0.977	0.969	0.973 [0.970–0.976]
filt only (patch)	7430	174	243	0.977	0.968	0.973 [0.970–0.976]
clf only [robust8-nr]	7430	160	81	0.979	0.989	0.984 [0.981–0.986]
clf only [robust8]	7408	141	93	0.981	0.988	0.984 [0.982–0.987]
clf only [robust6]	7384	156	170	0.979	0.978	0.978 [0.975–0.981]
clf only [sa32]	7426	135	75	0.982	0.990	0.986 [0.984–0.988]
clf→filt [robust8-nr]	7428	157	83	0.979	0.989	0.984 [0.982–0.987]
clf→filt [robust8]	7406	139	95	0.982	0.987	0.984 [0.982–0.987]
clf→filt [sa32]	7424	134	77	0.982	0.990	0.986 [0.984–0.988]
filt→clf [robust8-nr]	7428	157	81	**0.979**	**0.989**	**0.984 [0.982–0.987]**
filt→clf [robust8]	7406	139	93	0.982	0.988	0.985 [0.982–0.987]

The pipeline maintains recall, and in fact raises it (RQ1). On Svanström the production cell's recall is *higher* than bare (0.948 → 0.991) while precision rises 0.609 → 0.901. That precision gain is RGB-side: the bare detector's confuser false positives (the RGB-only row alone carries 1,845 of them) are removed by the `mlp_v5` filter, whose drone/confuser separability is established independently by the Model MRI (Section 4.2). The recall gain is a separate, routing effect under per-modality scoring: the router stops the system being scored on the modality that is blind on a given frame, so its misses leave the books (bare FN 172 → 27 at the router stage). No new detection is created downstream; the architecture cannot do that. On Anti-UAV the pipeline is no-harm by every cell, which is exactly the role of a saturated control.

In the shipped stack the filter suppresses the false positives; the reject class is a complementary net we trade away (RQ2). The shipped no-reject router routes every frame, so on Svanström it removes almost no bare false positives on its own (2019 → 1997); the `mlp_v5` filter does that work (2019 → 337 alone, → 314 composed). The reject-class `robust8` ablation tells the complementary half of the story: its router *alone* removes 82% of bare FPs (2019 → 354) by vetoing whole untrustworthy frames, a different failure mode than the filter's per-detection veto, which is why composing `robust8` with the filter reaches 187 FP. Each stage catches what the other misses: the filter cannot recover a frame whose trusted modality is wrong, and the reject class cannot veto a single confuser detection inside a trusted frame.

We ship the no-reject router and delegate false-positive removal to the filter (Section 4.1.4); `robust8` keeps that second net at the recall cost quantified throughout this section.

Composition order is a recall/precision dial, and for the no-reject router `filt`$\rightarrow$`clf` *is the better side of it.* `filt`$\rightarrow$`clf` keeps recall at the router's level (0.991 on Svanström) because the router re-routes on the already-filtered detections; `clf`$\rightarrow$`filt` trades ~3 pp of that recall (0.960) for marginal precision. With a reject class the order mattered for confuser suppression (reject vetoes whole frames *before* the filter, the historic reason `clf`$\rightarrow$`filt` was shipped), but the no-reject router never vetoes a confuser frame, so the two orders give *identical* confuser fire (Table 4.11); with that tie broken we ship `filt`$\rightarrow$`clf` for the recall (Svanström F1 0.946 vs 0.931, DUT 0.835 vs 0.790).

Why the no-reject router is shipped. Among the reject-class routers, the 32-feature `sa32` leads every in-domain cell by 0.2–1.8 pp, but it needs an expensive per-frame OpenCV pass, was trained against a superseded detector's calibration, and leaks scene statistics: its features discriminate *which scene* a frame came from rather than *whether* its detections are trustworthy, so its in-domain lead cannot be trusted to carry to surfaces it has not seen (Section 3.8.9). `robust8` matches it within a point on eight free features at a 404× cheaper decision, with `robust6` between them. On the confuser-rich Svanström flagship the reject class earns its keep (at the shipped `filt`$\rightarrow$`clf` order, `robust8` 0.964 vs the no-reject 0.946, at far lower confuser fire). The shipped router is nonetheless the no-reject `robust8-nr`: a single global router must serve paired, single-modality, and degraded-channel deployments alike, and the reject class collapses recall on the latter two (Tables 4.10, 4.12); averaged across surfaces the no-reject router leads on the *composed* router+filter F1 the deployment actually runs (0.850 vs 0.744). These are composed means, not router-only: the no-reject router suppresses no confusers on its own (that is the filter's job), so it is never deployed without the filter behind it. `robust8/robust6/sa32` are retained as ablations and the trade is detailed in Section 4.1.4.

Dual-modality routing matches the better modality on every surface (RQ3). Table 4.9 scores each modality against its *own* ground truth (never a union) and the routed system trust-aware on the same frames. The per-surface reversal is the point: IR dominates Svanström (0.940 vs RGB's 0.607), RGB dominates Anti-UAV (0.985 vs IR's 0.961); a fixed single-modality choice loses badly on whichever surface its sensor is weak. The router matches or edges the stronger modality on each *without being told which it is* (Svanström 0.946, just above IR-only's 0.940; Anti-UAV 0.984, level with RGB-only's 0.985). That is routing's value: not a number a single modality could exceed on its own home surface, but the ability to serve *both* regimes from one operating point, since which modality wins reverses with the conditions (night, glare, range) a deployment cannot choose in advance (Section 3.2.2).

Table 4.9: RQ3: single modality vs routed pipeline (bare per-modality rows vs the production cell). Each modality is scored against its own GT; the routed row is trust-aware. The modality ranking *reverses* between surfaces; the routed no-reject system matches the stronger modality on each without being told which it is.

configuration	Svanström F1	Anti-UAV F1
RGB only (`ft4`, bare)	0.607 [0.590–0.623]	0.985 [0.982–0.988]
IR only (`v3b`, bare)	0.940 [0.931–0.949]	0.961 [0.956–0.966]
Routed (production, `robust8-nr`, filt→clf)	**0.946 [0.938–0.950]**	**0.984 [0.982–0.987]**

A clean in-domain check: DUT Anti-UAV (*test split*). The two surfaces above are IR-dominant (Svanström) or saturated (Anti-UAV). The DUT Anti-UAV *test split* [41] adds an RGB-strong, confuser-free check: an all-drone RGB benchmark on which the RGB detector `ft4` is in-domain (DUT was one source in the composite `rgb_dataset` it trained on), paired with the thermal `v3b` detector fed grayscale-converted RGB, i.e. the no-thermal cross-domain fallback. **The grayscale-IR column measures this fallback mode, not the IR detector's native thermal performance**: on real thermal it reaches $F1$ 0.940 (Svanström, Table 4.9) and 0.961 (Anti-UAV); the much lower numbers here reflect a thermal model run on grayscale RGB, an input it was never trained on, and should not be read as the IR detector's capability. Only the official test split is used (no DUT frame here was a training image for the router or filters). Because no confuser is present, every filter veto can only remove a true positive, so the pipeline's one available contribution is precision. We report at `imgsz`=960, the resolution-optimal setting for this small-drone surface (Section 3.8.6; drones down to 25×9 px need the pixels).

Two readings. First, dropping the reject class is what makes the fused stack competitive on a clean RGB surface: the shipped `filt→clf[robust8-nr]` reaches F1 0.835 (recall 0.800), recovering the recall the reject-class `robust8` surrenders here by rejecting 24% of these confuser-free frames; the per-frame filters remain a small recall tax where they can only delete true positives. Second, the strongest single configuration is still RGB-alone: `ft4` reaches $F1$ 0.899 against its own ground truth, above the fused 0.835, because the cross-domain grayscale-IR channel — a fallback mode far below its native thermal $F1$ 0.940 (Table 4.9) — drags the fused number down. On an RGB-strong surface the fused stack buys a live second channel and a bounded precision floor rather than peak F1, and the no-reject router keeps that cost small.

Why the shipped router drops the reject class. The four-class `robust8` can output `reject` (trust neither modality), which on a clean drone surface discards recoverable recall — the DUT gap above (24% of confuser-free frames rejected) and

Table 4.10: Full-pipeline ablation, DUT Anti-UAV *test split* ($n = 2{,}200$ frames, 2,245 GT drones, RGB `ft4` + grayscale-IR `v3b`, both @960, IoU@0.5, trust-aware). The grayscale-IR channel is the IR detector's no-thermal *fallback* mode (thermal model on grayscale RGB), not its native thermal performance. The full test split is used (not strided). Row conventions as in Table 4.7 (*RGB/IR only* score one detector vs its own GT; the trust-aware rows sum both modalities). Confuser-free (all-drone): filters can only delete true positives, so here the pipeline buys precision, not FP suppression. `filt`→`clf[robust8-nr]` (the shipped no-reject router) is the production configuration; dropping the reject class recovers the recall `robust8` surrenders on this clean surface (at `filt`→`clf`, R 0.574 → 0.800, F1 0.702 → 0.835).

cell	TP	FP	FN	P	R	F1 [95% CI]
RGB only (`ft4`, vs RGB GT)	1917	104	328	0.949	0.854	0.899 [0.889–0.908]
grayscale-IR only (`v3b`, vs IR GT)	1113	375	1132	0.748	0.496	0.596 [0.576–0.616]
RGB `ft4` + grayscale-IR `v3b`	3030	479	1460	0.864	0.675	0.758 [0.747–0.769]
filt only (`mlp_v5`, RGB)	2820	443	1670	0.863	0.620	0.722 [0.715–0.739]
filt only (`aligned`, IR)	2796	418	1694	0.871	0.626	0.728 [0.715–0.736]
filt only (patch)	2931	470	1559	0.862	0.653	0.743 [0.732–0.754]
clf only [robust8-nr]	3024	464	522	0.867	0.853	0.860 [0.850–0.869]
clf only [robust8]	2586	302	1301	0.895	0.665	0.763 [0.749–0.778]
clf only [robust6]	2415	374	1808	0.866	0.572	0.689 [0.674–0.705]
clf only [sa32]	2651	430	1480	0.860	0.642	0.735 [0.721–0.750]
clf→filt [robust8-nr]	2558	372	988	0.873	0.721	0.790 [0.781–0.803]
clf→filt [robust8]	2234	247	1653	0.900	0.574	0.701 [0.686–0.717]
clf→filt [sa32]	2265	361	1866	0.863	0.548	0.670 [0.655–0.686]
filt→clf [robust8-nr]	2558	372	638	**0.873**	**0.800**	**0.835 [0.827–0.849]**
filt→clf [robust8]	2182	237	1564	0.901	0.574	0.702 [0.691–0.725]

the single-modality collapse of Table 4.12. The shipped `robust8-nr` drops that class (same eight features, an argmax route to RGB, IR, or both) and delegates all false-positive removal to the per-frame filter; averaged across drone surfaces it leads on composed F1 (0.850 vs 0.744). The cost is confuser fire the filter no longer has the reject net to compound (RGB frame fire 0.0011 → 0.014, still ∼1% and damped further by temporal voting); we judge that trade worth the recall, and `robust8` is retained as the ablation that quantifies what the reject class buys (Section 4.1.4).

4.3.2 Out-of-Distribution Confusers: the Benefit Side

On confuser-only surfaces no drone is present; every surviving detection is a false positive, and the headline is the frame fire rate. These surfaces are evaluated in the single-modality deployment regime they represent (an RGB-only camera for the RGB confusers; a thermal camera for the IR confusers), with the router gating the one live channel.

Three observations, drawn together in Figure 4.5. First, the *filter* is the decisive stage on RGB confusers: the per-frame `mlp_v5` cuts FP detections 835 → 39 where the patch CNN manages only 835 → 282, so the feature-reuse filter is both 37–72× faster than the CNN it replaced and ∼10× stronger on the surface that matters. Second, the reject class compounds with the filter, and the shipped stack trades part

Table 4.11: Confuser-surface FP reduction. RGB confusers: $n = 2{,}633$ (full test split) @640. IR confusers: $n = 4{,}000$ of 5,237 (even spread) @640. Fire = fraction of frames with ≥ 1 surviving detection. The shipped `filt→clf[robust8-nr]` equals `filt only (mlp)` (and equals `clf→filt[robust8-nr]` to the digit, since a no-reject router never vetoes a confuser frame, so composition order is immaterial here); the `robust8` ablation reaches 0.0011 (RGB) by adding the reject net, the suppression the shipped stack trades for recall.

cell	RGB confusers	IR confusers
bare	0.304 [0.287–0.321]	0.294 [0.280–0.309]
clf only [robust8-nr]	0.304 [0.287–0.321]	0.294 [0.280–0.309]
clf only [robust8]	0.049 [0.041–0.057]	0.273 [0.259–0.287]
clf only [robust6]	0.035 [0.028–0.043]	0.223 [0.209–0.235]
clf only [sa32]	0.003 [0.001–0.005]	0.263 [0.249–0.277]
filt only (mlp)	0.014 [0.010–0.019]	0.028 [0.023–0.033]
filt only (patch)	0.102 [0.091–0.114]	0.246 [0.233–0.260]
filt→clf [robust8-nr]	**0.014 [0.010–0.019]**	**0.028 [0.023–0.033]**
clf→filt [robust6]	0.0011 [0.000–0.003]	0.019 [0.014–0.024]
clf→filt [robust8]	0.0011 [0.000–0.003]	0.024 [0.019–0.030]

of that away: the `robust8` ablation stacks reject on top of the filter to reach 0.11% (3 of 2,633 frames), whereas the shipped no-reject `robust8-nr` leaves that work to the filter alone at 1.4% (39 of 2,633) — from nearly one frame in three at the bare detector to roughly one in seventy, the deliberate cost of the recall it recovers on clean and single-modality surfaces.

Third, the thermal-airplane gap is now *largely closed* by thermal-native filter training (Section 4.1.3): on the `IR_confusers` surface the shipped no-reject cell suppresses thermal-confuser fire from the bare 0.294 to 0.028 (−90%). The surface is still airplane-dominated and airplanes remain every filter generation's weakest category *relatively*, but they no longer "resist": the residual fire is the genuinely-ambiguous fraction no threshold separates from drones, not an untrained-confuser-class gap. The remaining cost is an in-distribution recall trade on airplane-like thermal drones (Section 4.7), not an open suppression hole.

4.3.3 Solo In-Distribution Surfaces

The in-distribution test splits and the CCTV val set are single-modality surfaces: they receive the bare detector row (Part A) and the filter-only ablation, plus the router gating the one live channel.

Three per-surface readings. On the IR test split the thermal-native aligned filter costs ∼2.6 pp recall for a precision nudge — it now vetoes a few genuinely airplane-like `ir_dset` drones no threshold separates (Section 4.1.3) — but stays recall-safe relative to the patch CNN's 5.0 pp cost on the same frames. On the RGB test split the feature-reuse filter's former coverage carve-out is now *closed* by the bird-split re-mine (`mlp_v5_v4`): recall recovers 0.691 → 0.887, within ∼0.4 pp F1 of the

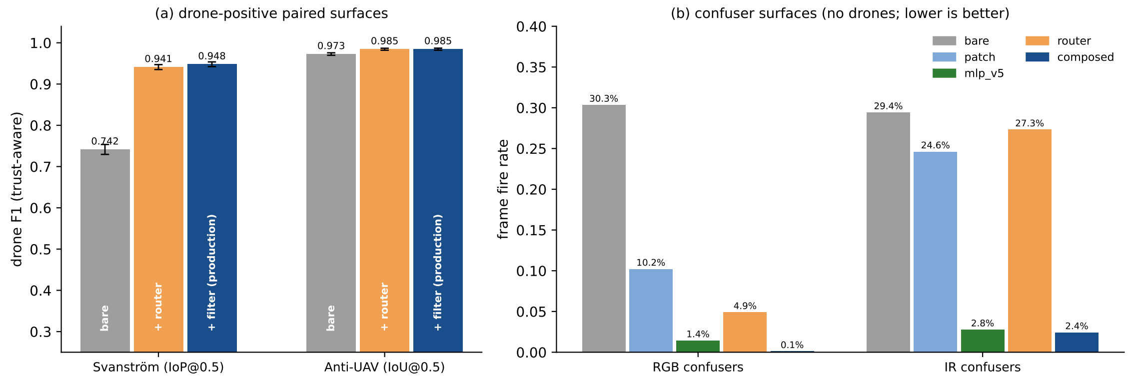


Figure 4.5: Pipeline ablation at a glance, drawn for the reject-class `robust8` cascade (the strongest-suppression configuration; the shipped no-reject `robust8-nr` trades part of the confuser suppression for recall, Table 4.11). (a) Drone-positive paired surfaces: the router supplies most of the F1 lift and the filter finishes it; Anti-UAV confirms no-harm. Whiskers are 95% bootstrap confidence intervals (the central range of the metric over 1,000 resamples of the frames; bars whose whiskers do not overlap differ by more than sampling noise). (b) Confuser surfaces: bare $\rightarrow$ patch $\rightarrow$ `mlp_v5` $\rightarrow$ router $\rightarrow$ composed frame fire rates, with near-total suppression on RGB (30.4% $\rightarrow$ 0.11% composed) and thermal-confuser fire now largely suppressed by the thermal-native filter (29.4% $\rightarrow$ 2.4% composed); the shipped no-reject stack delegates suppression to the filter alone and sits at 1.4% on RGB. Generated from `tier1_results.json` by `thesis_eval/gen_ablation_figure.py`.

Table 4.12: Solo-surface ablation (single modality vs own GT). IR test split: $n = 4{,}000$ of 9,612; RGB test split: $n = 4{,}000$ of 17,209; SelCom: full 311. The `clf` rows show the router on a single live channel: the reject-class `robust8` collapses recall (it routes against a dead second modality), while the shipped no-reject `robust8-nr` degrades gracefully to the bare detector, which is why single-modality deployments can run the full router. The filter's in-distribution recall cost is the honest counterweight to its OOD benefit.

surface	cell	P	R	F1 [95% CI]
IR test split	bare (`v3b`)	0.942	0.981	0.961 [0.955–0.967]
IR test split	clf [robust8]	0.947	0.923	0.935 [0.928–0.942]
IR test split	clf [robust8-nr]	0.942	0.981	0.961 [0.955–0.967]
IR test split	filt (aligned, thermal)	0.945	0.939	0.942 [0.934–0.949]
IR test split	filt (patch)	0.949	0.931	0.940 [0.933–0.947]
RGB test split	bare (`ft4`)	0.954	0.899	0.926 [0.919–0.933]
RGB test split	clf [robust8]	0.981	0.383	0.551 [0.532–0.570]
RGB test split	clf [robust8-nr]	0.954	0.899	0.926 [0.919–0.933]
RGB test split	filt (mlp)	0.960	0.887	0.922 [0.915–0.929]
RGB test split	filt (patch)	0.957	0.832	0.890 [0.881–0.898]
SelCom val	bare (`ft4`)	0.858	0.451	0.591 [0.532–0.639]
SelCom val	clf [robust8]	0.881	0.125	0.220 [0.161–0.276]
SelCom val	clf [robust8-nr]	0.858	0.451	0.591 [0.531–0.639]
SelCom val	filt (mlp)	0.950	0.451	0.612 [0.551–0.659]
SelCom val	filt (patch)	0.858	0.451	0.591 [0.532–0.639]

bare detector, so the −11.7 pp regression the earlier filter showed here is gone. On SelCom the filter cuts false positives 22 → 7 at *zero* recall cost, while the patch CNN, never having seen CCTV-scale crops, is bit-for-bit the bare detector — the abstention behaviour that motivated the feature-reuse design. Finally, a regime note: the reject-class `robust8` collapses recall on these single-modality surfaces (it routes against a dead second channel), but the shipped no-reject `robust8-nr` degrades gracefully to the bare detector instead, so single-modality deployments can run the shipped router safely; detector + filter remains the GUI's minimal single-camera configuration.

4.3.4 Per-Size Attribution: Where the Resolvable Floor Bites

Several claims in this thesis lean on drone *size*: the resolution floor of Section 3.8.6, the `retrained_v2` small-drone collapse, and the confuser filter's coverage carve-out. The unified cache stores every ground-truth and detection box, so per-size precision/recall replays exactly; buckets are ground-truth $\sqrt{\text{area}}$ in native-image pixels, with TP/FN bucketed by the ground-truth box and FP by the predicted box. A bucket's recall is only reported when it contains ground truth.

Table 4.13: Per-size recall, bare → +filter, on the two surfaces that carry size-dependent claims (production detectors, Tier-1 configurations). The earlier RGB-test small-drone carve-out is closed by the bird-split re-mine (`mlp_v5_v4`): +filter recall now tracks bare in every bucket. Svanström IR ground truth is dominated by sub-16 px drones, which is why `imgsz` matters there.

surface	detector	bucket	n_{gt}	bare R	+filt R
rgb_dataset test	`ft4`	<16 px	726	0.782	0.767
		16–32 px	607	0.865	0.844
		32–64 px	702	0.936	0.920
		≥64 px	1,417	0.956	0.951
Svanström	`ft4` (RGB)	<16 px	27	0.630	0.593
		16–32 px	994	0.897	0.823
		32–64 px	633	0.956	0.918
Svanström	`v3b` (IR)	<16 px	993	0.970	0.970
		16–32 px	643	0.998	0.998

Two attributions emerge. First, the confuser filter's former RGB-test regression (Section 4.1.3) was a *small-drone* recall cost that the production `mlp_v5_v4` closes: where the earlier filter collapsed sub-32 px recall, the bird-split re-mine restores coverage so +filter recall now tracks bare recall across every bucket (drones above 64 px were never affected). Second, the resolvable-floor story is quantified: Svanström's median drone is 29.8 px ($\sqrt{\text{area}}$, native 640×512 RGB) and its IR ground truth is majority sub-16 px (median 14.8 px), so a 640-pixel inference side leaves the median IR drone ∼15 native pixels across — the measured basis for the `imgsz=1280` requirement on the Svanström RGB channel (Section 3.2.3).

4.3.5 Background Failure Profile

Where do the remaining errors live: in the scene, or on the object? All 273 Svanström source sequences and the 9 drone video clips were manually tagged with a coarse background class from one representative frame each (`sky`: near-pure sky; `horizon`: sky with a visible ground/treeline/building strip; `ground`: clutter-dominant); the tag propagates to every cached frame of the sequence, and per-background scores replay from the cache. The tags are coarse, single-frame, and single-annotator, and sequences are split into drone-bearing and confuser-only before aggregating, because Svanström's pure-sky sequences are disproportionately its helicopter and airplane clips, a confound that would otherwise dominate the table.

Table 4.14: Background failure profile on Svanström (bare detectors, per modality vs own GT, Tier-1 rules). On confuser sequences the fire rate barely moves with background; on drone sequences precision is high on every background and recall dips on pure sky.

sequences	background	n frames	RGB FP-frame rate	RGB R	IR FP-frame rate
confuser-only	sky	1,294	0.667	—	0.006
confuser-only	horizon	888	0.741	—	0.017
confuser-only	ground	147	0.646	—	0.000
drone-bearing	sky	394	0.074	0.827	0.102
drone-bearing	horizon	1,114	0.033	0.944	0.073
drone-bearing	ground	163	0.000	0.933	0.153

The profile supports the thesis framing directly. On confuser-only sequences the RGB detector fires on 65–74% of frames *regardless of background*: the hallucination is driven by the confuser object, not by scene clutter, which is why a per-detection filter (which looks at the object) works and scene-level heuristics would not. On drone-bearing sequences the same detector is clean (0–7% FP-frame rate) on every background, and its recall dips on pure-sky sequences (0.827 vs 0.944 on horizon), consistent with the per-size table: the pure-sky clips carry the most distant drones. The thermal channel inverts the profile: near-zero fire on confuser sequences ($\leq$ 1.7%, airplane-dominated) and a mild ground-clutter penalty on drone sequences (15.3% FP-frame rate), which is one of the complementarities the trust router exploits. On the video clips the same breakdown is confounded with per-clip drone distance and is reported only as context in the results file.

4.3.6 The Filter Buys Back the Detector Floor

The production confidence floors (`rgb_conf`=0.25, `ir_conf`=0.40) exist for one reason: to keep the bare detector's false positives in check. With false-positive control delegated to the per-frame filter, that reason weakens. A detector-confidence sweep (0.05–0.50, crossed with the filter, replayed from a dedicated floor-0.05 cache) therefore asks whether *lowering the floor and letting the filter absorb the flood* beats the defaults.

On the deployment-anchor surface it does. On SelCom (full 311-image val), dropping the floor to 0.05 raises recall 0.451 → 0.678 (+22.7 pp) and the filter restores the precision the low floor surrenders (0.531 → 0.707), for $F1 = 0.692$ [0.654–0.730] against the default's bare 0.591 and filtered 0.612 [0.551–0.659]. The filter is *recall-transparent* here (filtered recall is identical to bare recall at every floor), which is precisely the mechanism the design promises: the detector proposes, the filter disposes, and nothing true is lost. The move is also confuser-safe: at floor 0.05 the RGB-confuser corpus floods the bare detector with 1,281 FP detections and the filter holds the line at 50 (96.1% suppression; the default operating point leaves 39). One bound applies: the gain over the *filtered* default is marginal at the CI level (the intervals overlap), while the gain over the bare default is clean.

Table 4.15: Detector-floor sweep × filter on SelCom (full 311 images, IoP@0.5, `imgsz=1280`). The filter is recall-transparent at every floor; lowering the floor converts directly into recall.

detector floor	bare $P/R/F1$	+`mlp_v5` $P/R/F1$
0.25 (default)	0.858 / 0.451 / 0.591	0.950 / 0.451 / 0.612
0.10	0.658 / 0.573 / 0.612	0.816 / 0.573 / 0.673
0.05	0.531 / 0.678 / 0.595	**0.707 / 0.678 / 0.692**

The mode does *not* generalise to a new global default, and the sweep maps its boundary precisely. On in-domain RGB the bare default remains the safe choice at every floor (the production `mlp_v5_v4` no longer carries a small-drone carve-out here, Section 4.1.3, but the low-floor mode buys nothing on this already-strong surface). On the thermal side the floor must stay at 0.40: dropping it balloons bare thermal-confuser FPs 1,051 → 1,911, and although the thermal-native filter now suppresses most thermal confusers at the default floor (Table 4.11), the sub-floor flood is dominated by the airplane-like detections it deliberately keeps recall-safe, so the low-floor mode is not worth its precision cost there. On the saturated Anti-UAV pipeline (a 500-frame screening subsample) every cell sits within CI of the default's 0.976; there is nothing to gain. The result is therefore an *operating mode*, not a setting: on hard out-of-distribution RGB surfaces where the detector is recall-starved, drop the RGB floor to 0.05–0.10 and let the filter own precision; everywhere else the defaults stand. One regime caveat travels with it: the trust router's features were mined at the 0.25 floor, so sub-floor router cells are deployed-behaviour estimates rather than retraining claims.

4.3.7 Runtime Cost

Table 4.16: Per-stage runtime (GTX 1050 Ti). The production stages are cheap enough to run on every frame; the predecessors they replaced were the reason gating architectures existed at all.

stage	predecessor	production	speedup
trust router	`fusion_no_fn` 38.3 ms/frame	`robust8-nr` 0.095 ms/frame	∼404×
confuser filter	patch CNN 59–112 ms/det	`mlp_v5` 1.3–2.1 ms/det	37–72×

Total pipeline overhead beyond the two detector forward passes is 1–4%. This is what dissolves the alert-gating compromise of the predecessor design (Section 3.8.4): when the filter costs two milliseconds, it simply runs always.

4.4 Temporal Evidence at the Segment Grain

The deployed system does not alert per frame: an N-of-M smoother (Section 3.8.5) aggregates per-frame decisions into alert events. The segment-level evaluation runs on the 19-clip real-video surface, the one canonical surface with consecutive frames (1,359 drone-clip frames across 9 clips; 1,250 confuser-clip frames across 10), using sliding 3-frame windows with a 2-of-3 vote per clip, in the dual-channel video regime (`ft4` on the RGB frame, `v3b` on its grayscale conversion, `is_grayscale`=1).

Table 4.17: Segment-level (2-of-3 windows) evaluation on the real-video clips. Fully out-of-distribution for every component. The per-frame filter thresholds are the benchmark-tuned production values; the last column is the relative confuser-fire cut from bare. The patch arm here is a per-frame approximation of the alert-gated original.

cell	window P	window R	window F1	confuser window fire (cut)
bare (both channels)	0.957	0.753	0.843	0.350 (—)
filt only (mlp)	0.954	0.489	0.646	0.213 (−39%)
filt only (patch, per-frame)	0.957	0.689	0.801	0.182 (−48%)
clf only [robust8-nr]	0.957	0.753	0.843	0.350 (−0%)
clf only [robust8]	0.967	0.520	0.676	0.101 (−71%)
clf only [robust6]	0.999	0.584	0.737	0.081 (−77%)
clf only [sa32]	0.967	0.582	0.727	0.120 (−66%)
clf→filt [robust8-nr]	0.954	0.489	0.646	0.213 (−39%)
clf→filt [robust8]	0.975	0.377	0.544	0.076 (−78%)
clf→filt [robust6]	0.998	0.412	0.583	0.059 (−83%)
clf→filt patch [sa32]	0.969	0.535	0.689	0.080 (−77%)

This surface is fully OOD for every component, and it is where the pipeline's costs show. Three results. *First, the trade is real on operational video, and the shipped no-reject router sits on the recall side of it:* `clf→filt[robust8-nr]` keeps window recall 0.489 (F1 0.646, the best composed F1 here) while cutting confuser fire 39%, whereas the reject-class compositions (`robust8/robust6`) push the cut to 78–83% but veto OOD drones aggressively (window recall 0.38–0.41), the video counterpart of the filter's coverage limit on fully out-of-distribution footage (Section 4.1.3). The per-frame filter thresholds were tuned on benchmark surfaces, so on fully OOD video they over-veto, which is exactly where the deployment's alert-gate placement matters. *Second, the gate position matters more than the gate:* the predecessor design consulted its filter only at the alert moment, and under that placement the same surface showed segment F1 *gains* (below); a per-frame veto applies the same discrimination thousands of times more often, so every false veto compounds. The measured per-frame numbers therefore *validate the GUI's alert-gate placement* for OOD video rather than contradicting the pipeline: routing per-frame (cheap, −71–77% fire at 10–17 pp F1) plus verification at the alert gate is the operating point the deployment uses. *Third, temporal voting is recall-neutral and precision-positive here* (window vs frame: $\Delta R \approx \pm 0.01$ on every cell, confuser fire consistently lower), confirming the smoother's role as a stabiliser rather than a recall recoverer when detections are already temporally consistent.

A filter-threshold sweep over the cached probabilities settles whether the veto cost is an operating-point artifact: it is not. Lowering the shared filter threshold from 0.25 to 0.01 recovers most of the composed cells' recall *relative to router-only* (robust8: R 0.377 → 0.507 against the router's 0.520; robust6: 0.412 → 0.577 against 0.584): at a near-zero threshold the per-frame filter costs only ∼1 pp recall on top of the router while still trimming window fire by 7–9% relative. But the true drones' filter probabilities on this surface are smeared across the $[0.01, 0.25)$ interval rather than concentrated near 1, so no threshold simultaneously keeps them and rejects confusers: the video-deployment choice is a near-zero per-frame threshold (cheap, marginal benefit) or the alert gate (the predecessor's measured-gain placement). The full sweep is `thesis_eval/results/video_thr_sweep.md`.

Design-evolution evidence (predecessor stack). The temporal smoother's value was first established on this surface with the predecessor stack (baseline RGB, `sa32`, alert-gated patch filter); those numbers are retained as the design-evolution record. At the segment grain that cascade *gained* drone $F1$ for all four RGB variants tested (+6.6 to +16.5 pp; baseline RGB 0.760 → 0.826) while cutting confuser FPR 39–81% (baseline: 0.512 → 0.162), and it compressed the spread between detector variants from 15.5 to 5.6 pp F1: the cascade absorbs detector variance. Per-frame metrics on the same runs *fell* for three of four variants, which is the wrong-unit-of-analysis lesson: the per-frame grain counts every wrong box, the deployed system counts alerts.

The per-category breakdown of the predecessor runs carries one durable lesson: the cascade's suppression concentrates on birds (45–57× on the worst variants), is moderate on helicopters, and is essentially inert on airplanes (∼0.21 FPR across all

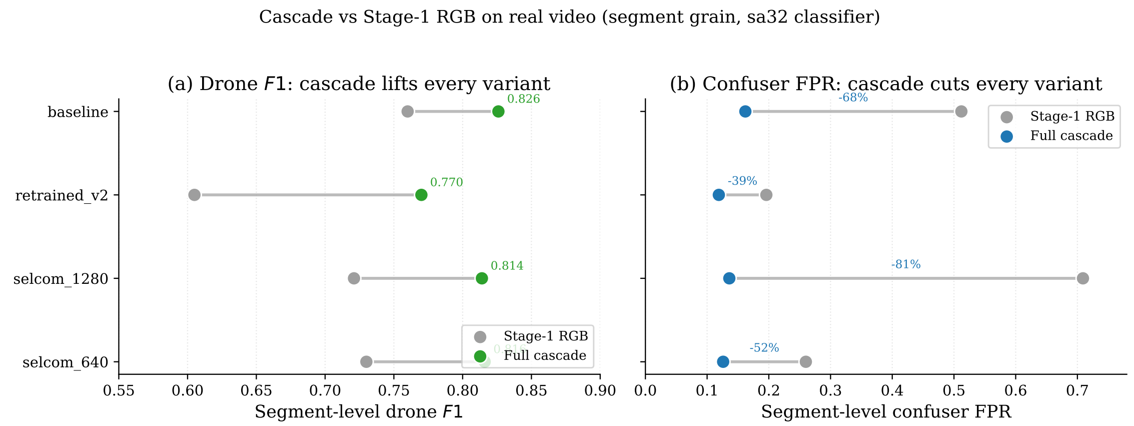


Figure 4.6: Design-evolution record (predecessor stack: baseline RGB, `sa32`, alert-gated patch): segment-level drone $F1$ and confuser FPR, Stage-1 RGB (grey) versus the full cascade, on the real-video diagnostic. (a) The cascade lifts drone $F1$ for every variant. (b) It cuts confuser FPR by 39–81%. The production-stack re-run of this surface is Table 4.17.

variants); the patch filter is genuinely uncertain on airplane crops (median confuser probability 0.540 vs 0.904 for birds). Airplanes remain the weakest category for every filter generation (Section 4.1.3).

4.5 The Grayscale Finding

The IR detector was trained exclusively on thermal infrared data. The grayscale-RGB mode (Section 3.8.8) exists as a hardware fallback: convert the RGB stream to single-channel grayscale, feed it to the thermal model. The finding is a recall result: run zero-shot on grayscale-converted visible light, the thermal-trained detector still *detects* drones when the RGB channel fails, at a recall competitive with a dedicated RGB detector, and with lower confuser fire. It was built as a conservative degradation mode and turned out to be a usable second detection channel.

4.5.1 The Fair Three-Way Comparison

On Svanström, at each model's canonical configuration and against the same ground truth:

Table 4.18: The grayscale three-way on Svanström (n = 4,000 of 28,710, IoP@0.5, bare detectors). The thermal-trained detector on grayscale-converted RGB performs within 2.7 pp F1 of the dedicated RGB detector (with higher precision and lower recall), while the raw-RGB control isolates the grayscale conversion as the load-bearing step.

configuration	P	R	F1 [95% CI]
RGB detector (`ft4`) on RGB, @1280	0.454	0.916	0.607 [0.590–0.623]
IR detector (`v3b`) on raw RGB, @640	0.362	0.126	0.187 [0.165–0.210]
IR detector (`v3b`) on grayscale RGB, @640	0.543	0.621	0.580 [0.561–0.599]

The control row settles the mechanism question at benchmark grade: fed the raw three-channel RGB, the thermal model collapses to $F1 = 0.187$; fed the *same frames* through a one-line grayscale conversion, it reaches 0.580, a tripling that isolates the single-channel intensity input as the load-bearing step. The zero-shot thermal model is then 2.7 pp F1 behind the dedicated RGB detector on visible-light frames, with the opposite error profile (more precise, less complete). On the real-video diagnostic, where the finding was first observed, the same mode more than doubles the raw-RGB control's aggregate F1 ($0.295 \rightarrow 0.636$), nearly halves bird-confuser FPPI ($0.261 \rightarrow 0.142$), and on the hardest bird-cluttered clip ties the best RGB detector ($F1$ 0.837 vs 0.840, a tie at that clip's sample size):

Table 4.19: Real-video six-mode diagnostic (the finding's discovery surface; 9 drone clips / 1,234 GT instances, 10 confuser clips / 1,250 frames, IoP@0.5). The IR-on-grayscale rows are the discovery; the raw-RGB row is the control.

	Drone (9 videos)			Confuser FPPI (10 videos)			
Mode	P	R	$F1$	All	Bird	Airplane	Helicopter
RGB baseline	0.771	0.749	**0.760**	0.512	0.960	0.451	0.278
RGB `retrained_v2`	**0.909**	0.453	0.605	0.196	0.139	0.385	0.133
RGB `selcom_1280`	0.649	**0.812**	0.721	0.709	1.528	0.493	0.333
RGB `selcom_640`	0.807	0.666	0.730	0.260	0.313	0.359	0.179
IR on grayscale-RGB	0.743	0.557	0.636	**0.158**	**0.142**	**0.283**	**0.103**
IR on raw RGB	0.647	0.191	0.295	0.150	0.261	0.158	0.079

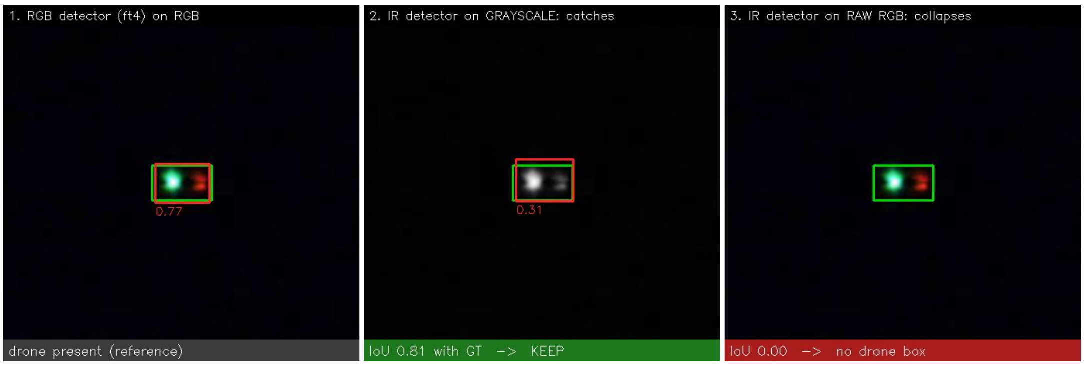


Figure 4.7: Cross-modal transfer, illustrated on one Anti-UAV night frame (each panel cropped around the drone). *Left*: the production RGB detector (`ft4`) on the RGB frame, for reference. *Middle*: the IR detector (`v3b`), trained zero-shot on thermal only, applied to the grayscale-converted frame; it lands the drone (IoU 0.81 with the green GT box). *Right*: the same IR detector on the raw 3-channel RGB collapses (IoU 0.00, no box). The transfer is carried by the single-channel intensity input; the colour information that distinguishes thermal from visible is what breaks it.

4.5.2 Mechanism and Bounds

The likely mechanism is shared low-level statistics: thermal IR and grayscale RGB are both single-channel intensity images that render small drones as silhouettes against sky, and a detector trained on thermal silhouette gradients retains them when colour, the one channel property that genuinely differs, is removed (Figure 4.7). The transfer is partial and patterned: on clean takeoff clips the dedicated RGB detectors win by 28–41 pp F1 per clip (texture-rich close-ups favour visible-light training), while on small-silhouette bird-cluttered scenes the thermal model's discrimination dominates. The IR model has also never seen a visible-light bird, yet its grayscale bird-FPPI (0.142) lands within sampling error of a detector explicitly hard-negative-mined for them (0.139): thermal-domain confuser training transfers along with the detection.

To our knowledge, no published study quantifies this specific transfer: a thermal-only-trained drone detector evaluated zero-shot on grayscale-converted visible-light video against drone ground truth. The adjacent literatures (modality hallucination, RGB-to-thermal translation, domain-adversarial alignment) all provide cross-modal information at training time (Section 2.6); here the transfer emerges from shared low-level structure alone.

4.5.3 The Fallback Value and the Honest Bound

The finding's practical value is as a *fallback*: grayscale mode gives a thermal-equipped system a usable second detection channel when the thermal camera is absent, recovering drone recall on visible-light video the RGB detector alone would otherwise have to carry. The recall it delivers is real but patterned, and the bound is the one fail: on clean texture-rich close-ups the dedicated RGB detectors win by 28–41 pp F1 per clip, while the transfer wins on small bird-cluttered silhouettes (the mechanism subsection above). Grayscale detection runs as the raw thermal model on grayscale input ($F1 = 0.580$ bare on Svanström, Table 4.18); it is a degradation channel, not a primary detection surface.

4.6 Threats to Validity

Internal validity. Three threats. First, the RGB detector's training stances and the patch filter's versions were iterated against Svanström evaluation results; some design decisions are implicitly biased toward Svanström performance. The real-video clips and the SelCom footage, neither consulted during those iterations, are the partial mitigation. Second, `imgsz` acts as a hidden hyperparameter; the protocol (Section 3.2.3) pins it per surface, and the one cross-resolution comparison the thesis needs (Svanström RGB@1280 vs IR@640) is flagged where it appears. Third, the trust classifier was trained against a specific RGB confidence calibration; swapping the RGB detector without re-mining classifier data risks silent degradation (the documented `retrained_v2_32feat` failure, Section 3.8.9).

Construct validity. IoP at 0.5 is more lenient than IoU at 0.5 for small-object detection; numbers here are not directly comparable to IoU-based literature without conversion (Section 3.2.1). Trust-aware scoring is the second construct choice, justified in Section 3.2.2; both rules are declared per table.

External validity. The IR detector and the trust classifier train on frames and sequences that overlap both paired evaluation surfaces; the overlap is quantified per component, and bounded by the held-out clean split, in Section 3.3: Anti-UAV conclusions are unchanged on 57,542 held-out frames (0.973 → 0.977 bare, 0.984 → 0.986 pipeline), the Svanström IR detector's solo numbers carry at most 7.3 pp of in-distribution inflation (part of it sequence difficulty, per the leakage-free RGB control), and the cascade leakage bound is small (the reject-class `robust8` moves −1.4 pp, 0.948 → 0.934; the shipped `robust8-nr` −3.5 pp, 0.946 → 0.911). The deployment evidence is one site (SelCom) and one camera class; generalisation to other CCTV installations is untested.

Sampling and uncertainty. Large surfaces are evaluated on ≈4,000-frame even-spread samples (Section 3.2.3); all tables print n and the source size, and headline cells carry 95% bootstrap CIs. Differences within overlapping CIs are treated as ties; this includes the grayscale three-way's 2.7 pp gap, which is reported as "slightly behind", not as parity. Per-clip real-video numbers (20–340 frames per clip) are indicative; aggregates are the reliable quantities.

Conclusion validity. Single-run point estimates are never compared at sub-CI granularity; the production-stack choices are justified on cost and OOD behaviour where in-domain differences are within noise (Section 4.3).

4.7 Limitations and Future Work

Edge latency. All evaluation ran on a GTX 1050 Ti; the cascade is interactive there, but Jetson-class edge latency is unmeasured. The per-stage budget (Table 4.16) makes the port plausible; it remains future work.

GUI integration of the aligned filter. The production design runs `mlp_v5_ir_aligned` per-frame on the IR branch; it is validated in the evaluation engine, while its operator-GUI wiring is an open engineering item at the time of writing.

The airplane gap (largely closed). Airplanes were every earlier filter generation's weakest category (patch: 52% catch on Svanström airplanes, below the decisiveness bar; the grayscale-harvested thermal filter left the thermal-confuser surface airplane-dominated). The production thermal-native head (Section 4.1.3) closes most of this: thermal-confuser fire falls 29.4% $\rightarrow$ 2.8% on-cache and 94% of held-out IR confusers are removed, by training on thermal airplane crops directly. The residual is no longer an untrained-confuser hole but an in-distribution recall trade (-3.7 pp on genuinely airplane-like `ir_dset` thermal drones no threshold separates from airplanes); choosing that operating point, and reducing it with more diverse thermal airplane-vs-drone data, is the remaining future-work item.

IR OOD recall. The IR detector's recall on independent, verified thermal drone footage outside its corpus families is unmeasured (the available community thermal sets could not be quality-verified and are not used); this is the evaluation gap the RGB parallel channel hedges against.

Track-level classification. The temporal smoother maintains per-track state and renders flight paths, but does not yet classify on track dynamics (velocity, persistence). Drones, birds, and aircraft have distinct flight signatures; a track-based classifier is a largely orthogonal evidence channel and the most promising single direction, particularly for the residual airplane-like thermal drones that appearance alone cannot fully separate.

Open evaluation items. The thermal-confuser per-category breakdown formerly registered here has since been run (a CPU replay of the existing cache): the *bare*-detector `IR_confusers` fire is airplane-driven (35.2% bare airplane fire vs 12.2% bird and 0% helicopter), which is why the thermal-native filter that now suppresses most of it (Section 4.1.3) had to be trained on thermal airplane crops specifically. The video-tuned filter-threshold question is closed by the sweep in Section 4.4.

Chapter 5

Conclusion

This thesis set out to make visual drone detection deployable: high recall and a low false-alarm rate at the same time, on hardware a site can afford. The position it defends is architectural: the base detectors are tuned purely for recall, and every discrimination they cannot learn without losing recall (confuser rejection, modality arbitration, false-positive filtering) is relocated into cheap, learned, downstream stages. The evaluation holds one protocol throughout (Section 3.2); every number below carries its configuration and its confidence interval in the tables it cites.

5.1 Answers to the Research Questions

RQ1 (capability). *Yes: the pipeline suppresses false positives while raising, not merely maintaining, detection performance.* On the discriminating Svanström benchmark the full production pipeline lifts drone $F1$ from 0.742 (bare detectors) to 0.946, with recall *rising* from 0.948 to 0.991 and precision from 0.609 to 0.905 (Table 4.7). On the out-of-distribution confuser corpus, where the bare detector fires on 30.4% of frames, the per-frame filter reduces fire to 1.4%; the reject-class `robust8` ablation drives it to 0.11% (three frames in 2,633, Table 4.11), the suppression the shipped no-reject stack trades for recall. The thermal-confuser surface, formerly the pipeline's weakest front, is largely closed by the thermal-native IR filter: the shipped no-reject stack cuts ~91% of thermal-confuser fire (29.4% → 2.8%, Table 4.11), and airplanes — once the resistant class — no longer dominate the residual (the honest held-out IR_confusers val/test figure is 94% removed, Section 3.8.8). On the saturated Anti-UAV control the pipeline does no harm ($F1$ 0.973 → 0.984, Table 4.8); on ordinary, confuser-free footage the false-alarm problem was already small at the detector, which is precisely why the thesis locates the problem in confuser scenes and solves it there.

RQ2 (attribution). *Each stage earns its place, and the stages are complementary rather than redundant.* The router and the filter address different false positives. For the shipped no-reject router the per-frame filter is the workhorse: on Svanström it removes 83% of bare false positives while the always-routing no-reject router removes almost none alone, and on the confuser corpus the filter is again decisive —

$\sim$7$\times$ fewer residual false positives than the patch-CNN predecessor (Table 4.11). The reject-class `robust8` ablation supplies the complementary mechanism, vetoing whole untrustworthy frames — the net the shipped stack trades for recall. Composition order is a recall/precision dial: filter-then-classify keeps the router's recall ($F1 = 0.946$) while classify-then-filter trades $\sim$1.5 pp of it for precision, so the choice is operational, not structural. Temporal aggregation carries the chapter's sharpest design lesson (Table 4.17): a *per-frame* filter veto with thresholds tuned on benchmark stills over-vetoes OOD drones, whereas the same veto at the *alert gate* gains F1 — the gate position, not the gate, is the variable, so the deployment routes per-frame and verifies at the alert gate. The runtime accounting closes it: both stages are cheap enough (1–4% total overhead; Table 4.16) to run on every frame, which is why the predecessor's alert-gating compromise disappears.

RQ3 (dual-modality value). *Yes: the modality ranking reverses across surfaces, and routing pays on both sides of the reversal.* On Svanström the thermal detector dominates the visible one ($F1$ 0.940 vs 0.607); on Anti-UAV the visible detector dominates (0.985 vs 0.961); the routed pipeline matches or beats the better single modality on each (Table 4.9). The accounting nuance matters: the thermal channel's value concentrates exactly on frames where the visible channel fails, so each modality is scored against its own ground truth and never unioned; under a union rule the same detections would mis-attribute the contribution by double-digit F1 (Section 3.2.2).

The methodological thread. Both contributions behind the components delivered measurable value. The *Model MRI*'s statistics-before-training discipline trained the filter family directly from measured drone/confuser separability (linear separability ≈95% RGB / 0.981 IR; drone detections as positives, confuser detections as negatives), and selected the router's eight features by leakage analysis. Re-running it on its own shipped corpus corrected two figures in an earlier report (Section 4.2). The *HITL data engine* carried the IR detector from $F1$ 0.503 to 0.967 across six revisions on a fixed test split; the one revision that bypassed the review loop (V5) cost 12.7 points of precision and is reported as the protocol's clearest negative result: the dataset state, not the model state, is what regressed (Section 4.1.2).

The findings. The grayscale result stands as measured: a thermal-only-trained detector, run zero-shot on grayscale-converted RGB, still detects drones when the visible channel fails, landing within 2.7 pp F1 of the dedicated RGB detector at matched configuration (0.580 vs 0.607), with higher precision and lower recall, and tying it on the hardest bird-cluttered clip. Its raw-RGB control collapses on both surfaces ($F1$ = 0.187 on Svanström, 0.295 on video), isolating the single-channel conversion as the load-bearing step (Section 4.5). Its practical payoff is a recall-preserving fallback when thermal hardware is unavailable.

5.2 Production Stack

The stack, as validated in this thesis: RGB detector `ft4` (the SelCom-plus-confuser fine-tune; the only confuser-injection recipe that passed every drone-recall regression gate); IR detector `v3b`; trust router `robust8-nr` (the eight free features selected by the leakage statistic, with the `reject` class dropped so the router always routes and the filter owns false-positive rejection), composed in the filter-then-router order (`filt`→`clf`, which keeps recall at the router's level); `robust8` with reject is retained as an ablation and is the recommended choice on confuser-rich paired streams; per-frame filters `mlp_v5` (RGB, the `v4` bird-split build, @0.25) and the thermal-native IR filter (`mlp_aligned_thermalonly`, CBAM-trained, @0.05); an N-of-M temporal smoother above them; inference at `imgsz=1280` on Svanström-like and SelCom-like content (960 is the measured deployment optimum for the SelCom camera) and 640 elsewhere; trust-aware scoring as the reporting rule. One validated *operating mode* extends it: on recall-starved OOD RGB surfaces, the RGB confidence floor can drop to 0.05–0.10 with the filter owning precision. The floors existed to control false positives, and that job has moved downstream (+10 pp F1 on SelCom at unchanged confuser safety; Section 4.3.6).

Two carve-outs accompany it, each diagnosed rather than hidden. The earlier `rgb_dataset` coverage gap — the shipped `mlp_v5` cost 11 pp F1 there — is now *closed*: the `v4` bird-split build lifts `rgb_dataset_test` F1 from 0.792 to 0.916 (recall 0.691 → 0.887), so the patch filter is no longer needed as a fallback on that surface (Section 4.1.3). *(i)* the shipped no-reject `robust8-nr` trades confuser suppression for recall: RGB-confuser frame fire rises to ∼1.4% and real-video-confuser fire ∼3× relative to the reject-class `robust8`, which is retained as the recommended router where confuser-rich paired video dominates (Section 4.1.4). *(ii)* The thermal-native IR filter is validated per-frame in the evaluation engine; its operator-GUI wiring is an open engineering item at the time of writing, as is Jetson-class edge latency.

5.3 Future Work

Four directions. A *track-based classifier* consuming the temporal smoother's per-track state (velocity, persistence, trajectory curvature) as a fourth evidence channel, the most promising single direction, and the one aimed at the airplane class that appearance-based stages handle least well. *Further hardening the airplane class*:

the thermal-native filter already cut the airplane-dominated thermal-confuser fire 29.4% $\rightarrow$ 2.8% (cache) / 94% (held-out), and additional OOD airplane crops as confuser-class training data would press the residual lower. *Completing the engineering*: GUI wiring of the aligned filter and TensorRT/ONNX edge benchmarking. And *extending the protocol*: the open-world `fusion_no_fn_v1.1` router remains selectable where missed drones are operationally cheaper than false alarms, and the dual-classifier extension (Section 4.1.4) is validated if a deployment needs the missing-modality regimes as first-class citizens.

Appendix A

Datasets in Detail

This appendix consolidates the per-source / per-clip dataset metadata that the methodology chapter summarises. Numbers come from the dataset YAMLs (`rgb_dataset/dataset.y` `IR_dset_final/dataset.yaml`), the documentation file shipped with the confuser corpus (`rgb_confusers_merged/dataset_documentation.md`), the Svanström paired-corpus meta file, and the YouTube extraction manifests.

A.1 Composite RGB Training Corpus

The full distribution by source prefix is given in Table 3.2; the split policy is 80 / 10 / 10 with sequence-disjoint partitioning where the source did not already supply its own splits. Six negative-source rows (VIRAT, UA-DETRAC, BDD100K) contribute the 21.9% background-negative fraction. The AirBird subset (10,000 images) is the bird hard-negative channel the baseline detector relies on; the composite corpus does not contain Svanström bird / airplane / helicopter splits, which are therefore OOD at evaluation time.

A.2 Confuser Hard-Negative Corpus

Per-source composition is in Table 3.3. Split policy:

- **Sources with existing splits** (Airplane Roboflow, New_Dataset Roboflow): original train / valid / test partitions preserved.
- **Sources without splits**: split by video / sequence ID at 80 / 10 / 10 to prevent leakage.
  - Svanström: 165 video sequences split by sequence ID.
  - raihanrsd: 58 video folders split by folder.
  - Helicopter Kaggle: 20 hash-based pseudo-groups.
- **Mislabeled frame removal**: 10 frames in the airplane / New_Dataset sources contained visible drones and were excised before training.

The mined-hard-negative process for `retrained_v2` ran the baseline detector on every confuser image and kept the 11,729 frames (43.4%) on which the baseline fired; those mined frames were merged with the composite RGB corpus to produce the final 183,751-image `retrain_dataset`. `hardneg_v3more` differs in using a smaller, less aggressive merge.

A.3 IR Detector Corpus

The per-prefix distribution is in Table 3.4. The Svanström IR partition (`svan`, 21,637 frames across the corpus' train/val/test splits: 17,314 train, 2,050 val, 2,273 test, plus 1,325 zoom-augmented copies under `czoom`) is the largest single Svanström appearance in any training corpus in this thesis and is the reason Svanström IR evaluation is in-distribution for the IR detector (Table 3.6).

A.4 Thermal Confuser Corpus (`IR_confusers`)

5,938 thermal frames containing aerial confusers and no drones: 4,281 airplane, 1,200 bird, and 457 helicopter frames (category encoded in the filename prefix), exported from Roboflow-hosted thermal confuser video; source streams disjoint from every `ir_dset_final` source family. The evaluated split holds 5,237 frames, sampled to 4,000 by even spread (3,043 airplane / 871 bird / 86 helicopter). All frames carry empty YOLO labels; every detection on this surface is a false positive by construction. The corpus contributes no training data to any model; it is the OOD stress surface for the IR branch (Section 3.1.4). The thermal-native head is trained with the GT-aware CBAM *train* split; its held-out gates are the disjoint *CBAM valid* split (drone recall recovered 0.717 $\rightarrow$ 0.967) and the held-out `IR_confusers` val/test re-mine (90 $\rightarrow$ 22 residual fires, 94% removed; Section 3.8.8).

A.5 Svanström Paired

28,710 frames, sampled at stride 3 from 279 source video sequences. Per-class instance counts: drone 11,695; airplane 6,090; bird 5,298; helicopter 5,627. Native resolution 640×512 (RGB) with a co-registered IR channel at the same resolution. Annotators drew loose boxes around small drones, motivating the IoP@0.5 scoring rule of Section 3.2.1. The dataset ships as source videos with MATLAB `groundTruth` annotation objects [18]; the frames and YOLO labels used here were produced by this project's own conversion script, so box geometry may differ marginally from the authors' MATLAB pipeline, and the original videos are retained alongside the converted corpus.

A.6 Anti-UAV RGBT

The converted YOLO test + val splits used for evaluation comprise 85,374 paired frames at 1920×1080 (RGB) and 640×512 (IR). Anti-UAV is a tracking benchmark in its native form; the per-frame detection evaluation reported in this thesis is computed by re-scoring the tracking benchmark's frame-level GT boxes with the IoU@0.5 matching rule.

A.7 DUT Anti-UAV

The DUT Anti-UAV detection dataset [41] ships 10,000 aerial RGB images with author-provided train / validation / test partitions. It enters this thesis twice. As a *training* source, 5,200 of its images contribute to the composite RGB corpus (the `dut` prefix in Table 3.2), so the production RGB detector `ft4` is in-distribution on it. As an *evaluation* surface, the official test split (the 2,200 frames / 2,245 ground-truth drones used in Table 4.10) is the confuser-free, all-drone RGB check of Section 4.3.1, evaluated at `imgsz=960` with the thermal detector in its grayscale-fallback mode. No DUT frame used at evaluation time was a training image for the trust router or the per-frame filters.

A.8 SelCom CCTV

Single-camera source: fixed perimeter installation, 1920×1080, 25 fps. Source clip ∼1:30, 2,076 extracted frames (1,953 positive + 123 explicit negative). Median drone $\sqrt{\text{area}} \approx 36.8$ px in the native frame (so ∼12 px at `imgsz=640`, ∼24 px at `imgsz=1280`). The 80 / 20 train / val split (20% pure-SelCom validation, seed 0) is preserved across the `ft1` / `mixed_ft1` / `mixed_ft2` / `mixed_ft2_1280` variants; the strongest CCTV-evaluated fine-tune is `Yolo26n_selcom_mixed_ft2_1280`, a comparison detector; the production RGB detector is `ft4` (Section 5.2).

A.9 YouTube Real-Video Diagnostic

Note: the evaluated drone-positive set is the 9-clip / 1,359-frame / 1,234-GT-instance subset of this table that re-ran cleanly through the corrected-scoring pipeline (`flock_of_birds_attack_drone` excluded; some clips hold slightly fewer frames on disk than at extraction time). All Chapter 4 video numbers, including the temporal evaluation, use that evaluated subset.

The 10 confuser clips were drawn from public YouTube sources visually selected to

Table A.1: Drone-positive YouTube clips. “Extracted” is the frame count used in evaluation after stride sampling; total native frame count is shown for context. “Source” is the YouTube video ID (prepend `https://youtu.be/`), recovered from the extraction manifest.

Clip	FPS	Total frames	Stride	Extracted	Source
`drone_and_bird_sky_and_trees_short`	30.0	506	0 (full)	114	`KoUgV6W`
`drone_attacked_by_bird_mountain_side_view`	30.0	400	3	133	`yAbMFQr`
`drone_over_mountain_attacked_by_birds`	30.0	325	3	108	`1qRERzV`
`drone_seagull_attack`	30.0	1,146	0 (full)	235	`nTdbEPx`
`drone_takeoff_from_ground_and_not_hand_short`	60.0	1,379	0 (full)	163	`85XdkfW`
`drone_takeoff_short`	60.0	637	0 (full)	116	`Qx1Hlot`
`drone_takeoff_short_trees_background_dji`	60.0	954	0 (full)	166	`J-gcjDU`
`flock_of_birds_attack_drone`	30.0	338	3	112	`OAjZ2bg`
`flock_of_seagulls_attack_drone_beach`	60.0	1,010	3	336	`DdkFYVs`
`two_birds_drone`	30.0	456	0 (full)	150	`ngMjVZf`
Total		7,151		**1,633**	

Table A.2: Confuser-only YouTube clips. All frames are confuser-positive only; every detection in these clips is a false positive by construction. “Source” is the YouTube video ID (prepend `https://youtu.be/`); two compilation clips were saved without the ID marker.

Clip	FPS	Total frames	Strid
`airplanes_airplanes_compilation`	25.0	6,248	2
`airplanes_distant_airplane_over_head_flying_away`	29.0	1,606	2
`birds_birds_flying_overhead_various_sizes_short`	30.0	201	1
`birds_birds_in_slow_motion_flying_various_sizes_compilation`	50.0	13,550	5
`birds_distant_birds_flying_in_the_sky_short`	30.0	462	2
`birds_flock_of_birds_flying_short`	30.0	259	1
`birds_flock_of_birds_flying_sunset`	24.0	492	2
`helicopters_helicopter_compilation`	25.0	13,871	2
`helicopters_helicopter_overhead_short`	25.0	260	1
`helicopters_helicopter_overhead_very_small_airplane_in_background`	—	—	—
Total		36,949	

span the three confuser categories at multiple ranges and viewing angles; the video IDs above are recovered from the `_Media_` marker the downloader embedded in the clip filenames, preserved in `datasets/confuser_videos/extraction_manifest.json`. The aggregate confuser-FPPI numbers in Section 4.3 use the 1,250-frame totals.

A.10 Licensing and Ethics

The public benchmark datasets (Anti-UAV, Svanström, the Roboflow community sets) are used under their published research licenses; the IR component sources (FLIR, DV5, CST, etc.) are similarly research-license open. The SelCom CCTV stream is deployment-partner proprietary and is not redistributable; only summary metrics and qualitative figures derived from it appear in this thesis. The YouTube clips are short, low-resolution excerpts used under fair-use for research and contain no personally identifying information; the encoded source IDs in the clip filenames are preserved so the originals can be located by an interested party. None of the corpora include human faces or location markers beyond what is visible in the original public datasets.

Appendix B

Models Evaluated

This appendix lists the models the thesis actually discusses: the production pick at each cascade stage (**production**) and the comparison or ablation variants that carry a finding in the main text. The complete experimental registry, including every intermediate, superseded, and archived checkpoint, is retained in the project repository for reproducibility. The IR detector's full six-revision evolution is reported separately in Table 4.3.

Model	Role	Note
RGB detectors (YOLOv26n)		
`ft4`	**production**	SelCom-plus-confuser fine-tune; the only confuser-injection recipe that passed every drone-recall regression gate.
`baseline`	comparison	Stage-1 reference; best small-drone recall at `imgsz=1280`; base for the SelCom fine-tunes.
`hardneg_v3more`	comparison	Hard-negative mining: lowers helicopter and airplane fire but leaves bird fire essentially unchanged.
`retrained_v2`	comparison	Aggressive confuser retrain; suppresses birds but collapses Svanström drone recall to $R = 0.306$.
`selcom_mixed_ft2_1280`	comparison	CCTV (SelCom) fine-tune weights at `imgsz=1280`.
IR detectors (YOLOv26n, thermal)		
`ir_v3b`	**production**	Production IR detector; $F1 = 0.967$ on the fixed IR test split; two-epoch corrective fine-tune of `Final`.
`Final`	comparison	Immediate parent of `v3b`; the full V2–V6 / `Final` trajectory is in Table 4.3.
Trust classifiers (XGBoost fusion / routing)		
`robust8-nr`	**production**	Shipped trust router: the eight `robust8` features with the `reject` class dropped (always routes RGB/IR/both; the filter owns false-positive rejection).

Model	Role	Note
`robust8`	ablation	Four-class router (`robust6` + two free features + a tuned per-class threshold); recommended on confuser-rich paired streams, retained as the reject-class ablation.
`robust6`	comparison	Six statistically-selected, leakage-free features; the base extended into `robust8`/`robust8-nr`.
`sa32`	comparison	Strongest hand-engineered (32-feature) classifier; best in-domain on Svanström, but leaks scene statistics so its lead is not trusted to generalise to unseen surfaces; superseded in production by `robust8-nr`.
`control40`	comparison	40-feature control: best raw drone metrics but the worst confuser rejection.
`fusion_no_fn_v1.1`	comparison	Open-world fallback; the most conservative variant on the OOD confuser zoo.
Confuser filters		
`mlp_v5` (v4 build)	**production**	Production RGB filter: a distilled MLP on the detector's own `p3`+`p5` features, run per-frame; the `v4` bird-split build closes the `rgb_dataset_test` coverage gap ($F1$ 0.792 → 0.916, recall 0.691 → 0.887).
`mlp_aligned_thermalonly`	**production**	Thermal-native IR confuser filter (CBAM-trained, @0.05); held-out CBAM recall recovered 0.717 → 0.967 at FP 6, with 94% of held-out thermal confusers removed.
`patch_v2`	comparison	Superseded MobileNetV3 patch filter; audited predecessor, the `v4` RGB filter removes its last fallback role.

Table B.1: Models referenced in this thesis: the production pick at each cascade stage and the comparison or ablation variants discussed in the main text. The complete experimental registry is retained in the project repository.

Appendix C

Model MRI Sample Report

The Model MRI renders a full human-readable report for every run (Section 3.7); the report for the production RGB filter's corpus is at `mri/results/v5_report_regen/report.md`, and Figure 3.8 reproduces its verdict block. For that corpus (19,334 drone / 13,597 confuser detections, bare-detector hallucination 54.4%, LDA separability 0.952) the instrument projected, *before any training*, a 97.4% confuser false-positive cut at 98.9% drone-recall retention, and recommended the filter on that basis. The shipped filter's measured behaviour on the canonical surfaces (Section 4.1.3, Table 4.11) matches that projection; the agreement between pre-training projection and post-training measurement is itself part of the instrument's validation.

Appendix D

Number Provenance and Reproducibility

Every headline number in this thesis is the value of a named cell in a canonical results file, and every such file is reproducible by a stated command. The map lives in the project repository as `runs/README.md`; this appendix reproduces it. Three layers make the chain auditable:

1. **Unified detection cache.** `thesis_eval/pipeline_cache_unified.py` runs the production detectors *once* per evaluation surface and stores, per frame, every detection (box, confidence, the 517-D feature vector, patch-classifier probability), the trust classifier's input features, and the per-modality ground truth. Caches are tagged with the detector weight identity.
2. **Deterministic replays.** Every evaluation in Chapter 4 is a CPU re-score of that cache (no detector forward pass), so any reader with the repository can re-derive any table in about a minute per surface. The replay scripts and their outputs are listed in Table D.1.
3. **Automated audit.** `thesis_eval/_audit_headline_numbers.py` asserts, cell by cell, that the numbers printed in this document equal the values in the canonical results files; the submission build of this thesis passes that audit.

The `runs/` directory additionally holds the production IR detector's complete Ultralytics training record (`runs/corrective_finetune/`: training arguments, loss and PR curves, confusion matrices, and acceptance reports for the `v3b` fine-tune), so the training side of the production stack is documented alongside the evaluation side. The relational evidence store `knowledge/` (CSV tables with generated views) records every evaluation and finding as a row carrying its source path and reproduction command.

Table D.1: Headline-number provenance: thesis claim → canonical results file (under `runs/`, frozen copies of `thesis_eval/results/`) → reproduction command. All *pipeline* replay commands are zero-GPU; the two rows marked `(GPU)` (the IR-detector trajectory and stage latencies) require the detectors and target hardware.

Thesis number	File (cell)	Reproduce
Svanström composed F1 0.742 → 0.946 (shipped `robust8-nr`, `filt`→`clf`)	`results_noreject/tier1_results.json` (`svanstrom.B_pipeline`)	`pipeline_eval_unified.py -only svanstrom`
Confuser fire 30.4% → 1.4% (shipped `robust8-nr`; 0.11% `robust8` ablation)	`results_noreject/tier1_results.json` (`rgb_confuser.C_confuser`)	`pipeline_eval_unified.py -only rgb_confuser`
Anti-UAV no-harm 0.973 → 0.984	`tier1_results.json` (`antiuav.B_pipeline`)	`pipeline_eval_unified.py -only antiuav`
Grayscale 3-way 0.607/0.187/0.580	`tier1_results.json` (D table)	`pipeline_eval_unified.py`
Temporal 2-of-3 windows (Table 4.17)	`temporal_results.json`	`temporal_replay.py`
Per-size buckets (Table 4.13)	`notes_round1_results.json` (`SZ_per_size`)	`notes_round1_replays.py`
Background profile (Table 4.14)	`failure_profile_results.json`	`failure_profile_aggregate.py`
Low-conf mode (Table 4.15)	`conf_sweep/conf_sweep_results.json`	`conf_sweep_replay.py`
IR HITL trajectory 0.503 → 0.967	`knowledge/evals.csv` (`ir_final_*`, one fixed split)	`eval/ir_version_comparison.py` (GPU)
Stage latencies (Table 4.16)	`knowledge/ledger.csv` (speed rows)	`eval/bench_speed.py` (GPU)
Scoring swing 2.8 pp (Section 3.2.2)	`notes_round1_results.json` vs `tier1_results.json` (routed `+filt` / `clf->filt`)	`notes_round1_replays.py`
Overlap audit counts (Table 3.6)	`clean_split/README.md` + `leakage_controlled.json`	`leakage_controlled_replay.py`
Clean-split results (Table 3.7)	`clean_split/clean_split_results.json` (sequence lists alongside)	`cache -only svanstrom_clean,antiuav_clean -full`, then `pipeline_eval_unified.py`

Appendix E

Glossary of Abbreviations and Shorthand

C-UAS Counter-Unmanned Aerial Systems. The defensive use-case context for this thesis.

UAV Unmanned Aerial Vehicle, used interchangeably with "drone" for consumer-class quadcopters in this thesis.

YOLO You Only Look Once. The single-stage object-detection family [5]; this thesis uses YOLOv26n via the Ultralytics framework [6].

CNN Convolutional Neural Network. The patch filter is a MobileNetV3-class CNN [42].

HITL Human-in-the-Loop. The dataset / model co-development protocol of Section 3.5.3.

OOD Out-Of-Distribution. Imagery drawn from a different population than the model's training distribution.

FPPI False Positives Per Image. The per-frame false-alarm count, used as the primary metric on confuser-only evaluation surfaces.

FPR False Positive Rate. In this thesis: the frame-level fraction of confuser-only frames with at least one detection.

IoU Intersection over Union. The standard detection-matching rule; used for Anti-UAV scoring.

IoP Intersection over Prediction. The matching rule used for Svanström RGB scoring (Section 3.2.1); IoP = $|B_{\text{pred}} \cap B_{\text{gt}}|/|B_{\text{pred}}|$.

S1 / S2 / S3 Cascade stages: S1 = RGB detector alone; S2 = + trust classifier; S3 = + patch filter alert gate.

ft4 `Yolo26n_selcom_confuser_ft4_1280`. The production RGB detector (SelCom+confuser fine-tune); the only confuser-injection config that passed every drone-recall regression gate (Section 3.4).

robust6 The six-feature trust classifier (per-modality detection confidence + box geometry), its features screened by the scene-fingerprint ratio (Section 3.8.9); the base that `robust8` extends.

robust8 The four-class trust classifier (Section 3.8.9): `robust6` plus two further free features and a tuned per-class decision threshold; retained as the reject-class ablation and recommended on confuser-rich paired streams.

robust8-nr The shipped production trust router (Section 4.1.4): the eight `robust8` features with the `reject` class dropped, so it always routes RGB/IR/both and the per-frame filter owns false-positive rejection.

sa32 `scene_aware_v3more_32feat`. The strongest hand-engineered comparison trust classifier (32-feature XGBoost); it leaks scene statistics (discriminates *which scene* a frame came from, not *whether* its detections are trustworthy), so its in-domain lead is not trusted to generalise, and it is superseded in production by `robust8-nr` (Section 3.8.9).

fnfn `fusion_no_fn_v1.1`. The open-world fallback trust classifier (40-feature).

control40 `control_v3more_40feat`. A previous-candidate 40-feature trust classifier; deprecated.

v3b The production IR detector weights: 2-epoch corrective fine-tune on top of the `Final` IR checkpoint.

v2 (patch filter) `confuser_filter4_{rgb,ir}_v2_backup.pt`. The MobileNetV3 patch filter; the audited predecessor, superseded in production by `mlp_v5`.

`mlp_v5` The distilled feature-space confuser filter (Section 3.8.11): an MLP on the detector's fused `p3`+`p5` features that supersedes the v2 patch filter on confuser-rich surfaces and runs per-frame; production is the `v4` bird-split build.

IR filter The thermal-native IR confuser filter (Section 3.8.8): a single head (`mlp_aligned_thermal`
CBAM-trained, @0.05) trained directly on thermal confuser crops.

selcom_1280 The SelCom CCTV RGB fine-tune `Yolo26n_selcom_mixed_ft2_1280` evaluated at `imgsz=1280`; a comparison/ablation detector (the production RGB detector is `ft4`, Section 5.2).

selcom_640 The same SelCom weights evaluated at `imgsz=640` (resolution-ablation variant).

`imgsz` Ultralytics short for input letterbox size. Drives small-object recall (Section 3.8.6).

`patch_thr` The decision threshold on p_{confuser} at the patch-filter alert gate. Production: 0.9 on Svanström-like inputs, 0.7 on real-video.

Appendix F

Superseded Components and Design History

This appendix collects the figures documenting components and operating points that were superseded in the production pipeline, gathered here so the main text of Chapter 4 stays focused on the shipped system. Each is referenced from its design-history discussion there.

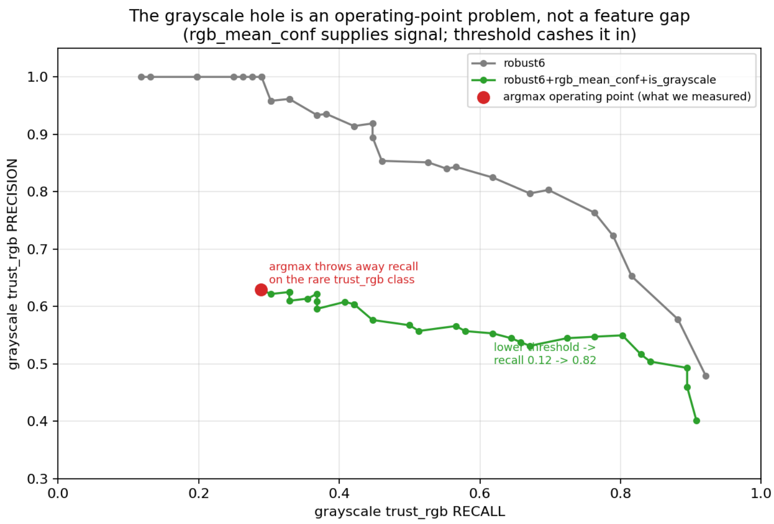


Figure F.1: Design-history: the grayscale `trust_rgb` hole was a decision-rule problem inside the four-class router, not a feature-space limit. Sweeping the per-class `trust_rgb` threshold τ trades precision for recall along a well-separated curve; the reject-class `robust8` operated at τ=0.20. The shipped `robust8-nr` drops the reject class and removes this knob entirely (the hole cannot form without a `reject` label). Source: `eval/compare_routing_pipeline.py`.

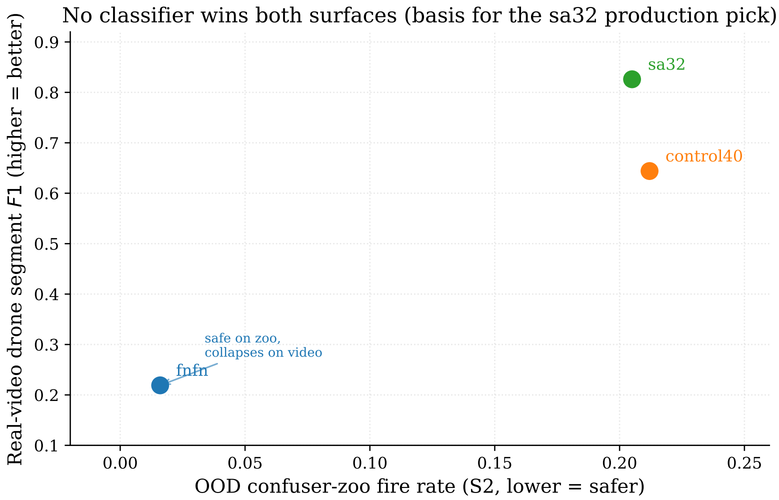


Figure F.2: No trust classifier wins both surfaces. On the OOD confuser zoo `fusion_no_fn_v1.1` is far the safest (S2 fire 0.016) but on real video it collapses drone $F1$ to 0.219; `sa32` inverts the ranking, dominating real-video drone $F1$ (0.826) at a higher zoo fire (0.205). This surface reversal motivated the `sa32` working pick, later superseded by the statistically-derived `robust6`/`robust8` (Section 3.8.9) and the shipped no-reject `robust8-nr` (Section 4.1.4); `sa32`'s scene-statistic leakage is why even its wins here are not trusted to generalise.

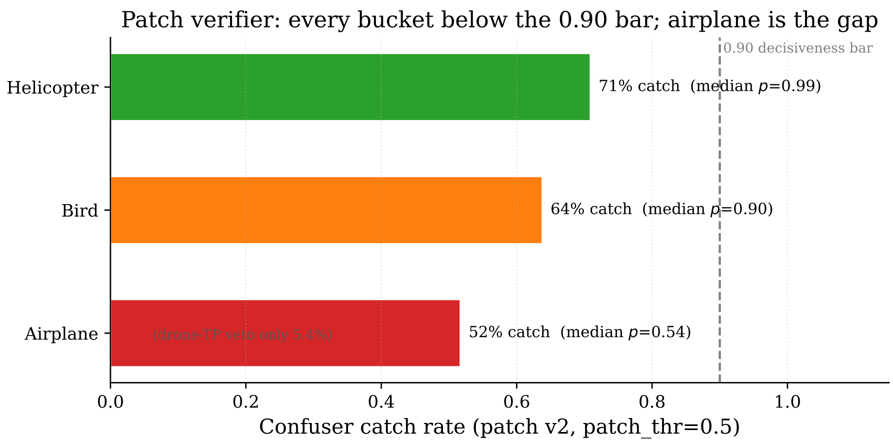


Figure F.3: Patch-filter (v2) per-bucket confuser catch rate at `patch_thr`=0.5, against the 0.90 "decisiveness" bar. Helicopters (71%) and birds (64%) are mid; airplanes (52%, median patch probability 0.540) are the actionable gap. All three buckets sit below the bar, the motivation for the feature-reuse successor.

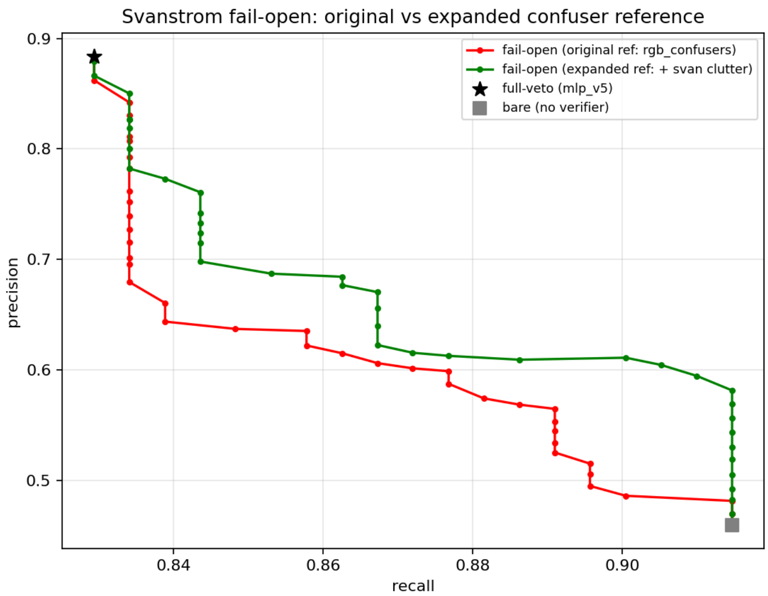


Figure F.4: Design-history: the recall/precision frontier on Svanström for the no-retrain fail-open gate (with the original `rgb_confusers` reference, red, versus a reference expanded with held-out Svanström clutter, green), against the full-veto operating point (star) and the bare detector (square). This gate was the candidate fix *before* the bird-split re-mine closed the coverage gap directly; it is retained only as the record of why a calibration patch was inferior to fixing the training coverage. Source: `eval/failopen_expanded_ref.py`.

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